SYNOPSYS®
Silicon to Software™

Route

Synopsys University Program

WEEK 12 Introduction to Routing
WEEK 13 Route & Route_opt
WEEK 13 Homework

WEEK 12 : Introduction to Routing

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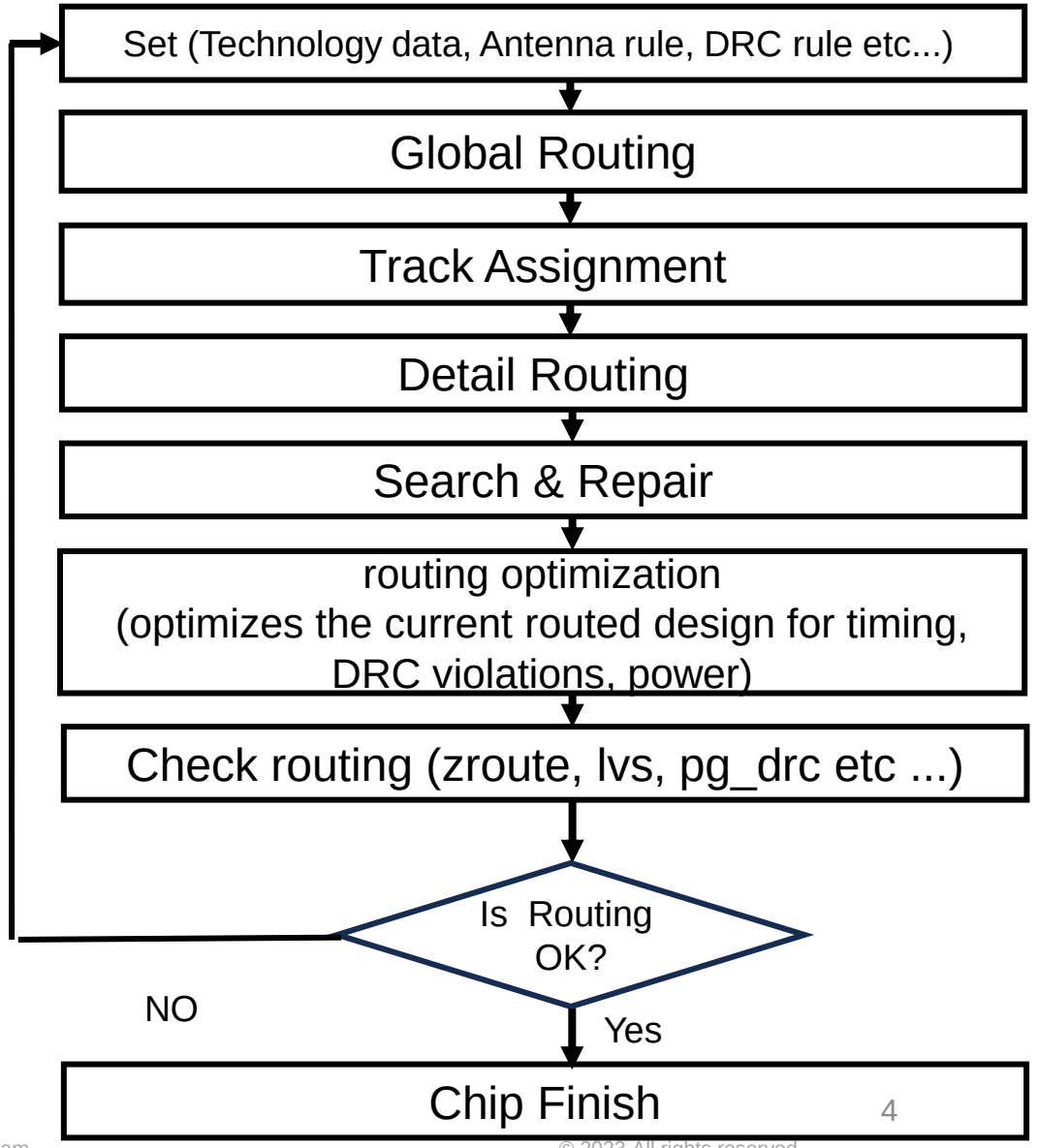
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Introduction to Routing

- Routing Overview
  - This module provides an overview of some of the key steps in C TS :
  - **Defining**
    - Tech file (.tf)
    - Netlist
    - Library files (.lib & .lef) & TLU+ file
    - All cells & ports or PADS should be legality placed with clock tree stru ctures & CTS DEF file
    - NDR rules
    - Routing Blockages
    - Technology data (Metal layers (lef, tech etc...), DRC rules, via creation rules, antenna rules, grid rules(Metal pitch) etc...)
  - **Overview to Routing (directory, input files, script)**
  - **Routing & Routing optimization**



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Introduction to Routing : Overview to Routing

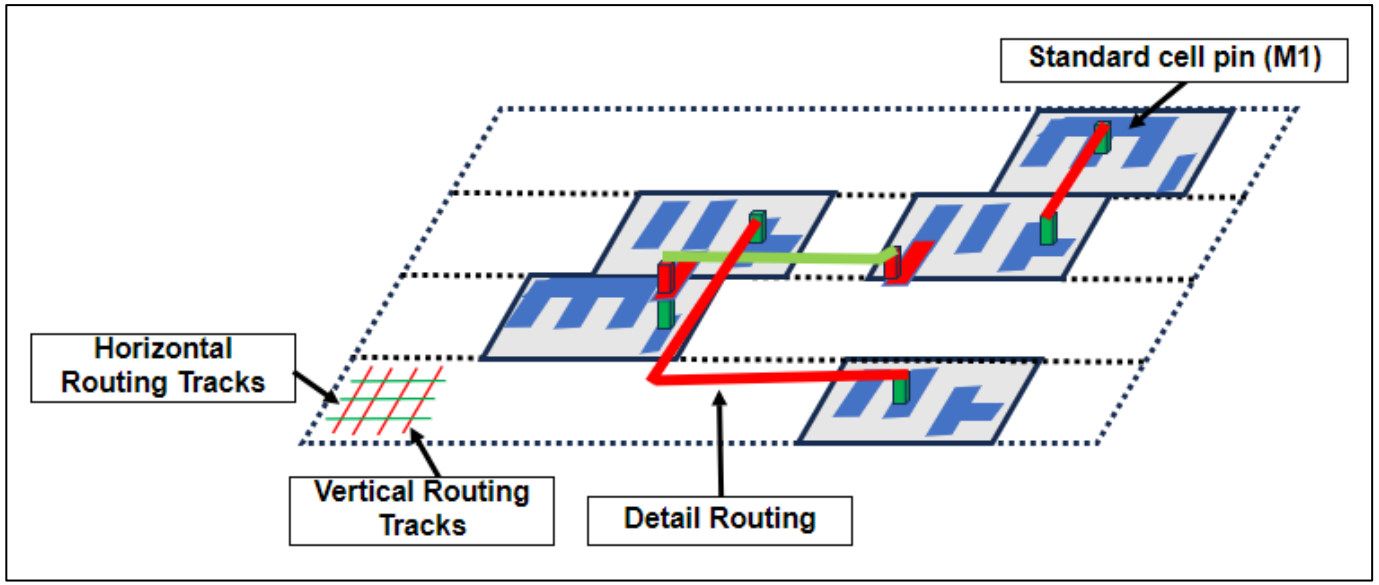
- **Importance of Routing**
  - Device (Gate) delay decreases
  - Interconnect resistance increases
  - Area component of interconnect layers increase, in an attempt to offset increasing interconnect resistance
  - Interconnect capacitance dominates total gate loading
  - Vertical heights of interconnect layers increase, in an attempt to offset increasing interconnect resistance

- **Objectives of Routing**
  - Clean Open/Short violation
  - Skew requirements
  - Routed paths must meet setup & hold Timing margin
  - DRVs max (Capacitance & Transition must be under the limit)
  - Metal traces must meet physical DRC requirements
  - Layout geometries should meet Current Density specification

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Introduction to Routing : Overview to Routing

- **What is Routing**



- Routing Make physical connections between signal pins using metal layers
- Routing is the step after CTS and optimization where exact paths for the interconnection of STD cells and Macros, IPs and I /O pins are determined
- Electrical connections using metals and vias are created in the layout, defined by the logical connections present in the Netli st
- After CTS, we have information of all the placed cells (STD cells, Macros, IPs, I/O PADs), blockages, clock tree buffers/inver ters and I/O pins

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Introduction to Routing : Overview to Routing

- **What is Routing**
  - **The ICC2 relies on this information to electrically complete all connections defined in the Netlist such that;**
    - Minimal DRC violations while Routing
    - The design is 100% routed with minimal LVS violations (short, open)
    - There are minimal SI related violations
    - There must be no or minimal congestion hot spots (cause of short or open net)
    - The Timing DRCs & QoR are met and good respectively

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Introduction to Routing : Overview to Routing

- **Inputs of Routing**
  - Netlist
  - All Cells, ports should be legally placed with clock tree structure & CTS nlib or DEF file
  - NDR rules
  - Routing blockages
  - Technolgy data (Metal Layers (lef, tech file ...) DRC rules(Antenna rules), via creation rules, grid rules (metal pitch)etc …)

- **Goals of Routing**
  - Minimize the total interconnect/wire length
  - Minimize the critical path delay
  - Minimize the number of layer changes that the connections have to make (minimize the number of vias)
  - Complete the connections without increasing the total area of the block
  - Minimizing the congestion hotspots
  - Meet the Timing DRCs and Timing QoR
  - SI driven (reduction in crosstalk noise and delta delays)

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Introduction to Routing : Overview to Routing

- **Constraints of Routing**
  - Set limits on routing to specific regions
  - Set constraints to number of layer to be used during routing
  - Set the Maximum length for the routing wires
  - Blocking routing in specific regions (routing blockage)
  - Set stringent guidelines for minimum width and minimum spacing
  - Set preferred routing directions to specific metal layers during Routing (Horizontal & Vertical Routing)
  - Constraining the routing Density
  - Constraining the Pin connection

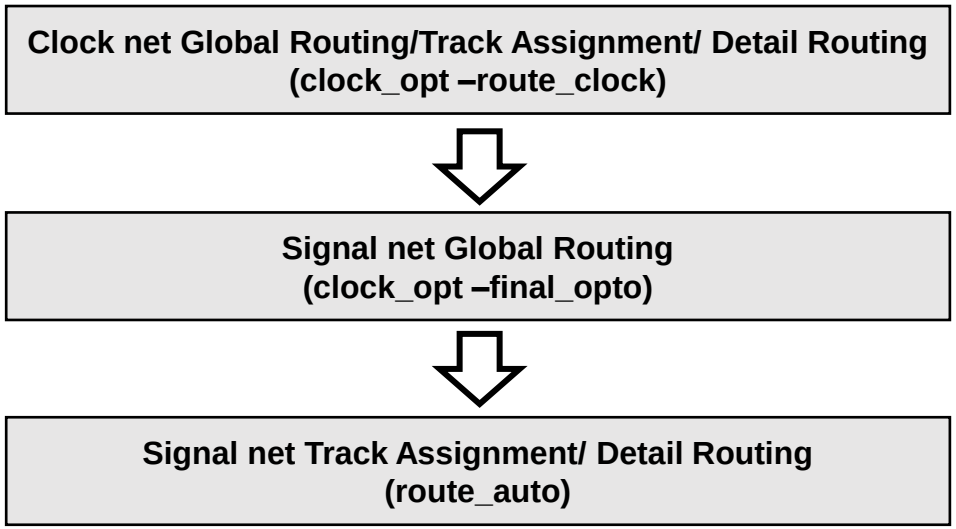
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Introduction to Routing : Overview to Routing

- **Routing Flow**
  - Clock nets should be detail routed (during clock_opt ***route_clock*** stage)
  - Signal nets are global routed during the clock_opt ***final_opto (PSYN)*** stage
  - Signal net routing must next be completed with

route_auto

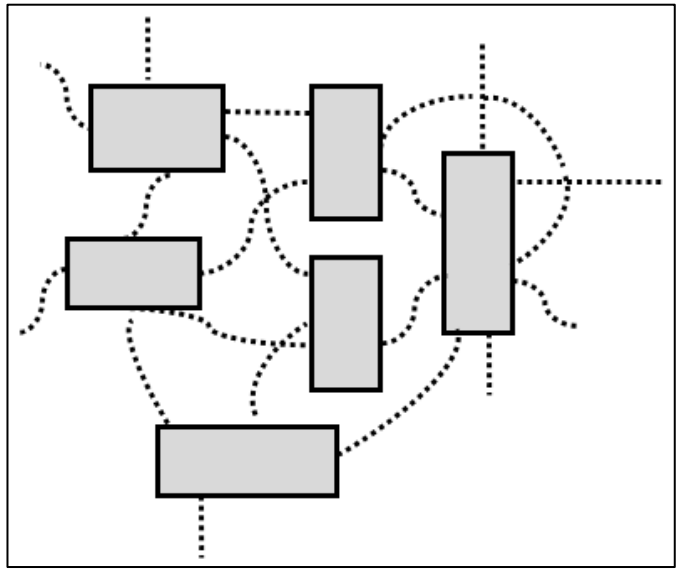
Performs all routing stages in one step. If the design is already global routed, the command runs track assignment and detail routing. Otherwise, the command runs global routing, track assignment, and detail routing.



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Introduction to Routing : Overview to Routing

- **Global Routing**

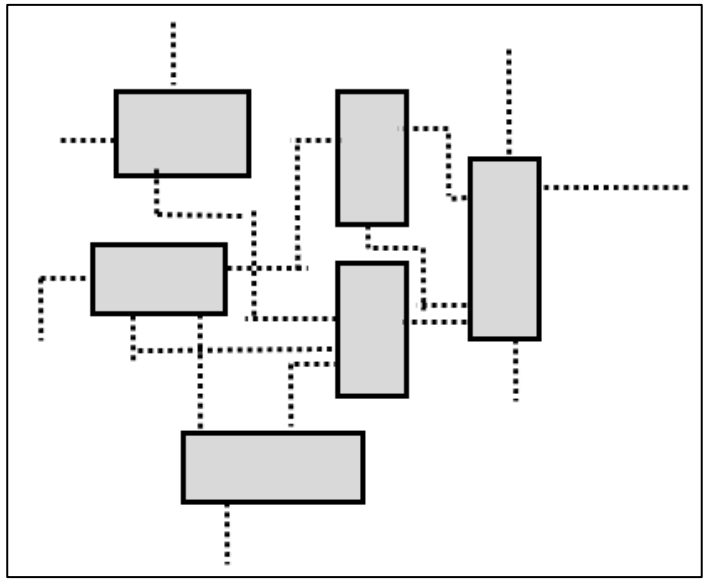


  - Identifying routable path for the nets driving/driven pins in a shortest distance
  - Does not consider DRC rules, which gives an overall view of routing and congested nets
  - Assign Layers to the nets
  - Avoid congested areas and also long detours
  - Avoid routing over blockages (routing blockage)
  - Avoid routing for pre-route nets such as (Power Rings/Stripes/STD rails)
  - Identify and assign net segments over the specific routable window called GRC(Global Route Congestion)

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Introduction to Routing : Overview to Routing

- **Detail Routing**

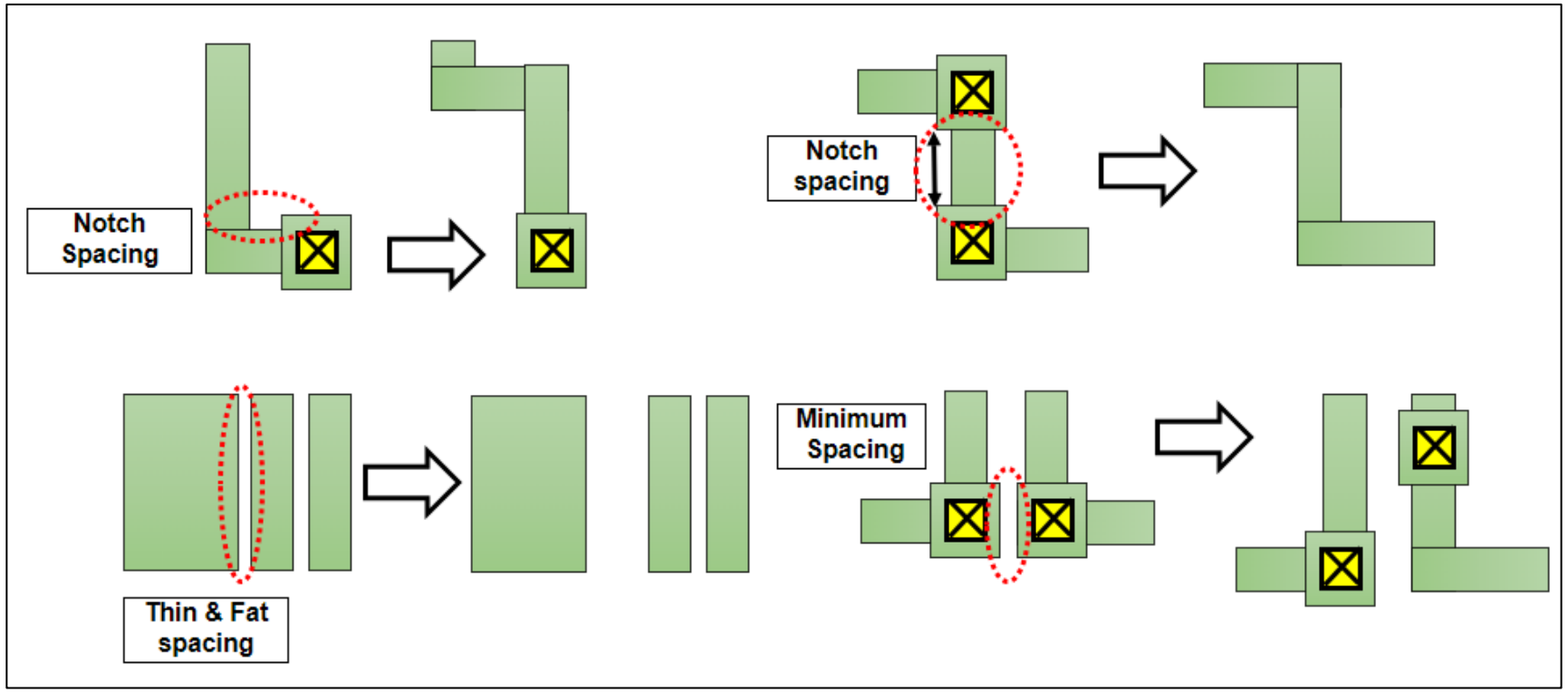


  - Detailed routing follows up with the track routed net segments and performs the complete DRC aware and timing driven rout ing
  - It is the final routing for the design built after the CTS and the timing is freeze (after Clock net routing)
  - Filler Cells are adding after Detailed Routing
  - Detail Routing is done after analyze the cause for congestion in the design, add density screen or change floorplan
  - The main goal of detailed routing is to complete all of the required interconnect without leaving open, shorts or spacing violat ions

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Introduction to Routing : Overview to Routing

- **Detail Routing**

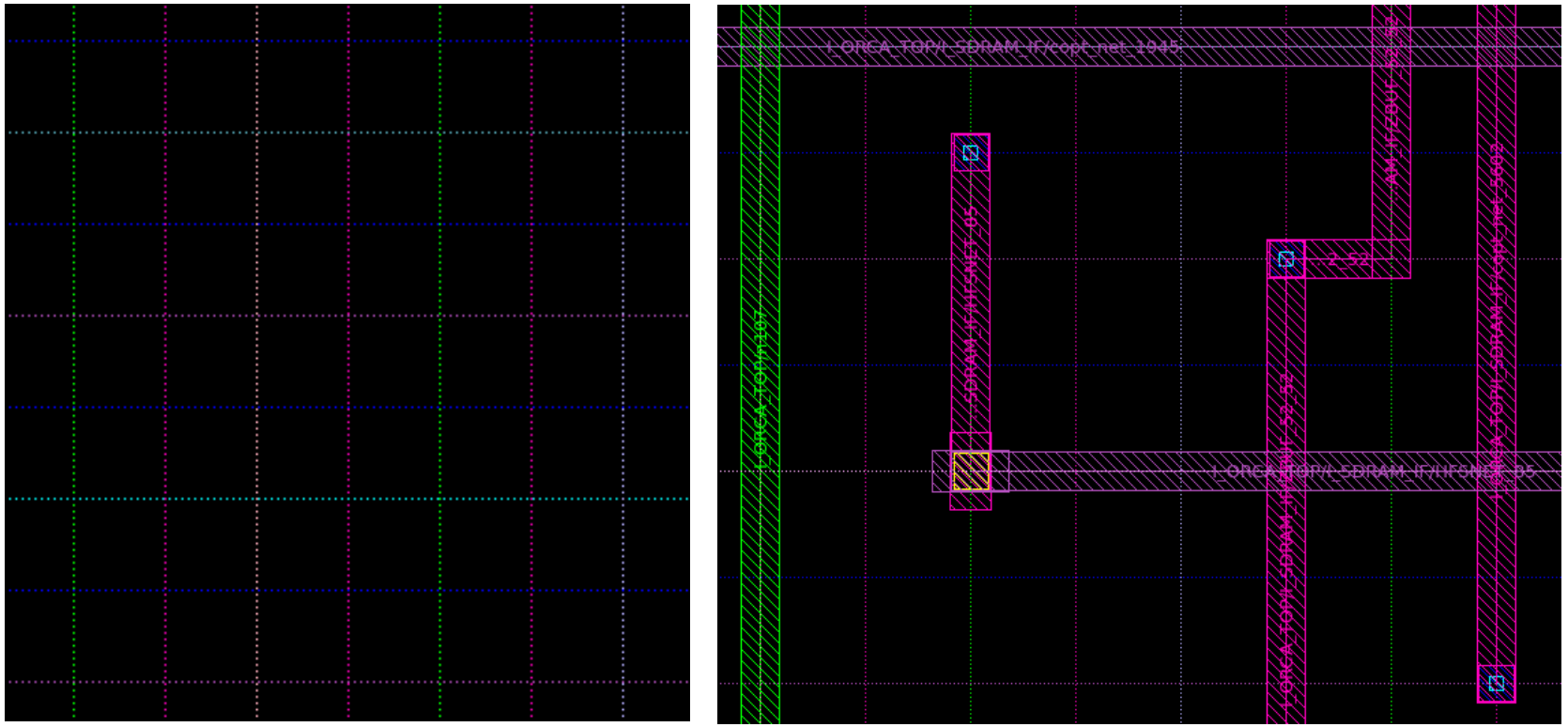


- Detail route fixes physical design rule violations
- Performs multiple iterations with varying repair box sizes

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Introduction to Routing : Overview to Routing

- **Track Assignment**



- Takes the Global Routed Layout and assigns each nets to the specific Tracks and Layer geometry
- It does not follow the physical DRC rules
- It will do the timing aware Track Assignment
- It helps in Via Minimization

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Introduction to Routing : Overview to Routing

- **Track Assignment**
  - Attempts to :
    - Make long, straight traces
    - Reduce the number of vias
  - After Track Assign, there should be no global routes or Global Route(GR) vias left

• **After Track Assignment(TA), all nets are routed (signal & Clock), but not very carefully**
• **There are many violations, particularly where the routing connects to pins**
• **Detail routing works to correct these violation**

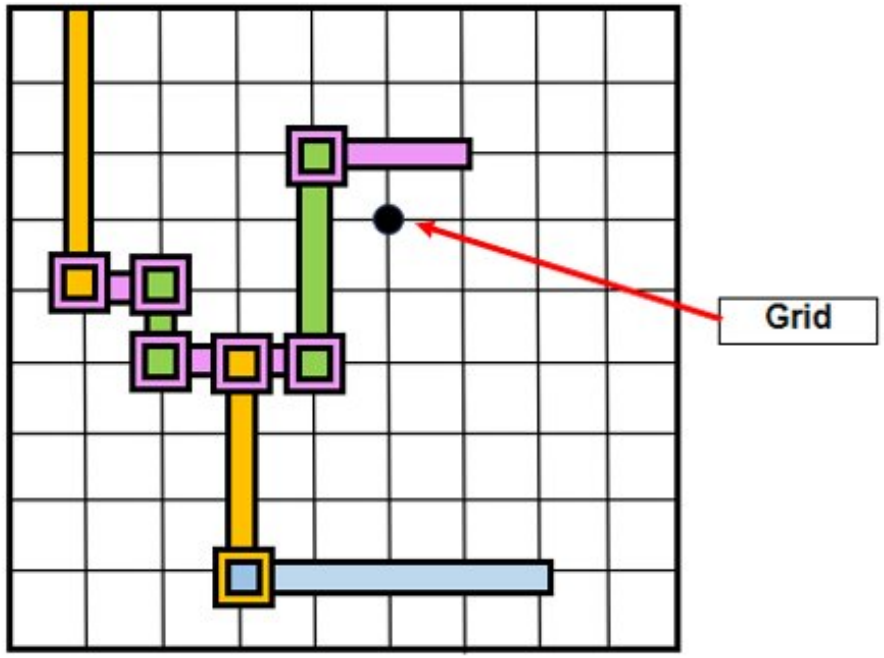
  - If track assignment can reduce the number of jogs and jumps in metal traces, this will generally improve timing (Each jump generally requires a via to jump to a higher or lower level Metal Layer)
  - Reducing the number of vias is generally a plus for reliability and yield since their failure rate is sightly higher than that of a simple, straight Metal Track in a modern, planarized process

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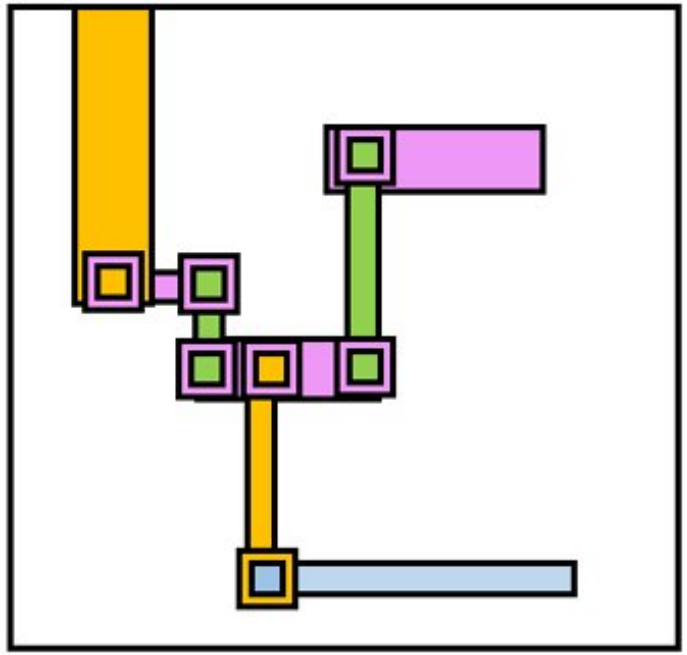
- **Types of Detail Routing**

❖ Grid Based Routing



- A grid is super imposed on the routing region
- Wires follow paths along the grid lines

❖ Grid Less Routing

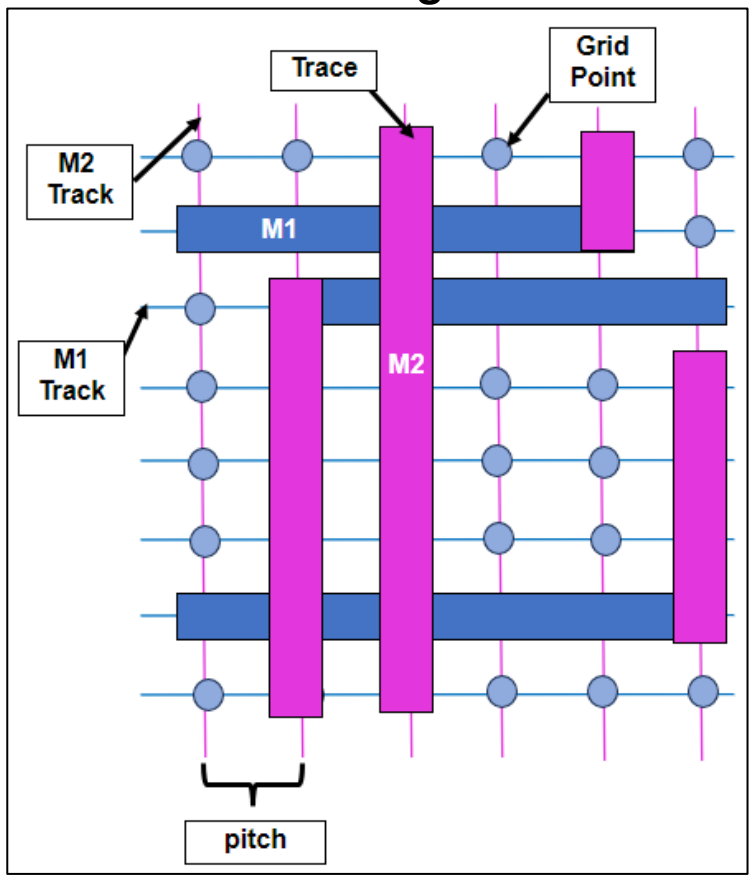


- Grid Less Routing does not follow the girdded approach
- It is generally much slower than grid based routing

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Introduction to Routing : Overview to Routing

- **Grid Based Routing**



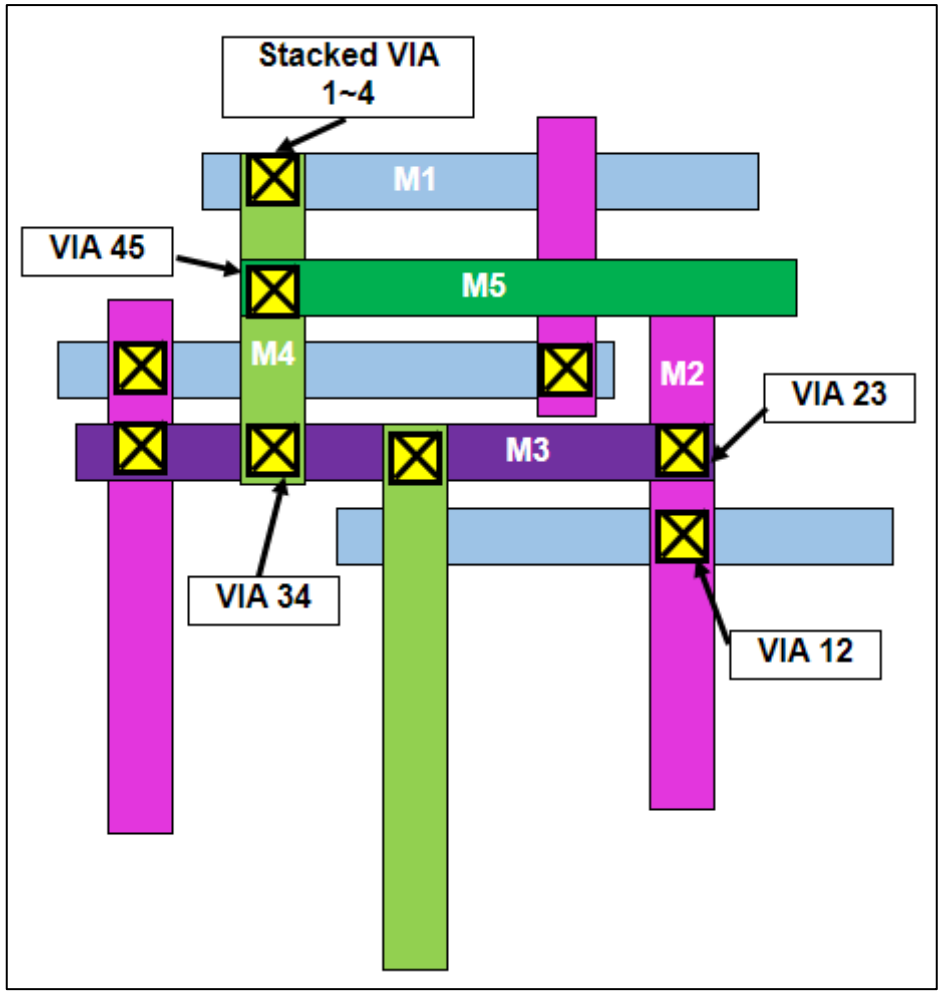
  - Metal traces (routes) are built along centered upon routing tracks on the grid points
  - Various types of grids are Manufacturing Grid, Routing Grid (pitch) and Placement Grid
  - Grid dimension should be multiple of Manufacturing Grid

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Introduction to Routing : Overview to Routing

- **Routing Preferences**
  - Typically Routing only in N/S E/W directions
  - Spacing checks with the adjacent layers
  - Via dimension rules
  - Slotting rules
  - Wire segments can cross wires on other layers
  - Power and Ground have their own layers, mostly the top layers
  - A segment cannot cross another segment on the same wiring layer
  - Layer Routing Directions (Each Metal Layer has own preferred routing Direction and are defined in a technology rule file)

    Ex) M1 – horizontal M2 – vertical M3 –horizontal ….
  - In some cases, we can avoid following preferred routing direction for Smart routing (non-preferred direction)



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Introduction to Routing : Overview to Routing

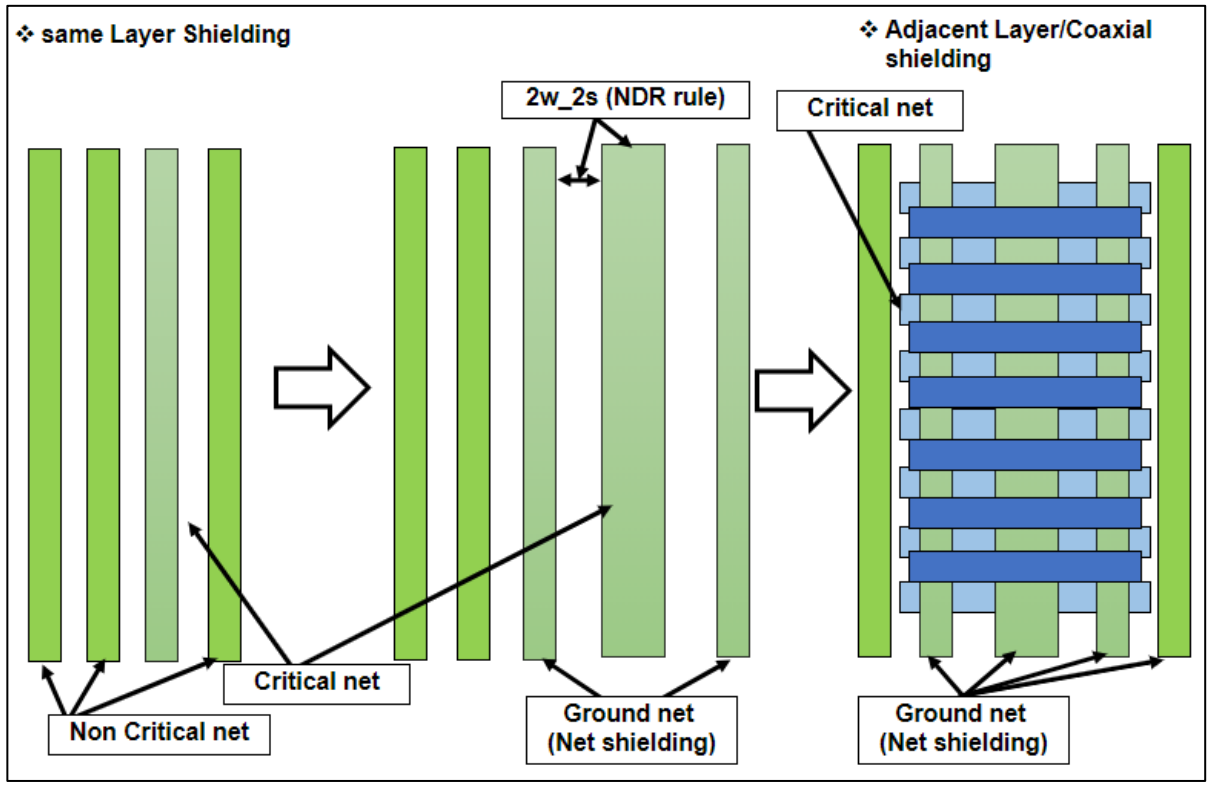
- **Detailed Routing**
  - Detailed Routing based on DRC type, ICC2 will either spread the wires or fill a gap (Theses fixes might cause some more violations like Notch filling can cause spacing violations because of increased width and wire spreading can cause violation in adjacent region)
  - ICC2 will analyze them and fix them (interative loop)

- **Search & Repair**
  - The search & Repair step is performed during detailed routing after the first iteration
  - In search & Repair, shorts and spacing DRC violations are located and rerouting of affected areas to fix all possible violatio n is executed

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Introduction to Routing : Overview to Routing

- **Post Routing Optimization**



- Signal Integrity (SI) Optimization by NDRs and Shielding for the sensitive nets
- Types of Shielding for sensitive nets
  - Same Layer shielding
  - Adjacent Layer / Coaxial shielding

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Introduction to Routing : Overview to Routing

- **Outputs of Routing**
  - Routing db file (nlib) or DEF file with no opens & shorts (check for LVS)
  - Timing report (Hold & Setup)
  - Timing DRC report
  - Congestion report
  - Skew & Insertion delay report
  - Geometric layouts of all nets
  - Netlist after routing

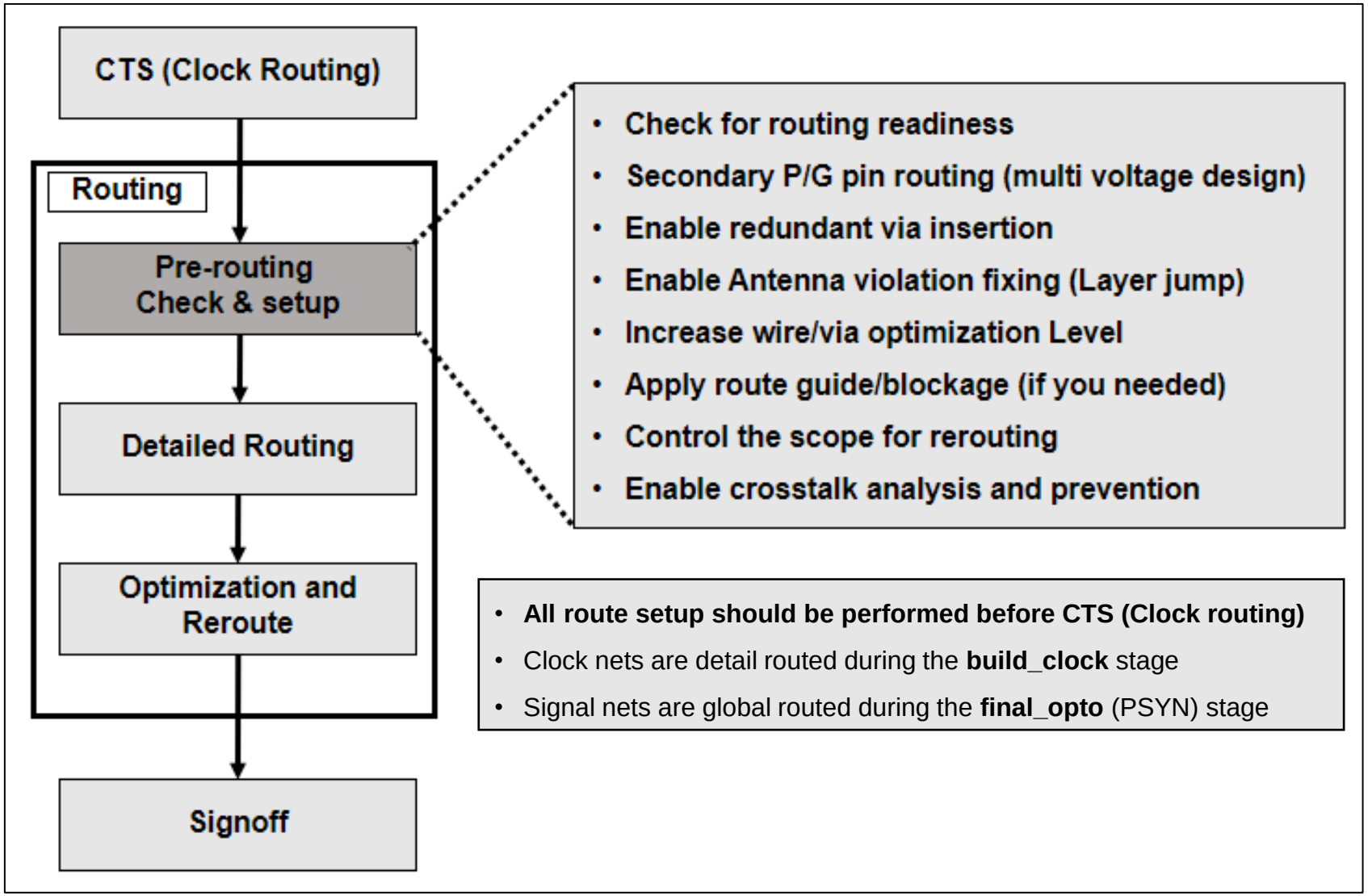
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WEEK 12 : Pre-routing Check & Setup

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Introduction to Routing : Pre-routing Check & Setup

- **Pre-Routing Checks and Setup for routing**



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Introduction to Routing : Pre-routing Check & Setup

- **Routability Check**
  - The routability check calls check_routability to confirm that the design is ready for routing
    - Blocked ports, out-of-boundary pins (sub-block design)
      - A logical port can have several physical pins
      - A port is considered blocked if none of its pins are accessible
    - Pin access edge rules honored
    - Minimum-grid violations

The **check_routability** command checks many issues known to degrade the routing performance or quality, including identification of blocked ports, out-of-boundary ports, minimum manufacturing grid violations, off-routing-track ports, and many miscellaneous sanity checks for routing objects. The command reports the results as part of the standard output and in an error data file. You can display the error data file in the GUI error browser. Some reports are available only in the standard text output and not in the GUI.

The **check_routability** command does not check the routing design rules. To check the routing design rules, use the **check_routes** command.

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Introduction to Routing : Secondary PG pin connection

- **Secondary PG pin connections (Multi voltage Design)**
  - Secondary PG pins (Backup P/G for always-on cells) should be routed first

```
route_group -nets {VDDH} ;# secondary_pg net
```

The **route_group** command routes a specified group of nets. You should use this command before signal routing to route critical nets such as clock nets or timing-critical nets. After signal routing, do not use this command to connect nets; use the **route_eco** command instead.

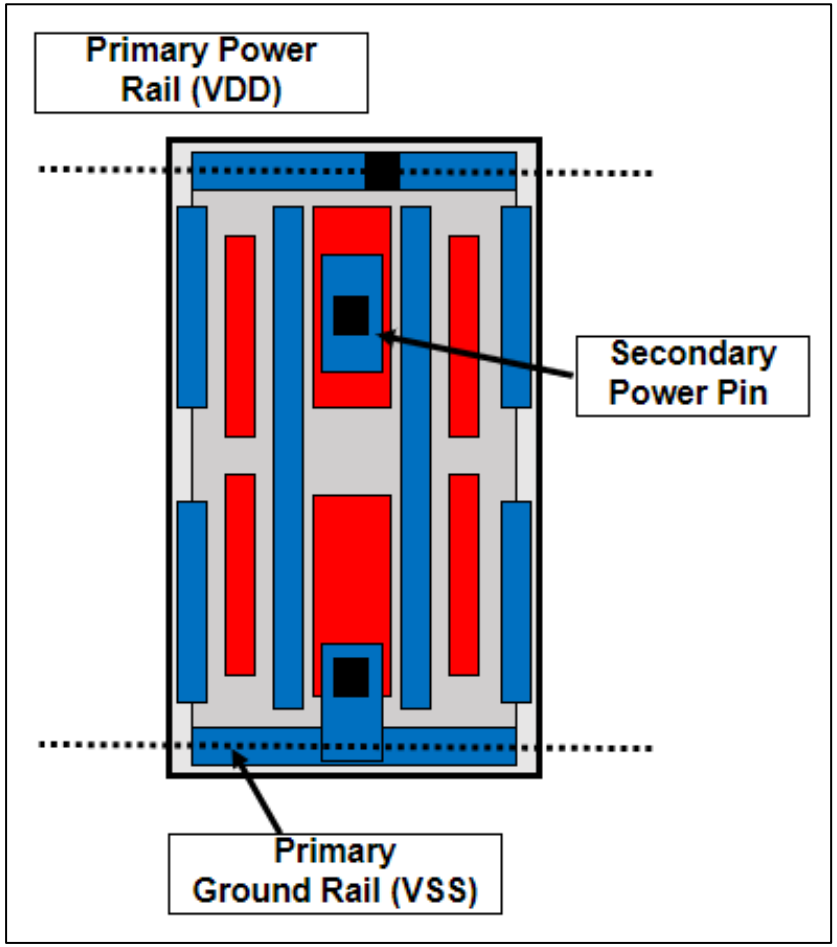
  - By default, there is no maximum to the number of PG pin connections from a Strap – choose your own maximum value based on your PG grid

```
set_app_options -value 4 \
        -name route.common.number_of_secondary_pg_pin_connections
```

This application option specifies the maximum number of secondary PG pins in a cluster.

The definition of a cluster is a set of connected secondary PG pins. Each cluster has one connection with a PG strap or ring. If the value is greater than 0, a cluster should not contain more than the specified number of secondary PG pins.

By default, the setting is 0, which means that there is no maximum.



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Introduction to Routing : NDRs for Secondary PG Routing

- **NDRs for Secondary PG Routing (Multi voltage Design)**
  - It is not uncommon to use NDRs for Secondary PG
    - Use only certain vias
    - Specify Metal width
    - Restrict routing to certain Layers

```
set_app_options -name route.common.separate_tie_off_from_secondary_pg \
        -value true ;# (default : false)
```

By default (**false**), tie-off connections and secondary PG power connections are considered the same nets, so nondefault routing rules, minimum and maximum layer constraints, and other constraints on the net apply to both tie-off connections and secondary PG power connections.

When you separate the tie-off connections from the secondary PG power connections by setting this option to **true**, you can apply nondefault routing rules, minimum and maximum layer constraints, and other constraints only to the secondary PG power connections.

```
#create routing rule
create_routing_rule secondary_pg \
        -width {M1 0.2 M2 0.2 M3 0.2} \
        -vias {...}

#set routing rule
set_routing_rule {VDDH} -rule seccondary_pg \
        -min_routing_layer M1   \
        -max_routing_layer M3

#secondary pg pin routing
route_group -nets {VDDH}
```

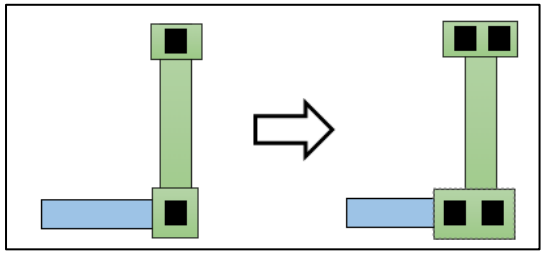

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Introduction to Routing : Redundant via insertion

- **Insert Redundant Vias on Signal Nets**
  - Replaces single-cut vias with multiple-cut via arrays (ex 2x1) or another single-cut via with a better contact code
    - By default, single-cut vias are replaced by two-cut via arrays

```
#Enable redundant via insertion (conversion occurs after evry detail route step)
set_app_options -name route.common.post_detail_route_redundant_via_insertion -value medium ;# (default : off)

#This application option enables automatic redundant via insertion after each detail routing step
```



```
#concurrent soft-rule based redunadnt via insertion increases conversion rate
set_app_options -name route.common.concurrent_redundant_via_mode -value reserve_space ;# (default : off)

#Near 100% redundant via insertion can be achieved with
set_app_options -name route.common.concurrent_redundant_via_mode -value insert_at_high_cost ;# (default : off)
```

This application option specifies the concurrent redundant via insertion mode for initial routing.

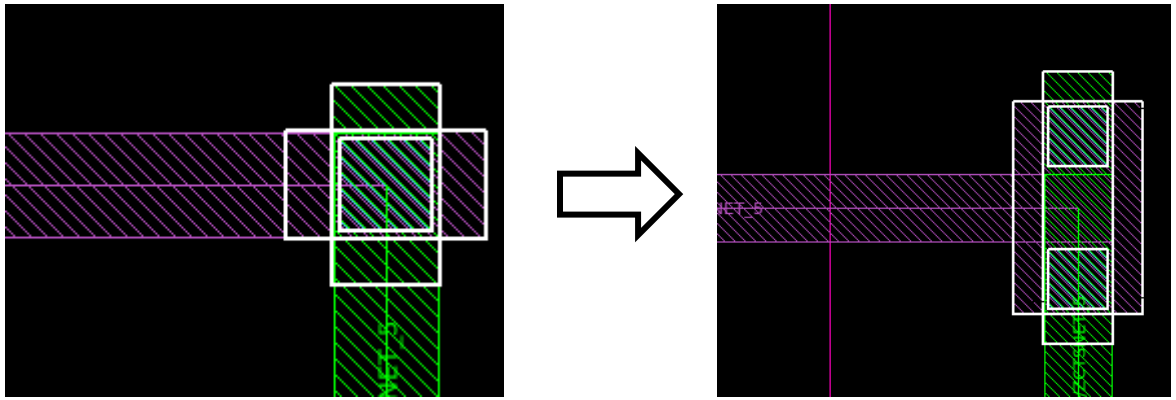
The valid modes are

- **off** (the default)
  Concurrent redundant via insertion is turned off.
- **reserve_space**
  Concurrent redundant via insertion reserves space for redundant vias but does not instantiate them. The space reservation is done by generating internal soft rules for redundant vias and performing soft-rule-based optimization. The actual instantiation is done later by explicitly calling the **add_redundant_vias** command or by setting the **route.common.post_detail_route_redundant_via_insertion** option to a value other than **off**.
- **insert_at_high_cost**
  In addition to the internal soft rules generated in **reserve_space** mode, concurrent redundant via insertion creates hard design rules for redundant vias and tries to insert them. Use this mode if only you need a nearly 100 percent conversion rate. It can lead to very long runtimes if the design is congested or the technology parameters are not friendly to redundant via insertion. This mode should be used with caution.

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Introduction to Routing : DFM (Design For Manufacturability)

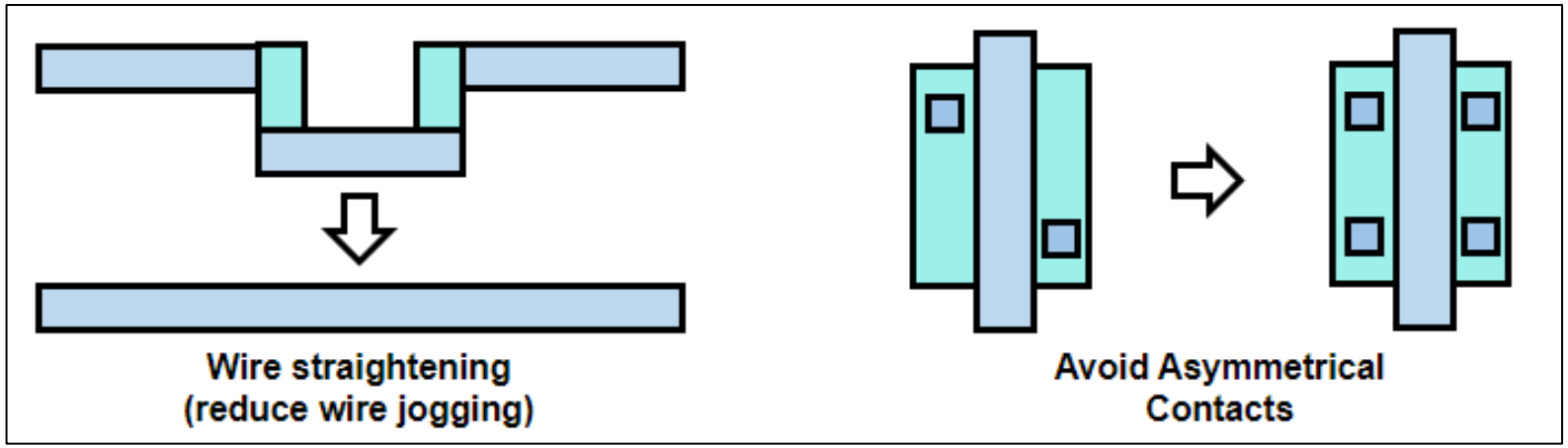
- **What is DFM?**
  - **DFM = Design For Manufacturability**
    - While the term "redundant via insertion" is commonly used, a better term to describe the optimization is a "via optimization"
      (This is because "via optimization" does not rely only on replacing single-cut vias with multiple-cut via arrays)
    - Single-cut vias can also be replaced with larger single-cut square vias, or with "bar" vias, or with vias with better metal overlaps, which ca n also improve the yield of vias
    - As the redundant via rate increases, it becomes more difficult to converge on routing design rules and you might see a reduction in signal integrity (Therefore you should use near 100% redundant via insertion only for those blocks that truly require such a high redundant via r ate)
    - In addition, achieving very high redundant via rates might require you to modify the floorplan utilization to allow enough space for the red undant vias



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Introduction to Routing : DFM (Design For Manufacurability)

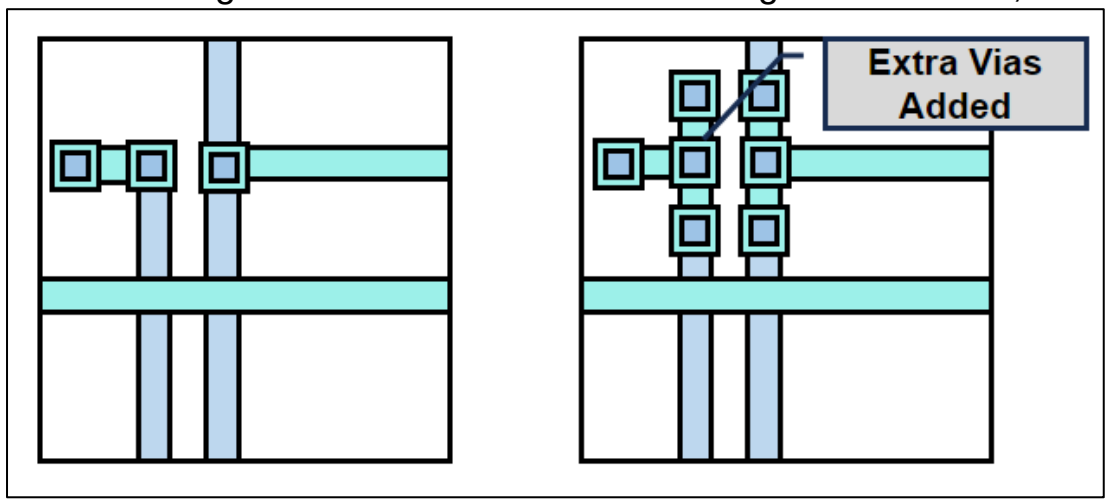
- **DFM (Design For Manufacturing)**
  - **During route, apply additional DFM and/or Design For Yield (DFY) rules**
    - Via reduction
    - Redundant via insertion
    - Wire straightening
    - Wire spreading



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Introduction to Routing : DFM (Design For Manufacurability)

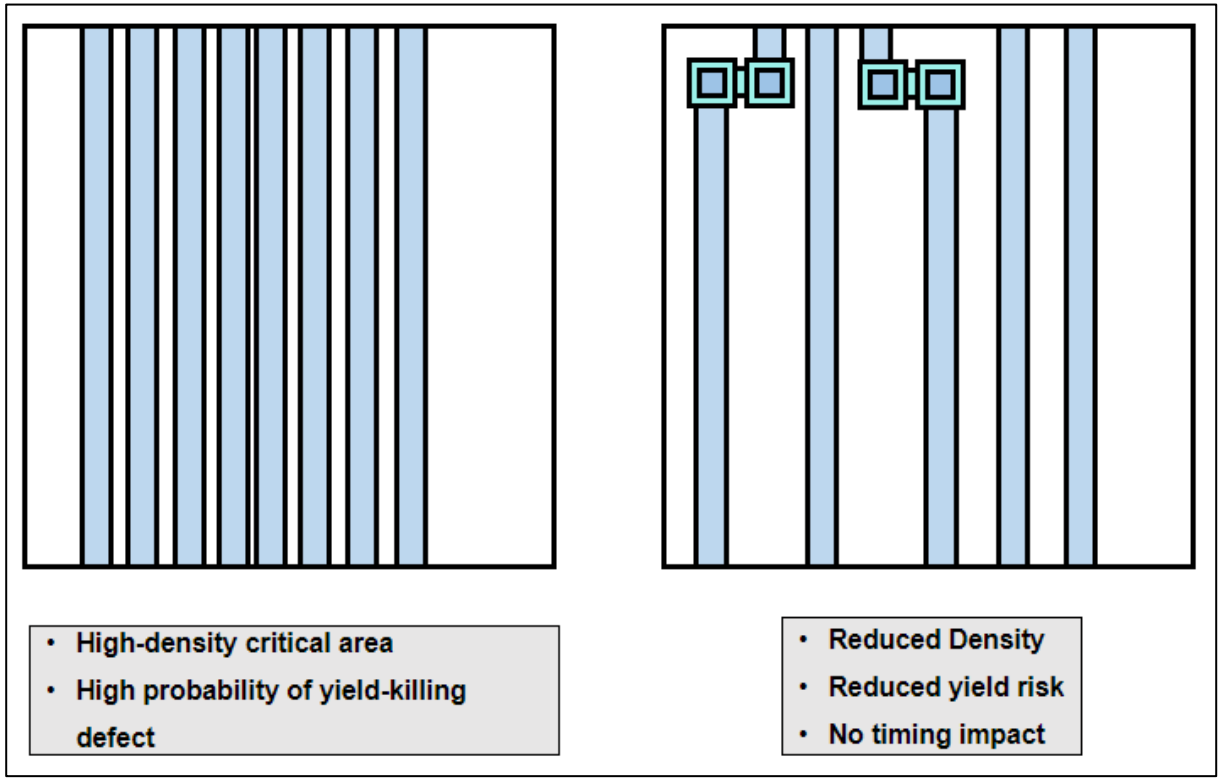
- **DFM (Design For Manufacturing)**
  - **Post-Route Via optimization, includes :**
    - Incremental routing for the minimization of vias
    - Replacement of single vias with multi-cut vias
  - **These operations are required for :**
    - **Reliability**
      - The ability to create reliable vias decreases with each process node
      - If a single via fails, it creates an open and the circuit is useless
    - **Replacement of single vias with multi-cut vias**
      - Electromigration hazards are even more significant in vias, which are essentially long, Narrow conductors



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Introduction to Routing : DFM (Design For Manufacurability)

- **DFM (Design For Manufacturing)**
  - **Wire spreading achieves :**
    - Lower capacitance and better signal integrity
    - Lower susceptibility to shorts or opens due to random particle defects



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Introduction to Routing : Wire & Via optimization

- **Enable Wire & Via Optimization**
  - **Via and wire optimization is recommended when adding redundant vias**
    - Attempts to make long straight traces and reduce the need for vias (using small jogs in non-preferred direction)

```
set_app_options -name route.detail.optimize_wire_via_effort_level \
        -value high ;# (default : low)
```

This application option specifies the effort level used to optimize wire length and via counts.

The valid values are

- **off**
  No extra effort to optimize wire length and via counts.
- **low** (the default)
  Low effort to optimize wire length and via counts.
- **medium**
  Medium effort to optimize wire length and via counts.
- **high**
  High effort to optimize wire length and via counts.

    - Alternatively, use optimize_routes to run this same optimization stand-alone after route_auto

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Introduction to Routing : Antenna Fixing

- **Antenna Fixing : Layer Jumping**
  - **Antenna rule violations can be fixed by layer jumping and by diode insertion**
    - Layer jumping is recommended during detail route step
    - Diode cell insertion is recommended post-route step
  - **Concurrent Layer jumping during detailed routing is enabled, by default**
    - Requires antenna rules provided as a .tcl file(ex antenna_rules.tcl)
    - The following antenna options are the Default settings

```
set_app_options -list {route.detail.antenna {true}}
set_app_options -list {route.detail.antenna_fixing_preference {hop_layers}}
set_app_options -list {route.detail.hop_layers_to_fix_antenna {true}}
```

  - **Concurrent Layer jumping during detailed routing is enabled, by default**
    - Usually performed during the final block preparation phase

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Introduction to Routing : Antenna Fixing

- **Antenna Fixing : Layer Jumping**
  - **Example of Antenna rules in ant.tcl**

```
remove_antenna_rules
define_antenna_rule -mode 4 -diode_mode 2 -metal_ratio 1000 -cut_ratio 20

define_antenna_layer_rule -mode 4 -layer "M1" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M2" -ratio 1000 -diode_ratio {0.06 0 400 40000}
```

  - **Option for "define_antenna_rule" command**

-mode mode

Defines the way antenna areas are computed. The valid values and their descriptions are as follows:

1 - Uses a polygon area, ignoring all lower-layer segments
2 - Uses a polygon area, including all lower-layer segments to the input pins
3 - Uses a polygon area, including all lower-layer segments
4 - Uses a side-wall area, ignoring all lower-layer segments
5 - Uses a side-wall area, including all lower-layer segments to the input pins
6 - Uses a side-wall area, including all lower-layer segments

•Note that only one rule can be defined for every mode for a given rule name. If two commands contain the same mode number and same rule name, the second command overwrites the first one

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Introduction to Routing : Antenna Fixing

- **Antenna Fixing : Layer Jumping**
  - **Option for "define_antenna_rule" command**

•*-diode_mode diode_mode*

•Defines the protection capability of the diode. By default, all output pins are considered to be a diode. The valid values and their descriptions are as follows:

0 - The diodes do not provide any protection
1 - The diodes provide unlimited protection

In this mode, the highest metal layer is not checked for antenna violations because the input-to-output connection is completed when the highest metal layer is formed

2 - Diode protection is limited; if more than one diode is connected, the largest value of the maximum antenna ratio for all diodes is used
3 - Diode protection is limited; if more than one diode is connected, the sum of the maximum antenna ratios for all diodes is used
4 - Diode protection is limited; if more than one diode is connected, the sum of the diode-protection values for all diodes is used to compute the maximum antenna ratio
5 - Diode protection is limited; the maximum diode-protection value for all diodes is used to calculate the equivalent gate area
6 - Diode protection is limited; the sum of the diode-protection values for all diodes is used to calculate the equivalent gate area
7 - Diode protection is limited; the maximum diode-protection value for all diodes is used to calculate the equivalent metal area
8 - Diode protection is limited; the sum of the diode-protection values for all diodes is used to calculate the equivalent metal area
9 - Diode protection is limited; scaling is based on maximum diode protection
10 - Diode protection is limited; scaling is based on total diode protection
11 - Diode protection is limited; scaling is based on maximum diode protection
12 - Diode protection is limited; scaling is based on total diode protection
19 - Has well type dependent data in addition to well independent data
20 - Has accumulated area threshold for floating metal or vias
21 - Diode protection is limited; this is a nwell antenna rule which safe guards by maintaining metal to nwell area ratio
22- Diode protection is limited; this is a TSV(Through Silicon Via) antenna rule which safe guards by the ratio of TSV area to the gate area that are electrically connected

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Introduction to Routing : Antenna Fixing

- **Antenna Fixing : Layer Jumping**
  - **Option for "define_antenna_rule" command**

-name rule_name

- **Specifies the name of the rule to be created. This is an optional option (When not specified, the global-antenna rule is created or modified)**

-metal_ratio metal_ratio

- **Specifies the maximum allowable ratio for metal area to gate size if the metal layer is not defined by using the define_antenna_layer_rule command**
- **It's required when the metal_pratio and metal_nratio options are not specified. If this value is zero, the ratio is ignored. The valid value is any non negative number (This can't be specified for -diode_mode 20)**

-cut_ratio cut_ratio

- **Specifies the maximum allowable ratio for cut area to gate size if the cut layer is not defined by using the define_antenna_layer_rule command**
- **It's required when the cut_pratio and cut_nratio options are not specified**
- **If this value is zero, the ratio is ignored**
- **The valid value is any non negative number (This can't be specified for -diode_mode 20)**

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Introduction to Routing : Antenna Fixing

- **Antenna Fixing : Insert Diode cell**
  - **To enable diode cell insertion, set the following app options:**

```
set_app_options -list {route.detail.diode_libcell_names "DIODE_cellname"}
set_app_options -list {route.detail.insert_diodes_during_routing ture}
```

  - **To prioritize diode cell insertion, use:**

```
set_app_options -list {route.detail.antenna_fixing_preference use_diodes}
```

This application option specifies the preferred technique used to fix antenna violations. Valid values are

- **hop_layers** (the default)

  Selects layer hopping as the preferred technique to fix antenna violations.

- **use_diodes**

  Selects diode insertion as the preferred technique to fix antenna violations.

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Introduction to Routing : Routing Guide

- **Routing Guide (Command)**
  - **Using a routing guide, you can change the default routing behavior in a particular area**

    (ex : change the preferred routing direction)

```
#create routing_guide

create_routing_guide -boundary {{0.0 0.0} {100.0 100.0}} \
        -layers {M3 M4} \
        -river_routing \
        -horizontal_track_utilization 50 -vertical_track_utilization 30
```

-horizontal_track_utilization percentage

- Specifies the horizontal track utilization for the routing guide
- You must specify an integer value between 0 and 100 for the percentage argument. If you do not specify this option or if you specify a value greater than 100, it is set to 100

-vertical_track_utilization percentage

- Specifies the vertical track utilization for the routing guide
- You must specify an integer value between 0 and 100 for the percentage argument. If you do not specify this option or if you specify a value greater than 100, it is set to 100

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Introduction to Routing : Routing Guide

- **Routing Guide**

-river_routing

- Specifies that the type of route guide is river_routing
- River routing refers to the routing of many non-shorting routes in roughly the same direction when routing is already restricted to just one or two layers
- When you use this option, you must also use the -layers option to specify the layers on which river routing is encouraged
- This option is mutually exclusive to -switched_direction_only, -max_patterns, and -access_preference options

❖ ***Macro, river routing will allow limited metal layers(ex : M5) to route in the non-preferred direction over the macro (If you use Preferred routing direction wold cause the router to create shorts over the macro, because Macro used M1 ~ M4 layers)***

-preferred_direction_only

- Creates a preferred-direction-only routing guide
- If you use the -layers option, the setting applies only to the specified layers; otherwise, it applies to all layers within the route guide boundary.

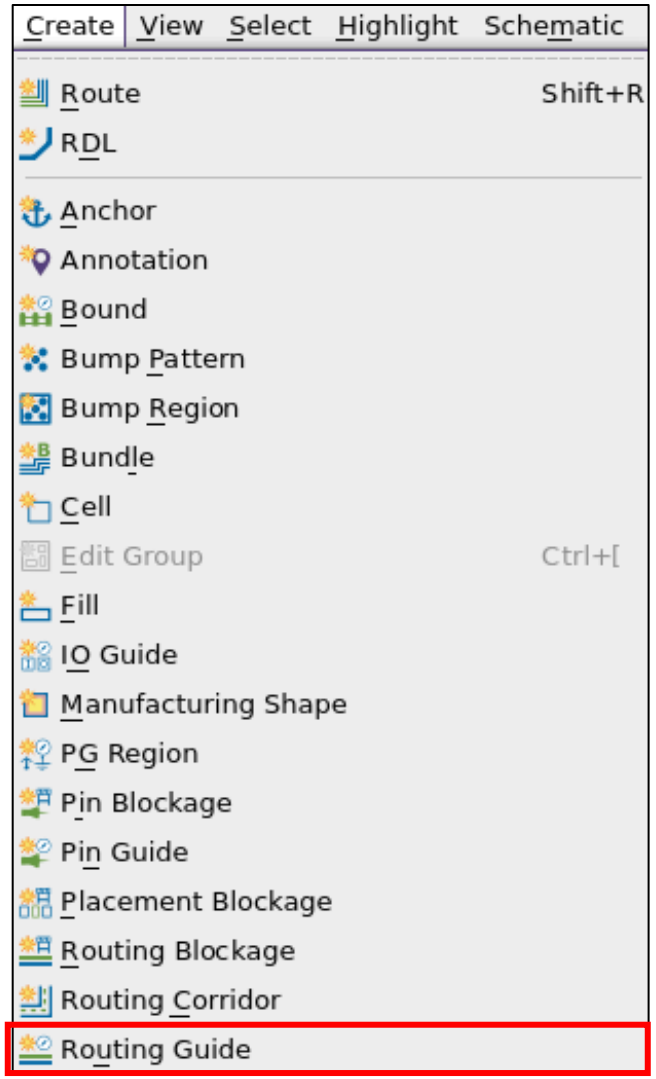
-switched_direction_only

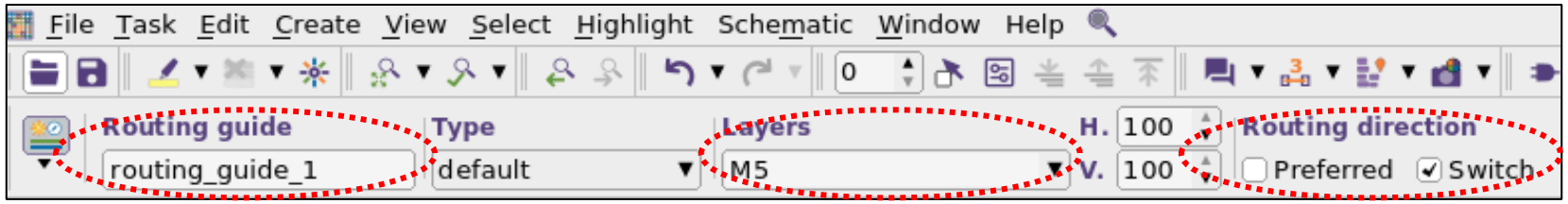
- Specifies that the type of route guide is switched_direction_only
- When you use this option, you must also use the -layers option to specify the affected layers
- This option is mutually exclusive to -river_routing, -max_patterns, and -access_preference options

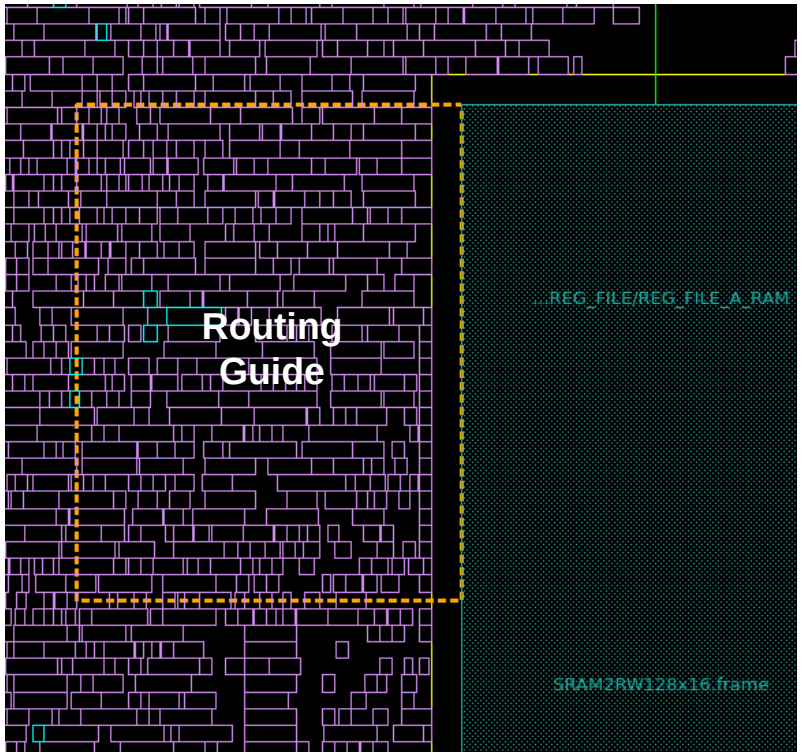
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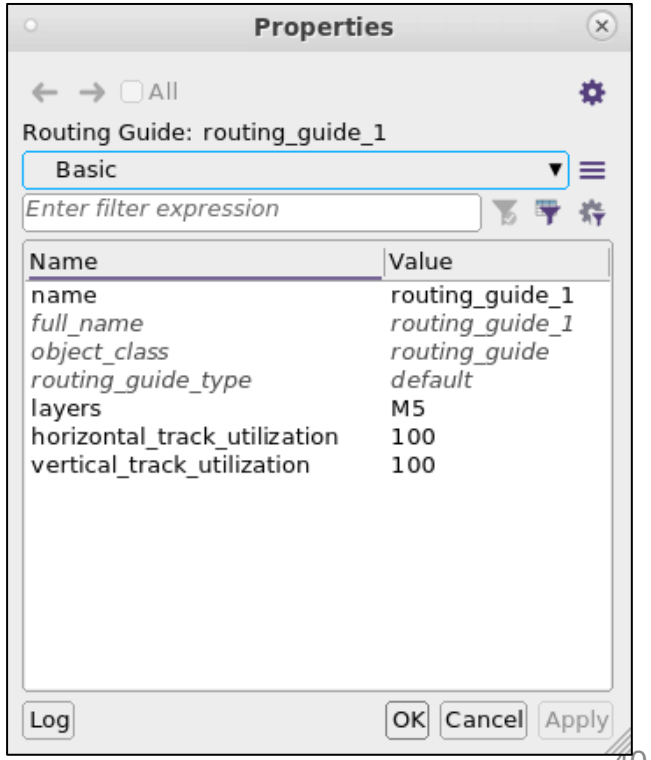
Introduction to Routing : Routing Guide

- **Routing Guide (Manual)**









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Introduction to Routing : Routing Blockage

- **Routing Blockage (Command)**
  - **To prevent signal routing in a certain area, define a routing blockage**

```
#create routing_blockage

create_routing_blockage \
        -layers "M3 M4 M5" \
        -boundary {{840.0040 571.1920} {878.7860 620.8160}} \
        -net_types "signal" \
        -zero_spacing \
        -reserve_for_top_level_routing
```

-net types
- Specifies the net types that must respect the generated routing blockages
- You can specify one or more of the folllowing values (analog_ground, analog_power, analog_signal, clock, deep_newell, deep_pwell, ground, nwell, power, pwell, reset, scan, signal, tie_high, tie_low)

❖ Default is all net types

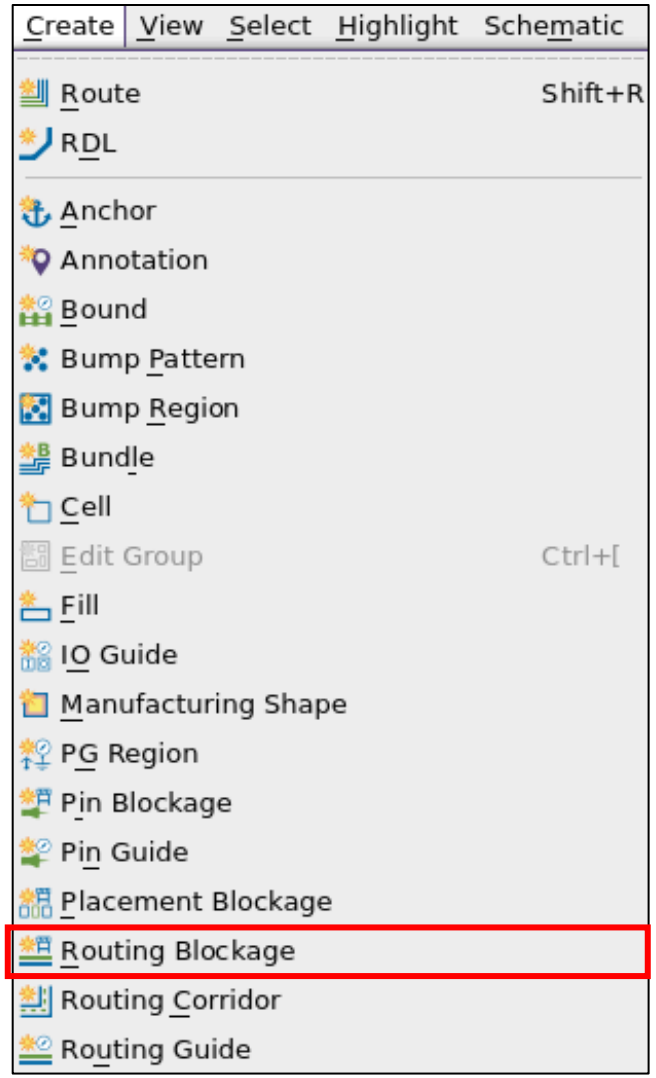
-zero_spacing
- Creates a zero-spacing routing blockages
- A zero spacing routing blockage disables the minimum spacing rule between the routing blockage boundary and the net shapes

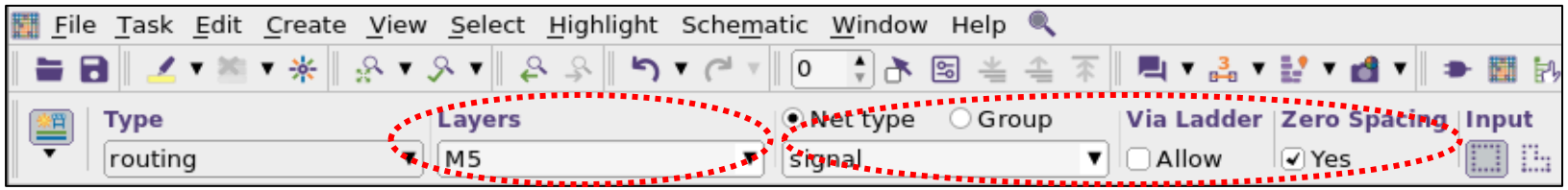
-reserve_for_top_level_routing
- Creates a routing blockage to reserve area for top-level routing
- Top-level routing uses the area reserved by this kind of routing blockages as a routing channel through the block

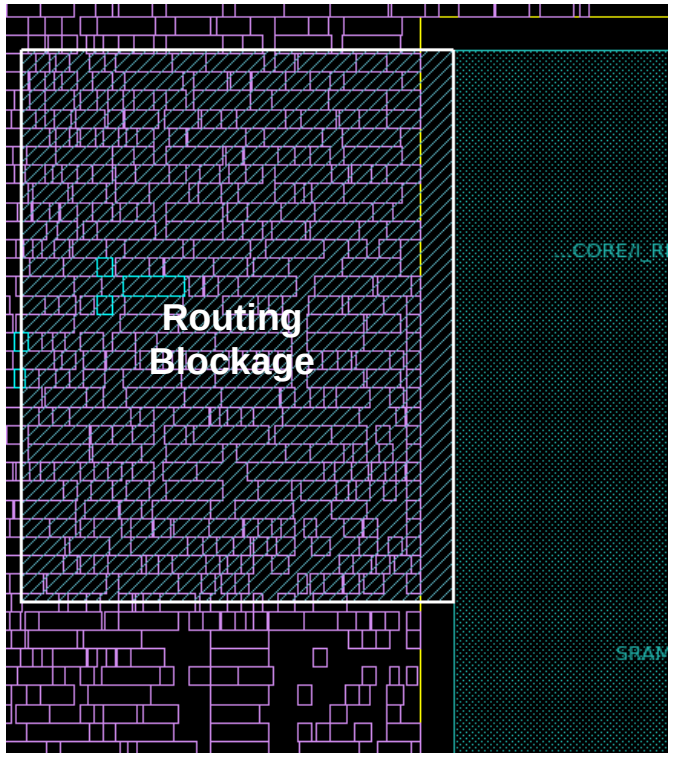
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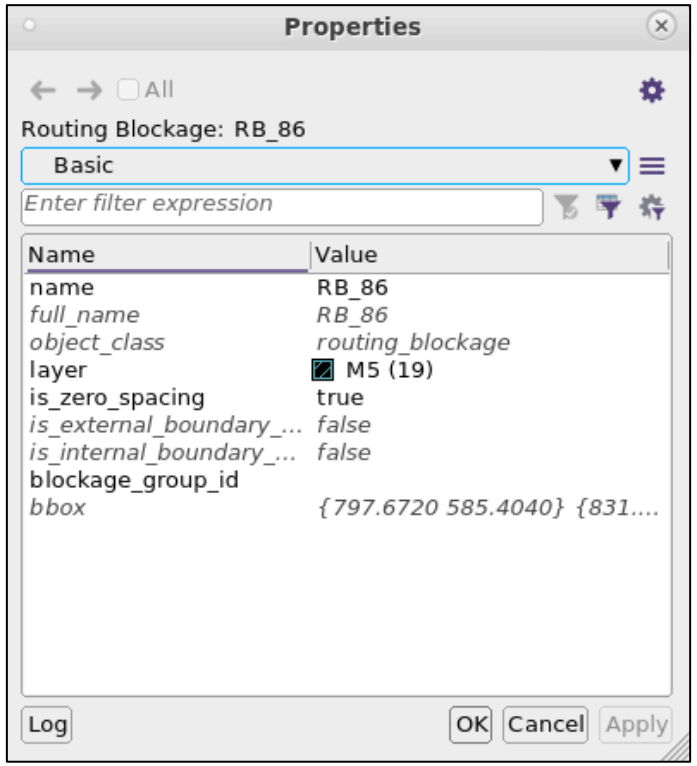
Introduction to Routing : Routing Blockage

- **Routing Blockage (Manual)**









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Introduction to Routing : Timing Driven Routing

- **Timing Driven Routing**
  - **By enable timing-driven routing, timing-critical nets are prioritized by the routers (defualt : false)**

```
#Timing Driven Routing app option

set_app_options -list {
  route.global.timing_driven true
  route.track.timing_driven true
  route.detail.timing_driven true
  }
```

  - **You can also control the effort of timing driven global routing with:**

```
set_app_options -name route.global.timing_driven_effort_level -value high ;# (default : high)
```

    - **Allowed values : low, medium, high (default)**
    - **High effort level emphasizes timing QoR over congestion while low effort is more congestion-driven (resulting in better physical DRC convergence)**

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Introduction to Routing : Crosstalk

- **Crosstalk Analysis**
  - **Delta delay affects setup & hold timing**
  - **Affects timing of the net, and therefore affects timing driven routing**

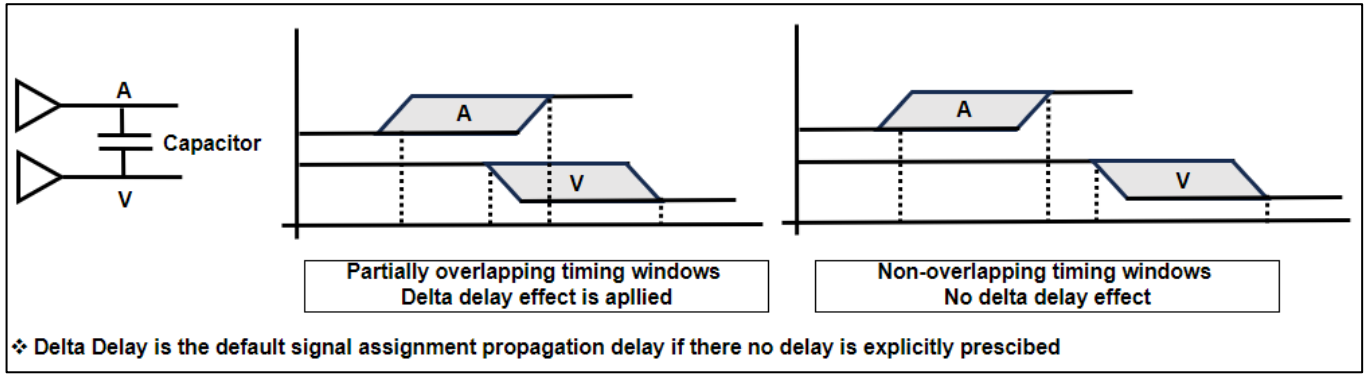
```
set_app_options -name time.si_enable_analysis -value true ;# (default : false)
```

- **This application option controls crosstalk analysis during timing analysis (By default, crosstalk analysis is not enabled)**
- **Turning this option on clears the existing delay data and any stored parasitics in the timer**
- **When set to true, this application also sets up the extraction of the routed nets along with their coupling caps**

  - **Delta delay calculation takes timing arrival windows into account**

```
set_app_options -name time.enable_si_timing_windows -value true ;# (default : true)
```

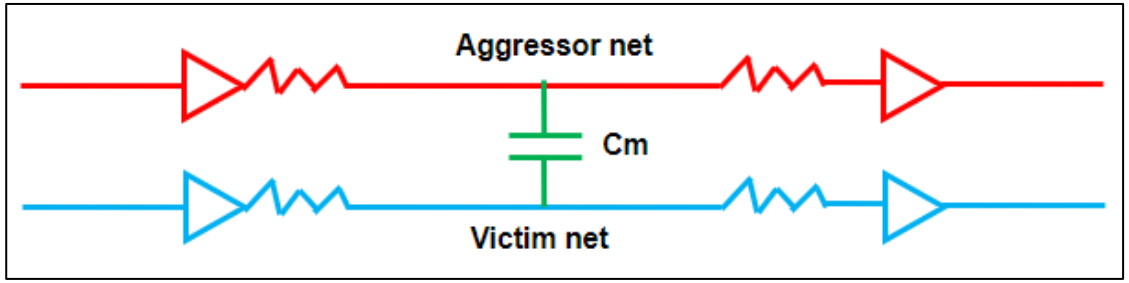
- **When timing window analysis is enabled, the overlap between arrival time of the signals on the victim net and aggressor nets is considered during crosstalk analysis**



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Introduction to Routing : Crosstalk

- **What is Crosstalk**
  - Aggressor is a net which creates impact on the other net
  - Victim is a net which is impacted by aggressor net
  - Crosstalk is the undesirable electrical interaction between two or more adjacent nets due to capacitive cross-coupling
  - Switching of the signal in one net (aggressor) can interfere neighbouring net (victim) due to cross coupling capacitance this is called crosstalk

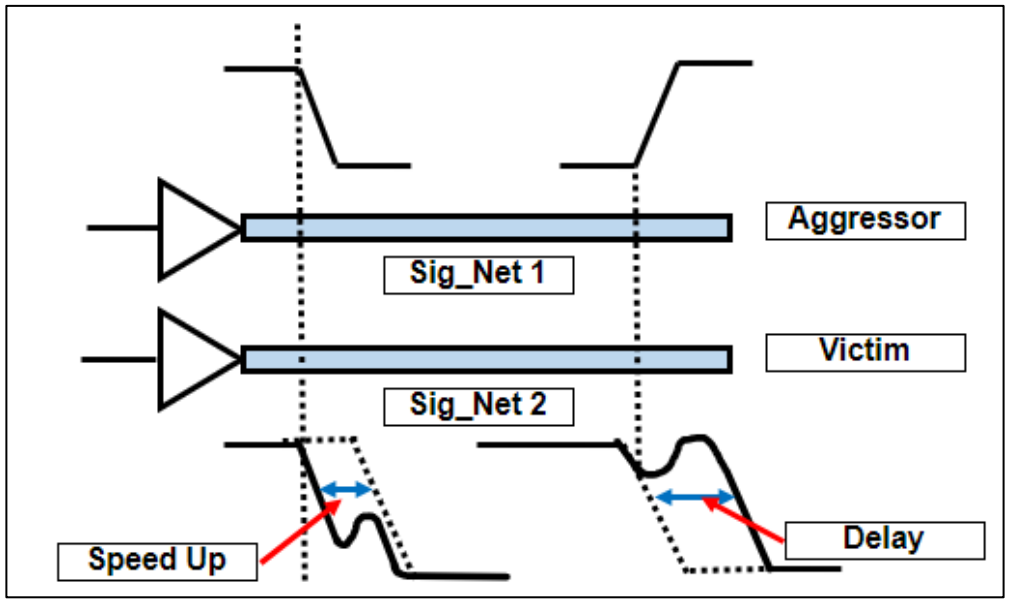


  - **Crosstalk Noise**
    - During the transition on aggressor net causes a noise bump or glitch on victim net (This noise is known as crosstalk noise)
    - In deep submicron technologies noise plays an important in terms of functionality or timing of device
  - **Crosstalk Delay**
    - It implies the delay happening in the output transition of victim due to transition of aggressor
    - Crosstalk delay depends on the switching direction of aggressor and victim net because of this either transition is slower or faster of victim net
  - **Crosstalk Glitch Analysis**
    - A steady signal net can have a positive glitch or negative glitch due to charge transferred by the switching aggressors through the coupling capacitance (If the glitch height is above the noise margin high (NMH), such a glitch is considered a potentially unsafe glitch)

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Introduction to Routing : Signal Integrity (SI)

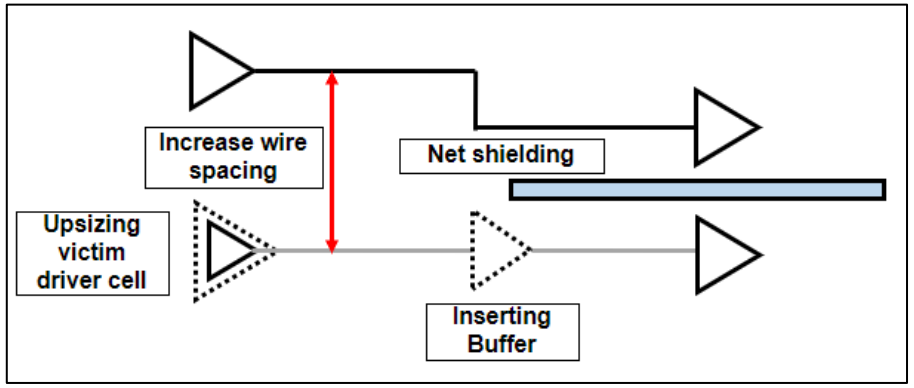
- **Signal Integrity (SI)**
  - **Signal integrity during routing is synonymous with crosstalk**
    - ❖ A switching signal may affect a neighboring net
    - ❖ The switching net is called the Aggressor
    - ❖ The affected net is called the "victim"
  - **The major effects**
    - **Signal slow down**
      - ❖ When the aggressor and victim switch in opposite directions
    - **Signal speed up**
      - ❖ When the aggressor and victim switch in the same directions

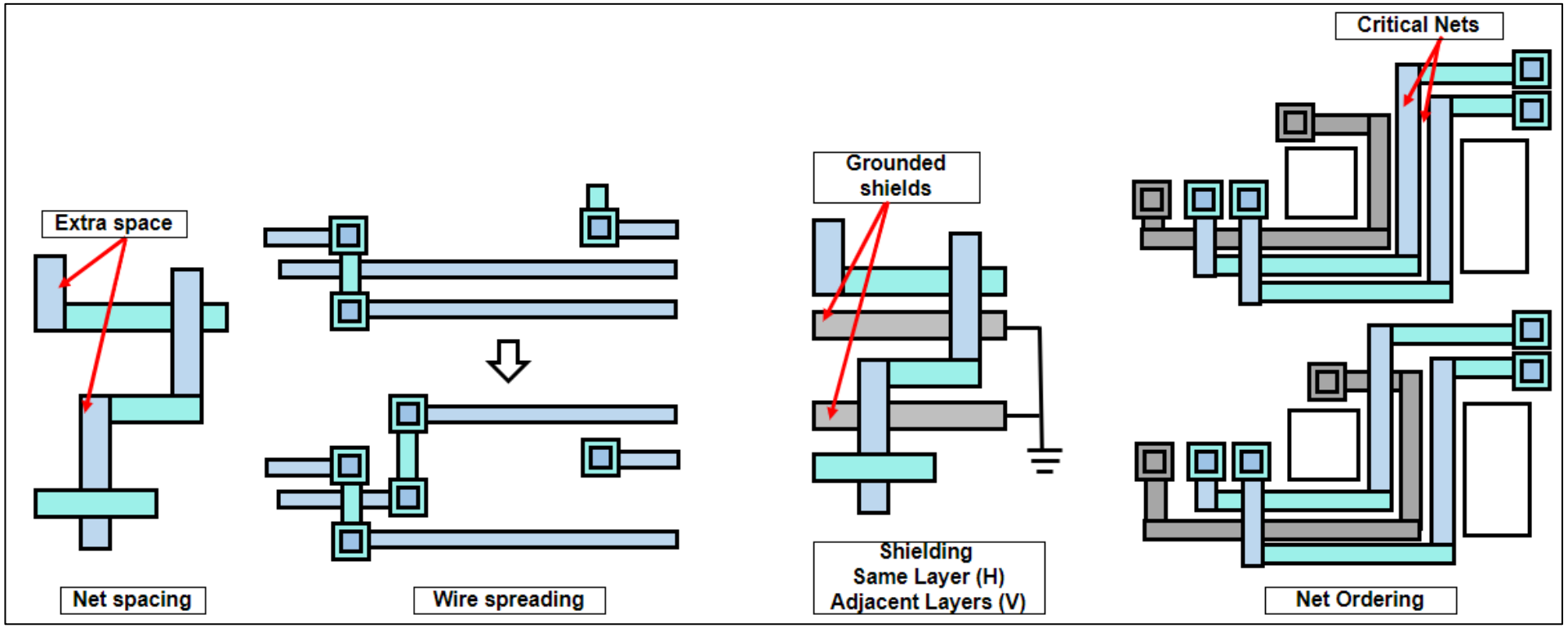


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Introduction to Routing : Signal Integrity (SI)

- **Signal Integrity (SI)**
  - **Crosstalk Prevention**
    - **Limit length of parallel nets**
    - **Wire spreading**
    - **Special nets shielding**
    - **Upsize driver cell or buffer**





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Introduction to Routing : Crosstalk Prevention

- **Crosstalk Prevention Routing**
  - **Crosstalk prevention happens during routing**

```
#routing
route_auto
```

  - **ICC2 prevention avoids putting long parallel nets on adjacent tracks (Focuses on timing-critical nets)**
  - **The current recommendation in route is to enable crosstalk prevention only during track assignment**
    - Enabling crosstalk during global routing has shown mixed results after routing

```
set_app_options -name route.global.crosstalk_driven -value true ;# (default : false)
set_app_options -name route.track.crosstalk_driven -value true ;# (default : false)
```

- **Crosstalk prevention happens during the global routing and track assignment stages of routing**
- **During global routing, timing critical nets are spread across GRCs, to avoid having many critical nets in one GRC, while having few or none in a neighboring GRC**
- **During track assignment, when actual metal traces exist, simple crosstalk modeling is used to estimate the effects of crosstalk on delay and static noise**
- **Timing-and noise-critical nets are prevented, wherever possible, from being routed on adjacent tracks**

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Introduction to Routing : Static Noise analysis

- **Static Noise**
  - **Noise analysis is also enabled with SI**

```
set timset_app_options -list {time.si_enable_analysis {true}} ;#3 (default : false)
```

  - **When you use the "report_noise" command without any options, it reports a single pin with the largest high and l ow noise slack (To report all the violators, use the –all_violators option)**

- **set_noise_margin**
  - This command specifies a noise margin for a lib pin, design port or pin
  - It can be used in the absence of library-specified noise immunity characteristics, or to override the library-specified characteristics by replacing them with noise margins

  - **report_noise**

```
noise_region: above_low
   pin name                    width         height          slack
-------------------------------------------------------------------
   I_ORCA_TOP/I_CONTEXT_MEM/CONTEXT_RAM0/A1[5]
                                0.28           0.24           0.10

noise_region: below_high
   pin name                    width         height          slack
-------------------------------------------------------------------
   I_ORCA_TOP/I_BLENDER/result_reg_30_/D
                                0.32           0.26           0.07
1
```

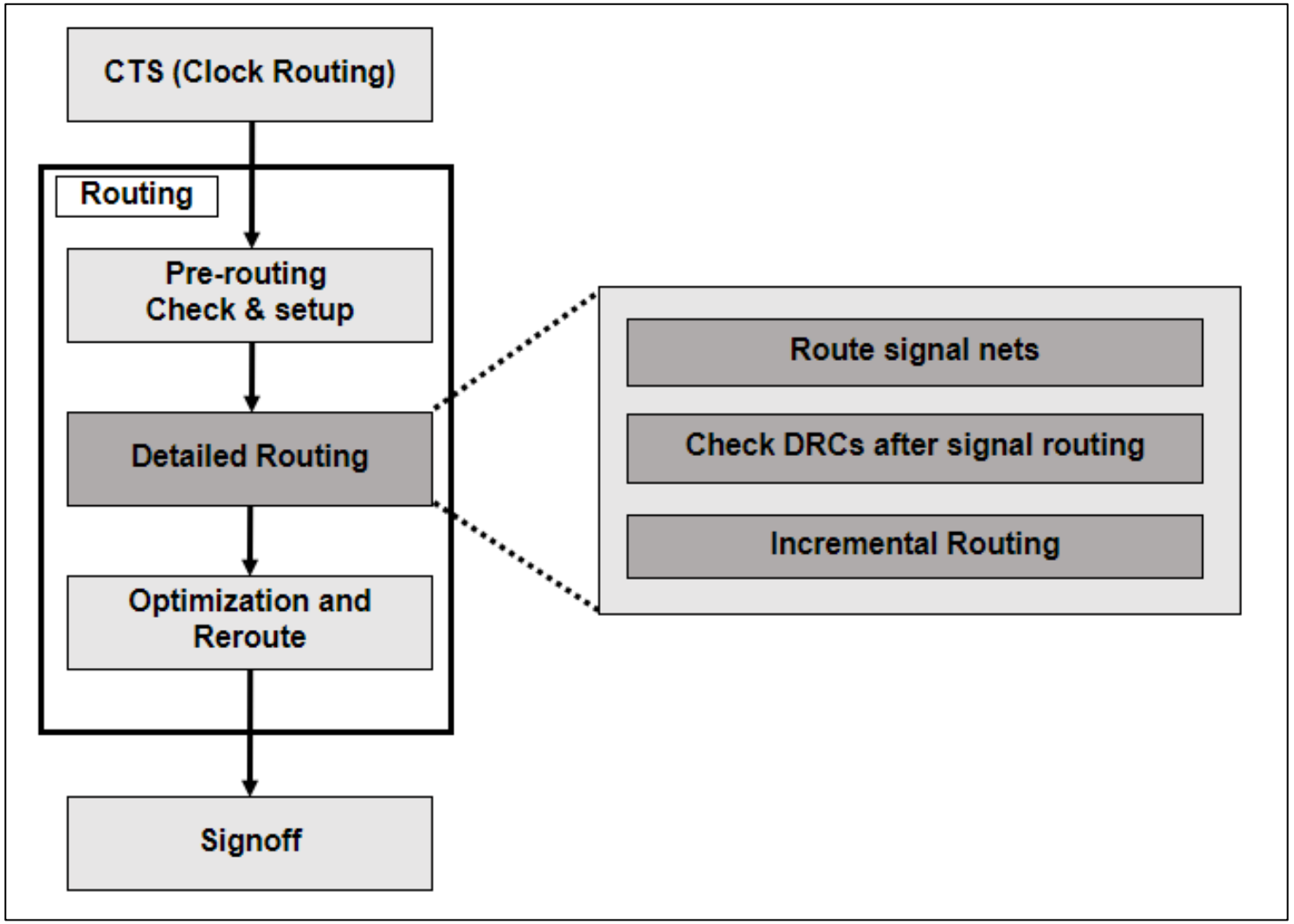

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WEEK 12 : Detailed Routing

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Introduction to Routing : Detailed Routing

- **Detailed Routing**



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Introduction to Routing : Detailed Routing

- **Route the Singal Nets**
  - **Route_auto**

```
#routing
route_auto
```

- **route_auto**
  - The route_auto command performs all routing stages in a single step
  - If the design is already global routed, the command runs track assignment and detail routing
  - Otherwise, the command runs global routing, track assignment, and detail routing

  - **Initial routing entails track assignment and detail routing no post-route optimization**
  - **To run the two phases individually:**

- **Track assignment**
- **Detail routing**

  - **If you Modify routing or crosstalk options, routing guides, or via mapping rules and re-route, as needed**

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Introduction to Routing : Detailed Routing

- **Detail Route iterations**
  - **By default value, "route_auto" command performs a maximum of 40 detail route iterations to fix DRC violations**
  - **If your technology requires more iterations, set the value using :**

```
#routing
route_auto -max_detail_route_iterations 50
```

-max_detail_route_iterations *num*

Specifies the maximum number of detail routing iterations. You can specify an integer between 1 and 1000.

By default, the maximum number of iterations for the detail routing command (40) is used.

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Introduction to Routing : Detailed Routing

- **Check Zroute DRC Violations after routing**
  - **Checks signal and clock routing for physical DRCs, NDRs, shorts, opens, antenna and voltage area violations (Does not check DRCs amongst pre-routed nets only) (ex) P/G grid structure)**
  - ❖ **For checking PG DRC use the command *"check_pg_drc"***

```
#check Zroute DRC
check_routes
```

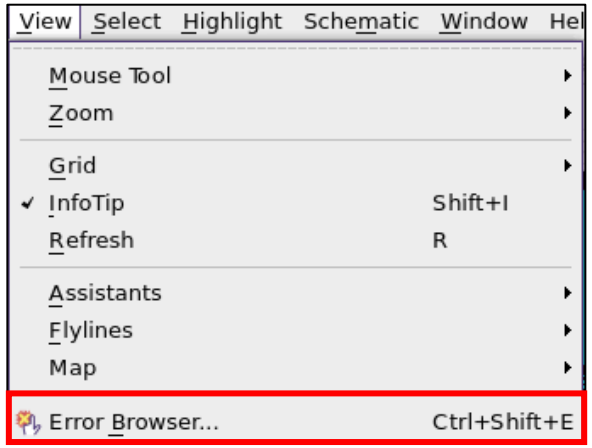
Verifies, checks, and reports routing statistics for a block. The statistics are based only on the signal routes; PG routes are not included. To report routing statistics that include PG routes, use the **report_design** command.

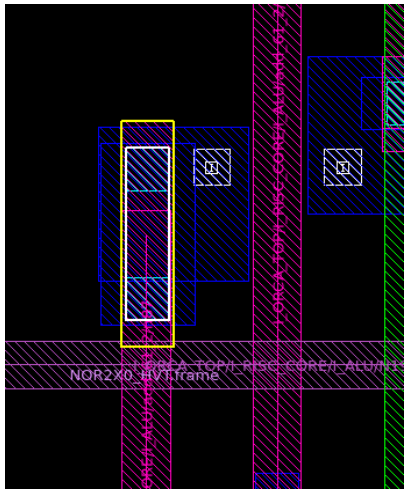
- **The check_routes command reports the following information for a routed design :**
  - **Open nets**
  - **DRC violation**
  - **Charge-collecting Antenna violations** (Antenna analysis is performed only if antenna rules are defined in the cell library)
  - **Voltage-area violations**
  - **Wire length**
  - **Contacts** (reports the total number of contacts, the total number of routed contacts, and the number of instances of each via master)
  - **Wires**
  - **Redundant via conversion rate & Total optimized via conversion rate**

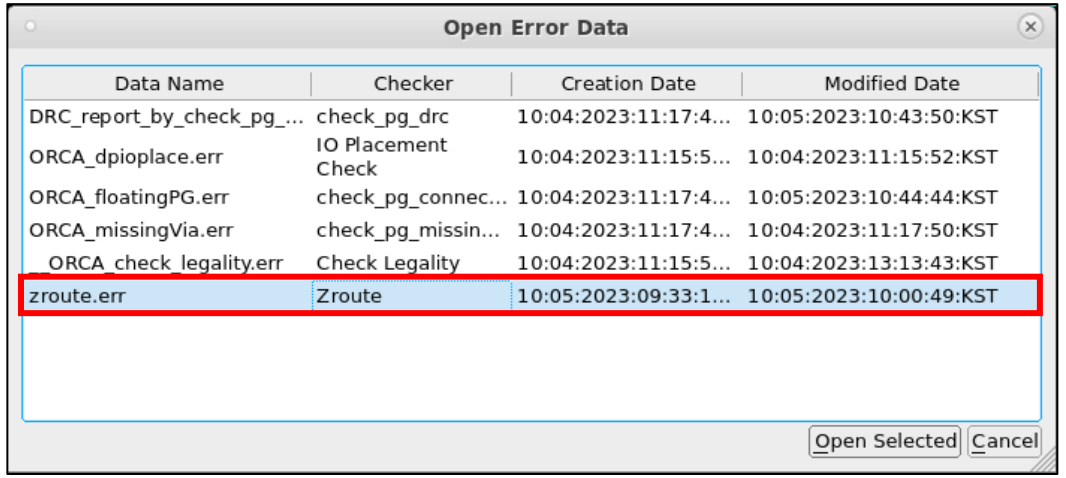
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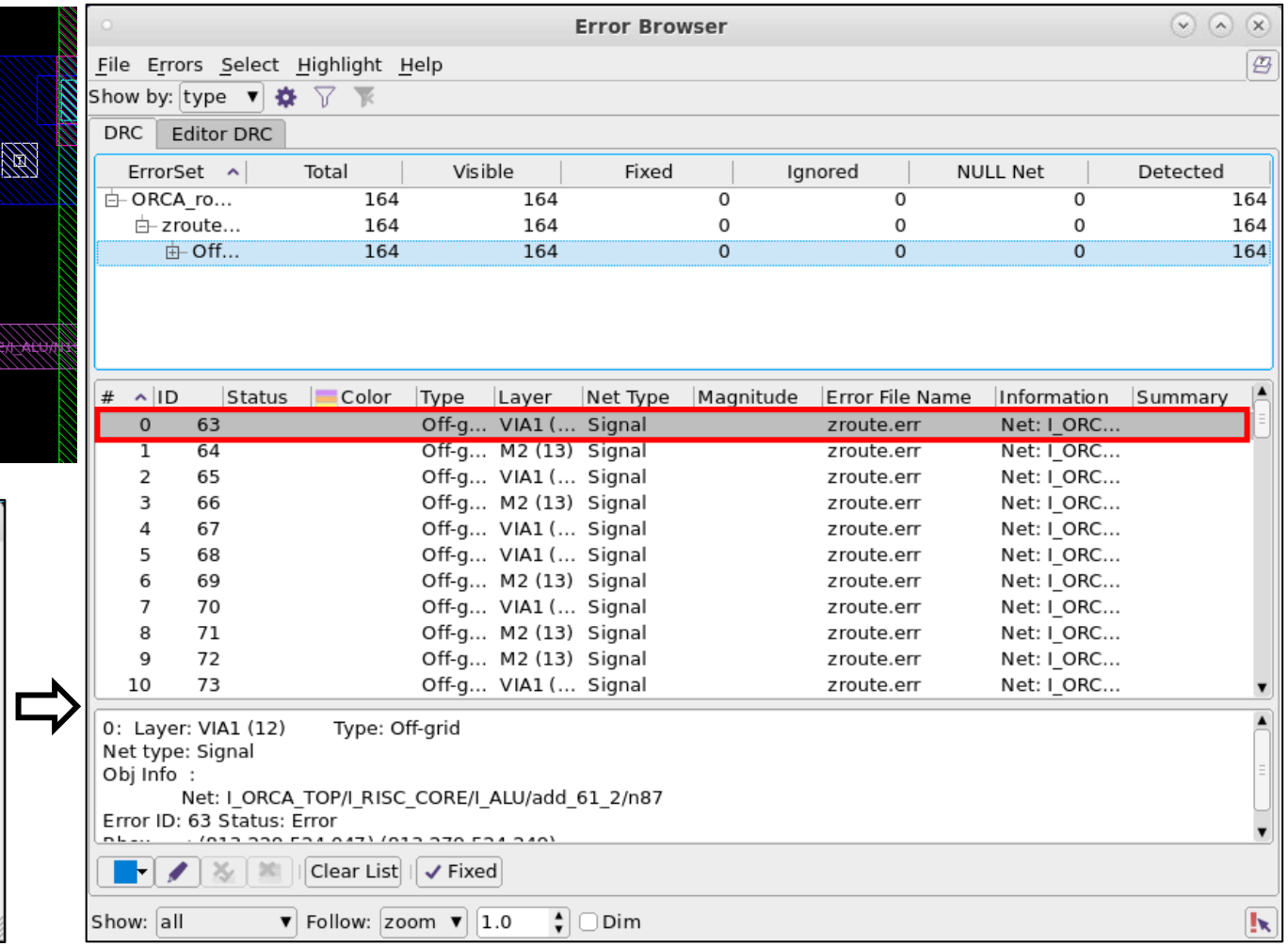
Introduction to Routing : Check Zroute after Routing

- **Check Zroute DRC Violations after routing**





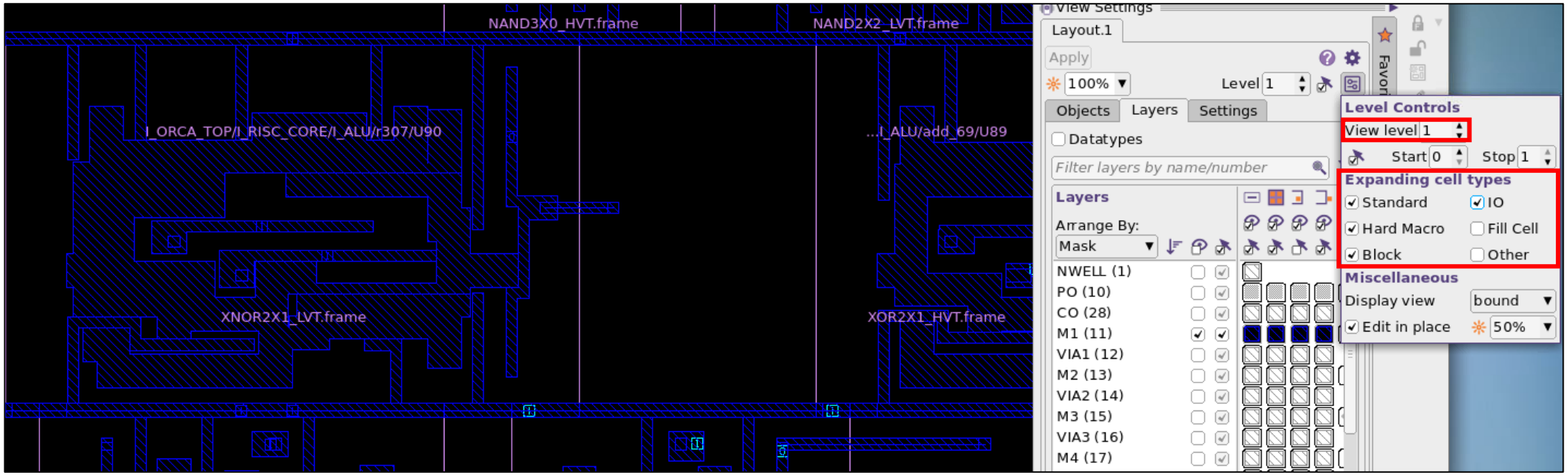




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Introduction to Routing : Check Zroute after Routing

- **Viewing Cell internal Structures**



- **By increasing the view level alone, cell internal structures will not be visible (It is important for understanding and analyze DRC violations)**
- **You need to enable "Expanding cell types" for Standard, Hard Macro and IO**

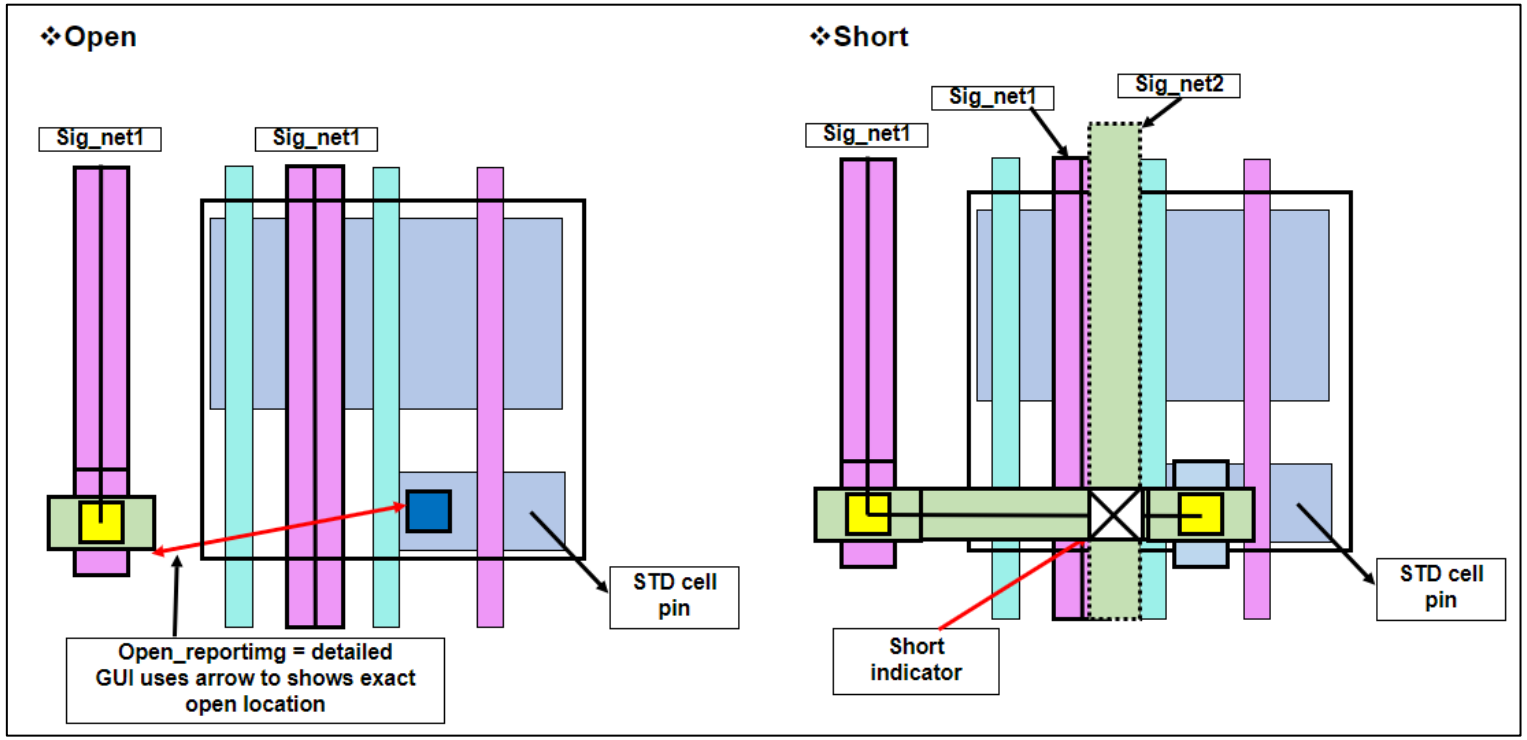
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Introduction to Routing : Check LVS after Routing

- **LVS (Layout Versus Schematic check)**
  - **Use "check_lvs" command to check for open, short, and floating nets**
    - Much quicker as it does not check for DRCs
    - Errors can be displayed using the Error Browser

```
#check LVS

check_lvs -checks all -open_reporting detailed \
        -check_child_cells true
```

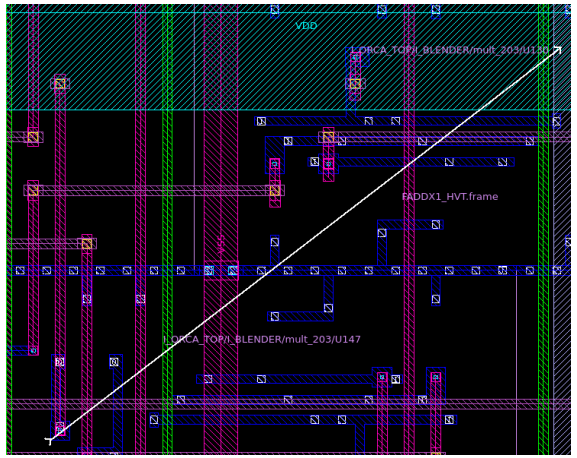


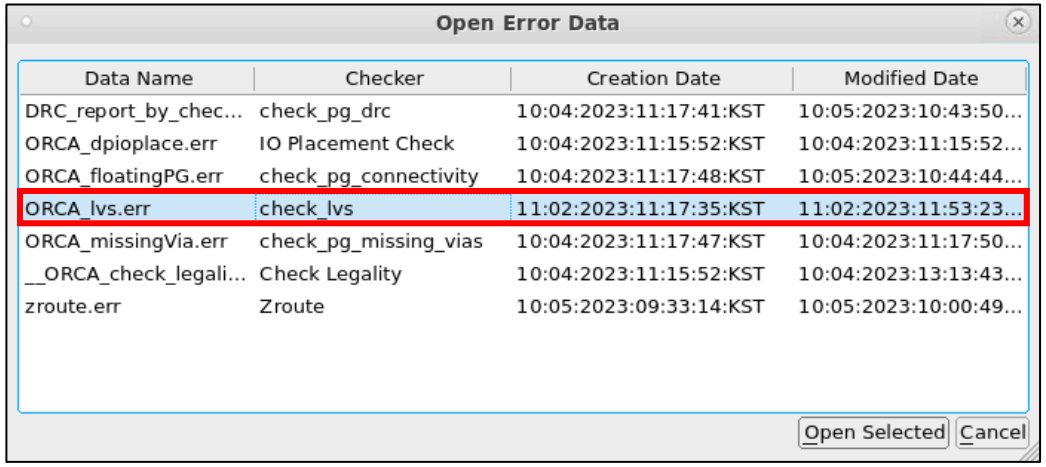
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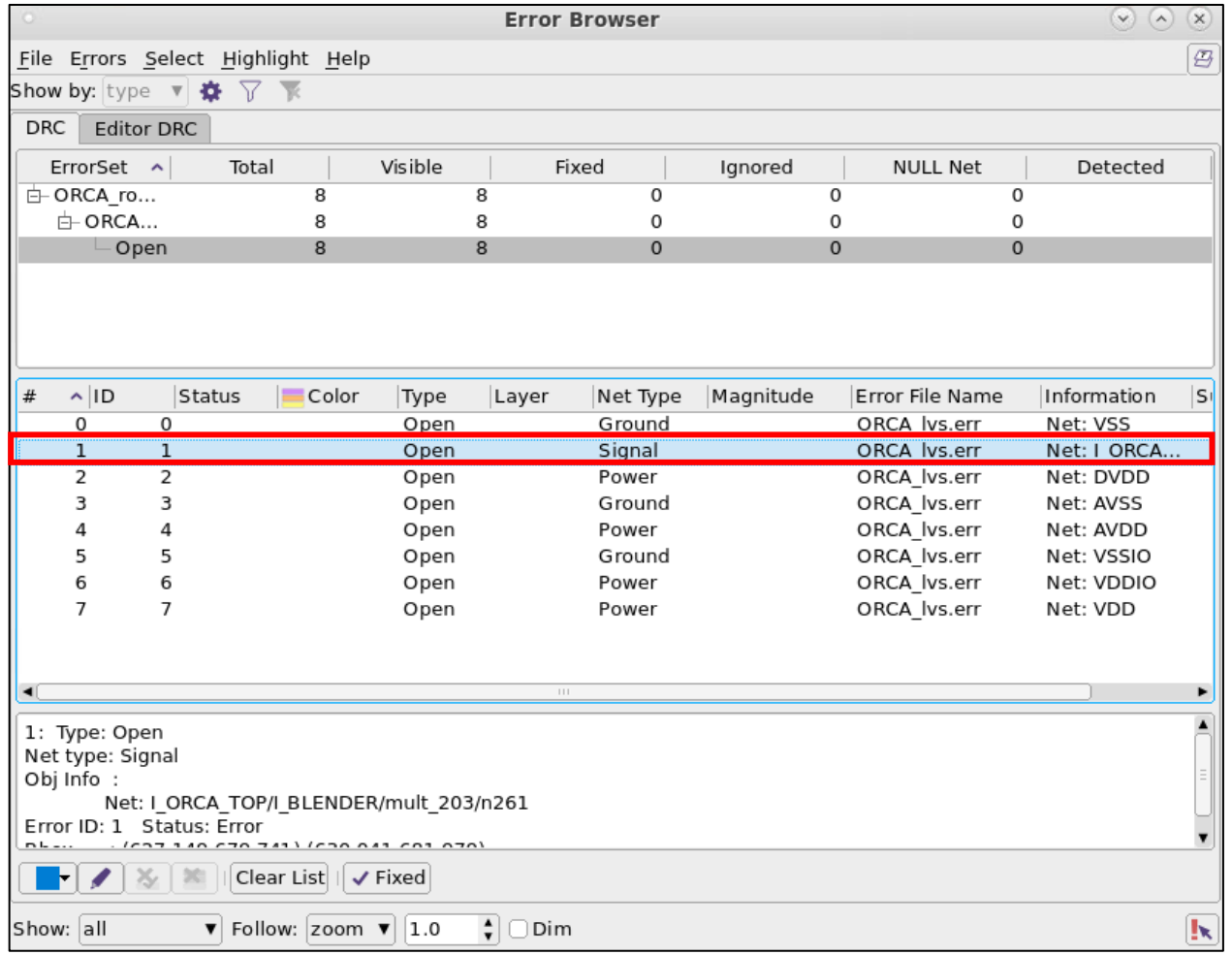
Introduction to Routing : Check LVS after Routing

- **Check LVS Violations after routing**









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Introduction to Routing : Detailed Routing

- **Detail Routing Loops**
  - **Run additional detail route iterations at any time to fix DRC violation**

```
route_detail \
        -incremental true
        -max_number_iterations 60
        -coordinates {{llx lly} {urx ury}} ;# you can specify the routing area multiple rectangels -
                                           ;# are supported using -coordinates option
```

- **Change Routing Options**
  - **Many detail route and common route options affect the behavior of the routing**
  - **Examine of various options**

```
report_app_options route.detail.*
report_app_options route.common.*
```

  - **For example, the following controls how much effort is spent on fixing shorts over Macros**

```
set_app_options \
        -name route.detail.repair_shorts_over_macros_effort_level \
        -value high} ;#off|low|medium|high

route_detail -incremental true ;# Fix DRC violations
```


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Introduction to Routing : Detailed Routing

- **Rerouting Problematic Nets**
  - **Nets that have DRC violations, huge delay due to excessive Layer hopping or Crosstalk violations might benefit fr om being rerouted**
    - Use ***"remove_routes"*** to remove the net delay routing
    - Use ***"route_eco"*** to reroute the net (global, track, detail)
    - **Example of remove & re-route net**

```
# remove routes before re-route
remove_routes -nets "I_ORCA_TOP/I_BLENDER/mult_203/n141" -detail_route

# re-route after remove routes
route_eco -nets "I_ORCA_TOP/I_BLENDER/mult_203/n141"
```

  - **Remember that you can lock a net's routing (if you're satisfied with it) so that it will not be touched by the router l ater**

```
set_attribute \
        -objects "I_ORCA_TOP/I_BLENDER/mult_203/n141" \;#dont_reroute_nets
        -name \;#physical_status
        -value fixed ;#locked
```

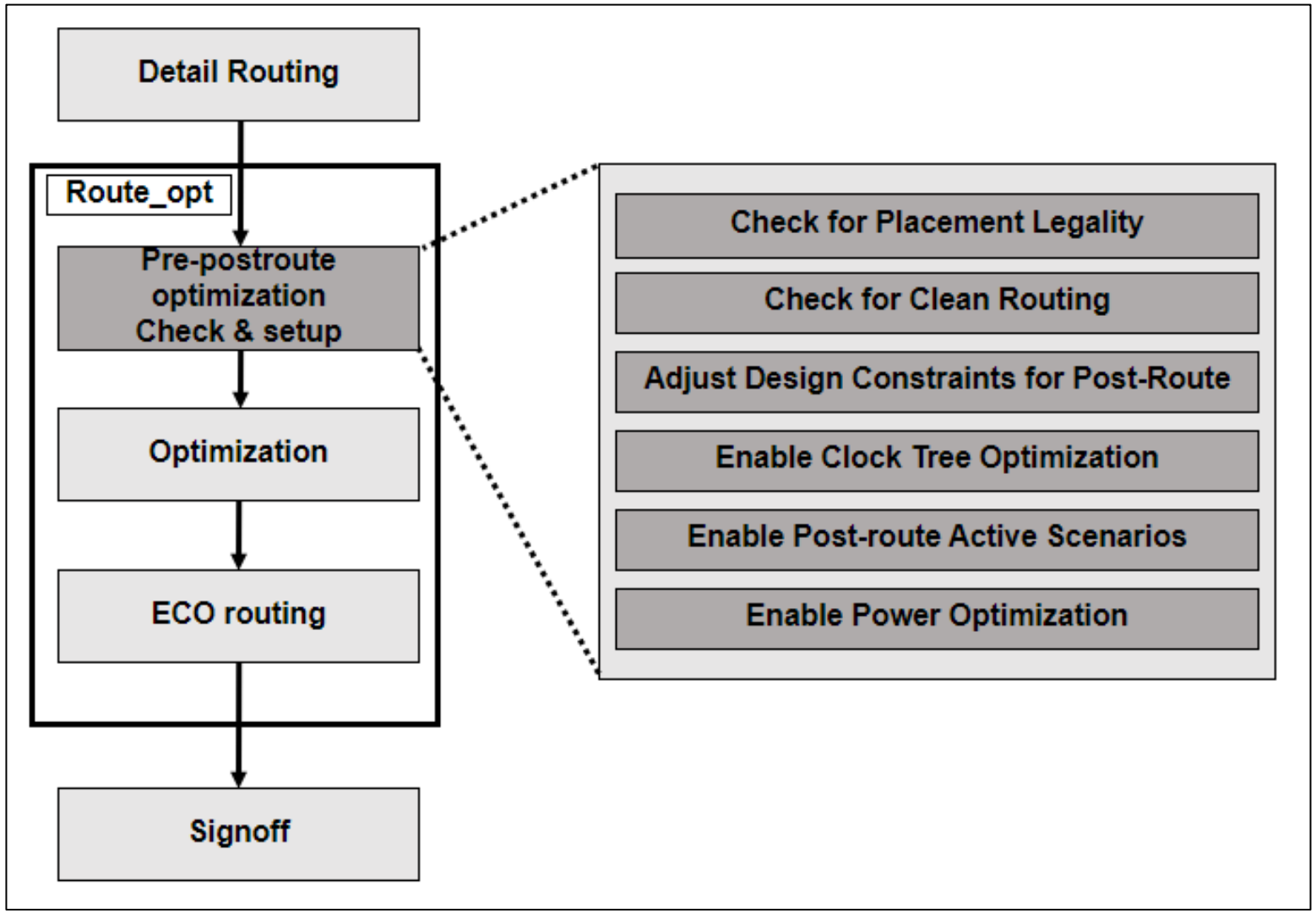

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WEEK 12 : Route optimization (post-route optimization)

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Introduction to Routing : Post-Route Optimization

- **Pre-Post-Route Checks and Setup for Post-Routing**



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Introduction to Routing : Post-Route Optimization

- **Goal of post-route optimization**
  - **The goal of the post-route optimization phase is to achieve design convergence with best out-of-box QoR**
    - Intelligent tradeoff of accuracy and runtime
    - Post-route solves many design violations very quickly and the last few problems with high accuracy
    - Optimizes a large number of design problems simultaneously
    - Solver-based optimization on QoR metrics of setup, hold, maximum transition, area, and power

- **Post_Route Optimization Flow**
  - **Clock & Signal nets have been routed**
  - **Afterwards, post-route optimization is performed**

```
#route_optimization
route_opt
```

    - **Concurrently optimizes:**
      - Timing & max-transition/capacitance (by default)
      - Clock tree, power (optionally)
    - **Updates routes affected by optimizations using ECO routing**

- **This command optimizes the current routed design for timing, electrical DRC violations, area, and power**
- **It performs legalization and ECO routing on the optimized design. By default, power optimization, concurrent clock and data optimization, and crosstalk reduction are not performed by this command**
- **Use application options in the "SEE ALSO" section to enable them :** ***ex) enable "route.track.crosstalk_driven"***

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Introduction to Routing : Post-Route Optimization

- **Check Placement Legality before Routing**
  - **Run "check_legality" to ensure design readness for "route_opt"**

```
****************************************
  Report : Legality
****************************************

VIOLATIONS BY CATEGORY:
   MOVABLE  APP-FIXED USER-FIXED  DESCRIPTION
         0          0          0  Two objects overlap.
         0          0          0  A cell violates a pnet.
         0          0          0  A cell is illegal at a site.
         0          0          0  A cell is not aligned with a site.
         0          0          0  A cell has an illegal orientation.
         0          0          0  A cell spacing rule is violated.
         0          0          0  A layer rule is violated.
         0          0          0  A cell is in the wrong region.
         0          0          0  Two cells violate cts margins.
         0          0          0  Two cells violate coloring.

         0          0          0  TOTAL

TOTAL 0 Violations.
```

  - **Check No moveable violations exist**

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Introduction to Routing : Post-Route Optimization

- **Check for Clean Routing**
  - **Apply "check_routes" to confirm that the design is ready for "route_opt"**
    - Clock & Signal nets are routed
    - No open nets
    - Reasonable physical DRC violations

```
Verify Summary:

Total number of nets = 17087, of which 0 are not extracted
Total number of open nets = 0, of which 0 are frozen
Total number of excluded ports = 0 ports of 0 unplaced cells connected to 0 nets
                                 0 ports without pins of 0 cells connected to 0 nets
                                 0 ports of 0 cover cells connected to 0 non-pg nets
Total number of DRCs = 164
Total number of antenna violations = 0
Total number of tie to rail violations = not checked
Total number of tie to rail directly violations = not checked
```


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Introduction to Routing : Post-Route Optimization

- **Adjust Design Constraints & Scenarios for Post Route optimization**
  - **Apply "compute_clock_latency" to update block's timing constraints**
    - Calculates and updates the I/O latencies of real and virtual clock objects for all active mode and corner combinations
    - This should have occurred before "clock_opt -from final_opto" step
  - **Apply post route design constraints like clock uncertainty, max_transition to match with signoff criteria**
  - **Apply all scenarios for post-route optimization**
  - **Align with scenario setup for signoff check**
  - **Enable leakage and dynamic scenario for power optimization as needed**

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Introduction to Routing : Post-Route Optimization

- **Enable Clock Tree Optimization in Post Route optimization**
  - **Enable CCD (Concurrent Clock and Data) optimization during route optimization**

```
set_app_options -name route_opt.flow.enable_ccd -value true ;# (default : false)
```

  - **Enable clock power or clock area recovery**

```
set_app_options -name route_opt.flow.enable_clock_power_recovery \
                -value auto ;# (none | auto | area | power)
```

- **This option turns on clock power recovery or clock area recovery in route_opt**
- **When power recovery mode is run, CCD will work on improving the power depending upon the scenario settings for power**
- **During power or area recovery, CCD performs clock cell sizing, clock repeater removal, clock repeater sizing and flip-flop sizing**

  - **Enable timing-driven post-route clock**

```
set_app_options -name route_opt.flow.enable_ccd_clock_drc_fixing \
                -vlaue auto ;# (auto | always_on | always_off)
```

- **This option specifies whether timing-driven post route clock drc fixing feature is turned on during the route_opt command**
- **When enable, this feature will fix clock transition and capacitance violations similar to clock drc fixing feature of the synthesize _clock_trees –postroute command, buf with minimal impact on the Timing QoR (both of setup & hold)**

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Introduction to Routing : Post-Route Optimization

- **Enable Power Optimization in Post Route optimization**
  - **Enable Power optimization**

```
set_app_options -name route_opt.flow.enable_power \
                -value true ;# (default : false)
```

  - **This option specifies whether the route_opt command enables power optimization**
  - **By default, power type is determined based on scenario settings as shown in the table below**

  - **Power optimization, once enabled, is based on the scenario configuration**
    - ❖ **Active scenario with leakage + dynamic : total power optimization**
    - ❖ **Active scenario with leakage only : leakage power optimization**
    - ❖ **If neither leakage or dynamic power-activated scenarios exist, no power optimization occurs**

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WEEK 13 : LAB for Route & Route_opt

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Routing : Introduction to Route.tcl

- Review and check the variables in each scripts
  - **Review 02_mcmm.tcl**

```
## Setup
###################################
#remove scenarios modes corners before setting

remove_scenarios -all
remove_modes -all
remove_corners -all

# Read TLU+ files
# TLU+ file is given by Foundary Company
# This file contains Metal capacitance at different spacing and width in the form of a lookup -
# talbe which provides high accuracy and runtime (This file can be used to extract RC value from interconnects)

read_parasitic_tech -name {minTLU} -tlup $data(tech_tlupbc)
read_parasitic_tech -name {maxTLU} -tlup $data(tech_tlupwc)

# source mcmm(multi corners multi modes).tcl

source -e ./icc2_common_env/mcmm.tcl

#read_sdc after create mcmm

#read_sdc(synopsys design constraints)
#SDC contain information such as clock definitions, input and output delays, false paths, and timing exeptions

#scan_shift mode
current_mode scan_shift
read_sdc $data(shift_sdc) ;# ${libdir}/SDC/ORCA_shift.sdc

#func mode
current_mode func
read_sdc $data(func_sdc) ;# ${libdir}/SDC/ORCA_func.sdc

#scan_capture mode
current_mode scan_capture
read_sdc $data(capture_sdc) ;# ${libdir}/SDC/ORCA_capture.sdc
```

Refer to "introduction_to_CTS" pages 93~95

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Routing : Introduction to Route.tcl

- Review and check the variables in each scripts
  - **Go to the Script Directory**

```
  < user@linux script >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/06_route/script
  < user@linux script >> ls
01_design_setup_for_route.tcl                 09_route_auto.tcl                 17_remove_clock_latency_and_ideal_network.tcl
02_mcmm.tcl                                   10_search_and_repair.tcl          18_set_max_transition.tcl
03_set_antenna_rule.tcl                       11_check_timing_and_DRC.tcl       19_set_app_option.tcl
04_set_optimization_cell.tcl                  12_report_route_and_exit.tcl      20_route_opt.tcl
05_remove_clock_latency_and_ideal_network.tcl 13_design_setup_for_route_opt.tcl 21_search_and_repair.tcl
06_set_max_transition.tcl                     14_mcmm.tcl                       22_check_timing_and_DRC.tcl
07_set_app_option.tcl                         15_set_antenna_rule.tcl           23_report_route_opt_and_exit.tcl
08_create_custom_via_for_routing.tcl          16_set_optimization_cell.tcl      full_script
```

  - **Review 01_design_setup_for_route.tcl**

```
#set current step & before step
set step route
set before_step psyn
#source design_setup.tcl
source -e ./icc2_common_env/design_setup.tcl

#source timer.rpt script - report runtime for placement
source ./icc2_common_env/sec_runtime.tcl

sec_StartTimer copy_lib

#set_host_options -max_cores 16

#open lib
open_lib ../05_cts/nlib/ORCA_${before_step}.nlib
#copy lib
copy_lib -from ../05_cts/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
#close & open current lib
close_lib -all
open_lib ./nlib/ORCA_${step}.nlib
#open block
open_block ./nlib/ORCA_${step}.nlib:ORCA.design
#link_block
link_block -rebind

sec_ReportResourceUtil copy_lib
```

The name of the step is different for each script

```
##connect_pg_net (CORNER_CELL & FILLER_CELL)

#CORNER CELL
connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_corner_cell_*/VSSIO"]
connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_corner_cell_*/VSS"]

connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_corner_cell_*/VDDIO"]
connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_corner_cell_*/VDD"]
#FILLER CELL
connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_filler_instance_*/VSSIO"]
connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_filler_instance_*/VSS"]

connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_filler_instance_*/VDDIO"]
connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_filler_instance_*/VDD"]
```


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Routing : Introduction to Route.tcl

- Review and check the variables in each scripts
  - **Review 03_set_antenna_rule.tcl**

```
#set anttena rule
source -echo ./icc2_common_env/saed32nm_ant_1p9m.tcl
```

  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/route/icc2_common_env/saed32nm_ant_1p9m.tcl**

```
# ########################################################

remove_antenna_rules
define_antenna_rule -mode 4 -diode_mode 2 -metal_ratio 1000 -cut_ratio 20

define_antenna_layer_rule -mode 4 -layer "M1" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M2" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M3" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M4" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M5" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M6" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M7" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M8" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M9" -ratio 1000 -diode_ratio {0.06 0 8000 50000}


define_antenna_layer_rule -mode 4 -layer "VIA1" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA2" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA3" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA4" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA5" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA6" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA7" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA8" -ratio 20 -diode_ratio {0.06 0 200 1000}
```

Define antenna rule for each layer

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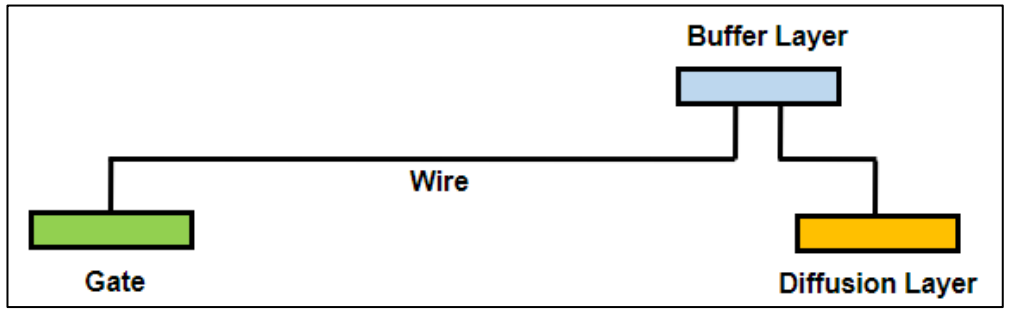
Introduction to Routing : Antenna Effect

- **Antenna Effect**
  - **The antenna effect in VLSI is the charge collection effect that occurs during the fabrication process, expecially pla sma etching**
  - **Problems that can occur due to Antenna Effect**
    - Reduced life expectancy of a device
    - Changing threshold voltage
    - Increased gate leakage
  - **Antenna Rules**
    - In order to avoid problems above, there is Antenna Rules determined
    - Antenna Rules are indicating Max Area of Metal to connect to a gate, avoiding Anttena Effect
    - Depending on the breakthrough voltage of the transistor Max ratio of the wire area to gate area may vary from 100:1 to 5000:q (Thicker g ate of the transistor, higher Ratio)

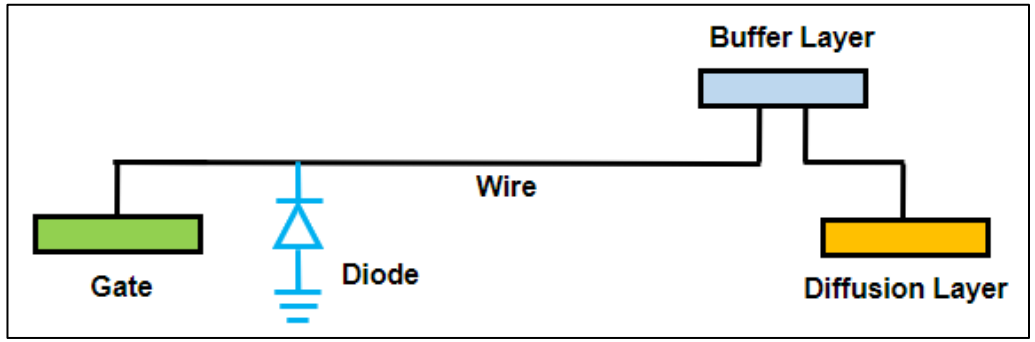
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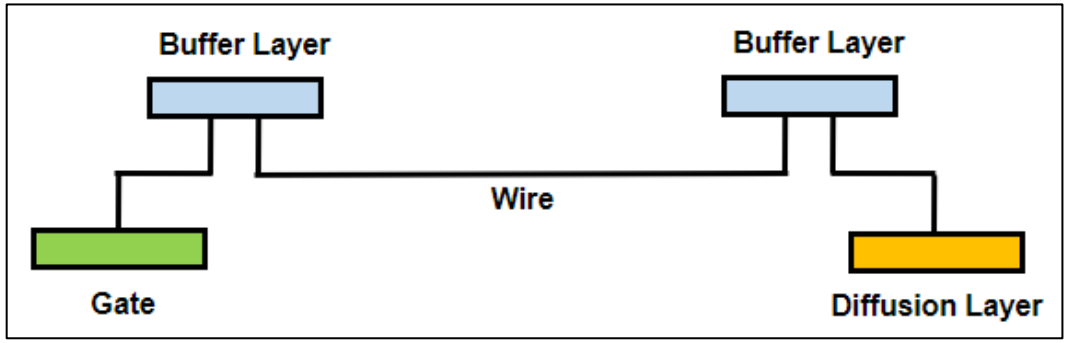
Introduction to Routing : Antenna Effect

- **Antenna Effect**
  - In this picture you can see a long wire, connected to the gate, forming gate breakthrough and it is not connected to the diff usion layer, antenna effect happens during plasma etching



  - **How to prevent Antenna Effect?**
    - Connect diode to the wire
    - Divide the wire into two parts and to connect wire and a gate with the buffer layer





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Routing: Introduction to Route.tcl

- Review and check the variables in each scripts
  - **Review 04_set_optimization_cell.tcl**

```
#set include hold Timing optimization cell
set_lib_cell_purpose -exclude hold [get_lib_cells]
set_lib_cell_purpose -include hold [get_lib_cells "*/NBUFFX2_HVT saed32hvt/DELLN1X2_HVT"]
```

  - **Review 05_remove_clock_latency_and_ideal_network.tcl**

```
#remove clock_latency & ideal_network

set mode "func scan_shift scan_capture"

foreach MODE $mode {
current_mode $MODE
echo "current mode is $MODE"
remove_clock_latency [get_clocks *]
remove_clock_latency [get_flat_pins -filter "is_clock_pin == true"]
remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]]]
set_propagated_clock [all_clocks]
}

compute_clock_latency
```


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Routing : Introduction to Route.tcl

- Review and check the variables in each scripts
  - **Review 06_set_max_transition.tcl**

```
#set max_transition for all Scenarios
current_mode func
set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]

current_mode scan_capture
set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]

current_mode scan_shift
set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
```

  - **Review 07_set_app_option.tcl**

```
################################
#set_app_options before routing
################################

set_app_options -list {opt.tie_cell.add_to_highest_hierarchy {false}} ;# During optimization, tie-cells will be inserted to drive constant pins if tie-cells
 were available for optimization purpose
set_attribute -objects [get_blocks -all -lib_cells saed32hvt:TIEH_HVT.frame] -name dont_touch -value false ;#set TIEH cell dont_touch value false
set_attribute -objects [get_blocks -all -lib_cells saed32hvt:TIEL_HVT.frame] -name dont_touch -value false ;#set TIEL cell dont_touch value false

set_app_options -list {
  route.global.timing_driven true ;# enables or disables timing-driven global routing
  route.global.crosstalk_driven true ;# enables or disables crosstalk-driven global routing
  route.track.timing_driven true ;# enables or disables timing-driven track assignment
  route.track.crosstalk_driven true ;# enables or disables crosstalk-driven track assignment
  route.detail.timing_driven true ;# This app option specifies whether timing-driven routing is enabled
  route.detail.force_max_number_iterations false } ;# This app option controls whether the max number of iterations must be run when DRCs do not converge

set_app_options -list {route.detail.generate_extra_off_grid_pin_tracks {true}} ;# This app option specifies whether to generate extra off-grid routing track
s for pin connections
set_app_options -list {route.detail.generate_off_grid_feed_through_tracks {high}} ;# This app option specifies the effort level used to generate extra off-g
rid routing tracks for feedthrough nets between blockages
```


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Routing : Introduction to Route.tcl

- Review and check the variables in each scripts
  - **Review 08_create_custom_via_for_routing.tcl**

```
#FIX XDFF ADDER VIA (Create custom via for routing)
create_via_def -is_default -lower_layer M1 \
               -upper_layer M2 -cut_size {0.020 0.020} \
               -lower_enclosure {0.015 0.015} -cut_layer VIA1 \
               -upper_enclosure {0.015 0.015} VIA12_thin
```

  - **Review 09_route_auto.tcl**

```
##########################
#route_auto (detail route)
##########################

#detail_route

sec_StartTimer route

route_auto ;# The route_auto command performs all routing stages in a single step (global routing, track assignment, detail routing)

sec_ReportResourceUtil route

#set app option for second route_auto
set_app_options -list {place.legalize.stream_place {true}} ;# Stream placement improves the largest displacements in the design (For each
large-displacement cell, it attempts to move out other cells to make room, such that a single very large displacement is replaced by a str
eam of small displacements)

#optimize_routability
optimize_routability ;#  The  optimize_routability  command  helps  to  improve routability of a design by spacing cells further apart wit
h the use of the keepout  margins  or by flipping cells

#second route_auto

sec_StartTimer route2

route_auto ;# The route_auto command performs all routing stages in a single step (global routing, track assignment, detail routing)

sec_ReportResourceUtil route2

set_app_options -list {place.legalize.stream_place {false}}
```


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Routing : Introduction to Route.tcl

- Review and check the variables in each scripts
  - **Review 10_search_and_repair.tcl**

```
#Search & Repair
route_detail -incremental true -initial_drc_from_input true ;# This command performs detail routing on a block
compute_clock_latency ;# This command is used primarily to correct your block's timing constraints after CTS
update_timing
```

  - **Review 11_check_timing_and_DRC.tcl**

```
##connect_pg_net (CORNER_CELL & FILLER_CELL)
source -e ./user_script/pg_connection.tcl

#check_route after route_auto
check_route ;# Verifies, checks, and reports routing statistics for a block

#check_lvs (short, open)
check_lvs ;# Checks for inconsistencies in the physical layout of the current design

#report_timing & QoR

report_timing -significant_digits 5
report_qor -summary -significant_digits 5
```

```
##connect_pg_net (CORNER_CELL & FILLER_CELL)

#CORNER CELL
connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_corner_cell_*/VSSIO"]
connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_corner_cell_*/VSS"]

connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_corner_cell_*/VDDIO"]
connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_corner_cell_*/VDD"]
#FILLER CELL
connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_filler_instance_*/VSSIO"]
connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_filler_instance_*/VSS"]

connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_filler_instance_*/VDDIO"]
connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_filler_instance_*/VDD"]
```

  - **Review 12_report_route_and_exit.tcl**

```
#update timing
update_timing -full ;#Updates timing information on the current design

sec_StartTimer report

#write "reports files" of psyn at ${rundir}/reports/psyn/* directory
source ./user_script/route_rpt.tcl

sec_ReportResourceUtil report

#Save & Exit
save_lib -all
exit
```


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Route_opt : Introduction to Route_opt.tcl

- Review and check the variables in each scripts
  - **Go to the Script Directory**

```
  < user@linux script >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/06_route/script
  < user@linux script >> ls
01_design_setup_for_route.tcl                 09_route_auto.tcl                  17_remove_clock_latency_and_ideal_network.tcl
02_mcmm.tcl                                   10_search_and_repair.tcl           18_set_max_transition.tcl
03_set_antenna_rule.tcl                       11_check_timing_and_DRC.tcl        19_set_app_option.tcl
04_set_optimization_cell.tcl                  12_report_route_and_exit.tcl       20_route_opt.tcl
05_remove_clock_latency_and_ideal_network.tcl 13_design_setup_for_route_opt.tcl  21_search_and_repair.tcl
06_set_max_transition.tcl                     14_mcmm.tcl                        22_check_timing_and_DRC.tcl
07_set_app_option.tcl                         15_set_antenna_rule.tcl            23_report_route_opt_and_exit.tcl
08_create_custom_via_for_routing.tcl          16_set_optimization_cell.tcl       full_script
```

  - **Review 13_design_setup_for_route_opt.tcl**

```
#set current step & before step
set step route_opt
set before_step route
#source design_setup.tcl
source -e ./icc2_common_env/design_setup.tcl

#source timer.rpt script - report runtime for placement
source ./icc2_common_env/sec_runtime.tcl

sec_StartTimer copy_lib

#set_host_options -max_cores 16

#open lib
open_lib ./nlib/ORCA_${before_step}.nlib
#copy lib
copy_lib -from ./nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
#close & open current lib
close_lib -all
open_lib ./nlib/ORCA_${step}.nlib
#open block
open_block ./nlib/ORCA_${step}.nlib:ORCA.design
#link_block
link_block -rebind

sec_ReportResourceUtil copy_lib
```

The name of the step is different for each script

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Route_opt : Introduction to Route_opt.tcl

- Review and check the variables in each scripts
  - **Review 14_mcmm.tcl**

```
## Setup
###################################
#remove scenarios modes corners before setting

remove_scenarios -all
remove_modes -all
remove_corners -all

# Read TLU+ files
# TLU+ file is given by Foundary Company
# This file contains Metal capacitance at different spacing and width in the form of a lookup -
# talbe which provides high accuracy and runtime (This file can be used to extract RC value from interconnects)

read_parasitic_tech -name {minTLU} -tlup $data(tech_tlupbc)
read_parasitic_tech -name {maxTLU} -tlup $data(tech_tlupwc)

# source mcmm(multi corners multi modes).tcl

source -e ./icc2_common_env/mcmm.tcl

#read_sdc after create mcmm

#read_sdc(synopsys design constraints)
#SDC contain information such as clock definitions, input and output delays, false paths, and timing exeptions

#scan_shift mode
current_mode scan_shift
read_sdc $data(shift_sdc) ;# ${libdir}/SDC/ORCA_shift.sdc

#func mode
current_mode func
read_sdc $data(func_sdc) ;# ${libdir}/SDC/ORCA_func.sdc

#scan_capture mode
current_mode scan_capture
read_sdc $data(capture_sdc) ;# ${libdir}/SDC/ORCA_capture.sdc
```

Refer to "introduction_to_CTS" pages 93~95

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Route_opt : Introduction to Route_opt.tcl

- Review and check the variables in each scripts
  - **Review 15_set_antenna_rule.tcl**

```
#set anttena rule
source -echo ./icc2_common_env/saed32nm_ant_1p9m.tcl
```

  - **Review $WORK_DIR/2_BACKEND/1_pnr/rundir/route/icc2_common_env/saed32nm_ant_1p9m.tcl**

```
# ########################################################

remove_antenna_rules
define_antenna_rule -mode 4 -diode_mode 2 -metal_ratio 1000 -cut_ratio 20

define_antenna_layer_rule -mode 4 -layer "M1" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M2" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M3" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M4" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M5" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M6" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M7" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M8" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M9" -ratio 1000 -diode_ratio {0.06 0 8000 50000}


define_antenna_layer_rule -mode 4 -layer "VIA1" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA2" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA3" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA4" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA5" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA6" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA7" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA8" -ratio 20 -diode_ratio {0.06 0 200 1000}
```

Define antenna rule for each layer

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Route_opt : Introduction to Route_opt.tcl

- Review and check the variables in each scripts
  - **Review 16_set_optimization_cell.tcl**

```
#set include hold Timing optimization cell
set_lib_cell_purpose -exclude hold [get_lib_cells]
set_lib_cell_purpose -include hold [get_lib_cells "*/NBUFFX2_HVT saed32hvt/DELLN1X2_HVT"]
```

  - **Review 17_remove_clock_latency_and_ideal_network.tcl**

```
#remove clock_latency & ideal_network

set mode "func scan_shift scan_capture"

foreach MODE $mode {
current_mode $MODE
echo "current mode is $MODE"
remove_clock_latency [get_clocks *]
remove_clock_latency [get_flat_pins -filter "is_clock_pin == true"]
remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]]]
set_propagated_clock [all_clocks]
}

compute_clock_latency
```


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Route_opt : Introduction to Route_opt.tcl

- Review and check the variables in each scripts
  - **Review 18_set_max_transition.tcl**

```
#set max_transition for all Scenarios
current_mode func
set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]

current_mode scan_capture
set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]

current_mode scan_shift
set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
```

  - **Review 19_set_app_option.tcl**

```
################################
#set_app_options before routing
################################

set_app_options -list {opt.tie_cell.add_to_highest_hierarchy {false}} ;# During optimization, tie-cells will be inserted to drive constant pins if tie-cells
 were available for optimization purpose
set_attribute -objects [get_blocks -all -lib_cells saed32hvt:TIEH_HVT.frame] -name dont_touch -value false ;#set TIEH cell dont_touch value false
set_attribute -objects [get_blocks -all -lib_cells saed32hvt:TIEL_HVT.frame] -name dont_touch -value false ;#set TIEL cell dont_touch value false

set_app_options -list {
  route.global.timing_driven true ;# enables or disables timing-driven global routing
  route.global.crosstalk_driven true ;# enables or disables crosstalk-driven global routing
  route.track.timing_driven true ;# enables or disables timing-driven track assignment
  route.track.crosstalk_driven true ;# enables or disables crosstalk-driven track assignment
  route.detail.timing_driven true ;# This app option specifies whether timing-driven routing is enabled
  route.detail.force_max_number_iterations false } ;# This app option controls whether the max number of iterations must be run when DRCs do not converge

set_app_options -list {route.detail.generate_extra_off_grid_pin_tracks {true}} ;# This app option specifies whether to generate extra off-grid routing track
s for pin connections
set_app_options -list {route.detail.generate_off_grid_feed_through_tracks {high}} ;# This app option specifies the effort level used to generate extra off-g
rid routing tracks for feedthrough nets between blockages
```


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Route_opt : Introduction to Route_opt.tcl

- Review and check the variables in each scripts
  - **Review 20_route_opt.tcl**

```
#######################
#post-route optimization
#######################

sec_StartTimer route_opt

route_opt ;# This command optimizes the current routed design for timing, electrical DRC violations, area, and power

sec_StartTimer route_opt

optimize_routes
```

  - **Review 21_search_and_repair.tcl**

```
#Search & Repair
route_detail -incremental true -initial_drc_from_input true ;# This command performs detail routing on a block
compute_clock_latency ;# This command is used primarily to correct your block's timing constraints after CTS
update_timing
```

  - **Review 22_check_timing_and_DRC.tcl**

```
##connect_pg_net (CORNER_CELL & FILLER_CELL)
source -e ./user_script/pg_connection.tcl

#check_route after route_auto
check_route ;# Verifies, checks, and reports routing statistics for a block

#check_lvs (short, open)
check_lvs ;# Checks for inconsistencies in the physical layout of the current design

#check_pg_drc
check_pg_drc

#report_timing & QoR

report_timing -significant_digits 5
report_qor -summary -significant_digits 5
```


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Route_opt : Introduction to Route_opt.tcl

- Review and check the variables in each scripts
  - **Review 23_report_route_opt_and_exit.tcl**

```
############################################################
#report Timing & constraints for route_opt
############################################################

#update timing
update_timing -full ;#Updates timing information on the current design

sec_StartTimer report

#write "reports files" of psyn at ${rundir}/reports/psyn/* directory
source ./user_script/route_opt_rpt.tcl

sec_ReportResourceUtil report

#Save & Exit
save_lib -all
exit
```


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LAB WEEK 1 : Route (route_auto)

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Routing : Use ICC2 (IC Compiler II tool)

1. **Invoke tool icc2_shell**
   - **Recommend the icc2/2021.06-SP5 version**

>> icc2_shell –gui | tee -i ./07_route.log ;# (| tee -I ./07_route.log) is the command to output log

```
 < user@linux 06_route >> icc2_shell -gui | tee -i ./07_route.log



                            IC Compiler II (TM)

                Version S-2021.06-SP5 for linux64 - Jan 14, 2022 -SLE

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                        https://solvnetplus.synopsys.com)

Loading user preference file /u/kjinsoo/.synopsys_icc2_gui/preferences.tcl
icc2_shell>
```

2. **Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/01_design_setup_for_route.tcl**
   - **set step & source design_setup.tcl**

```
icc2_shell> set step route
route
icc2_shell> set before_step psyn
psyn
icc2_shell> source -e ./icc2_common_env/design_setup.tcl
```

It is also good to Read Design by source 02_floorplan.tcl, but when you proceed for the first time, it is recommended that you open the script and apply it one phrase at a time

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Routing : Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/01_design_setup_for_route.tcl

- **Open_lib & copy lib**

```
icc2_shell> source ./icc2_common_env/sec_runtime.tcl
icc2_shell>
icc2_shell> sec_StartTimer copy_lib
1
icc2_shell> open_lib ../05_cts/nlib/ORCA_${before_step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/05_cts/nlib/
ORCA_psyn.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32sram.ndm' (FILE-007)
{ORCA_psyn.nlib}
icc2_shell> copy_lib -from ../05_cts/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
Information: Copying from library "ORCA_psyn.nlib" to "./nlib/ORCA_route.nlib".
{ORCA_route.nlib}
icc2_shell> close_lib -all
Closing all libraries...
1
```


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Routing : Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/01_design_setup_for_route.tcl

- **Close reference lib & open & link current block**

```
icc2_shell> open_lib ./nlib/ORCA_${step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/06_route/nli
b/ORCA_route.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32sram.ndm' (FILE-007)
{ORCA_route.nlib}
icc2_shell> open_block ./nlib/ORCA_${step}.nlib:ORCA.design
Information: User units loaded from library 'saed32hvt' (LNK-040)
Opening block 'ORCA_route.nlib:ORCA.design' in edit mode
{ORCA_route.nlib:ORCA.design}
```

```
icc2_shell> link_block
Using libraries: ORCA_route.nlib EXPLORE_physical_only saed32hvt saed32io_fc saed32lvt saed32pll saed32rvt
saed32sram
Visiting block ORCA_route.nlib:ORCA.design


icc2_shell> link_block -rebind
Warning: Block 'ORCA_route.nlib:ORCA.design' is already linked. (LNK-067)
0
icc2_shell> sec_ReportResourceUtil copy_lib
```


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Routing : Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/11_check_timing_and_DRC.tcl

- **Connect_pg_net for CORNER Cell & IO FILLER Cell (VDDIO, VSSIO, VDD, VSS)**

```
icc2_shell>
icc2_shell> #CORNER CELL
icc2_shell> connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_corner_cell_*/VSSIO"]
icc2_shell> connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_corner_cell_*/VSS"]
icc2_shell>
icc2_shell> connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_corner_cell_*/VDDIO"]
icc2_shell> connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_corner_cell_*/VDD"]
icc2_shell> #FILLER CELL
icc2_shell> connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_filler_instance_*/VSSIO"]
icc2_shell> connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_filler_instance_*/VSS"]
icc2_shell>
icc2_shell> connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_filler_instance_*/VDDIO"]
icc2_shell> connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_filler_instance_*/VDD"]
```

- **Filler cells and Corner cells are not included in the Netlist, and are not defined Pg connection in UPF file, so you must define them**
- **If the power connection information is not defined, DRC violation may occur**

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Routing: Use ICC2 (IC Compiler II tool)

3. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/02_mcmm.tcl (same as placement step)

- **Remove all (scenarios, modes, corners) and recreate mcmm**

```
icc2_shell> remove_scenarios -all
12
icc2_shell> remove_modes -all
3
icc2_shell> remove_corners -all
4
```

- **Read TLU+ files**

```
icc2_shell> read_parasitic_tech -name {minTLU} -tlup $data(tech_tlupbc)
1
icc2_shell> read_parasitic_tech -name {maxTLU} -tlup $data(tech_tlupwc)
1
```

- **Source mcmm.tcl (create mcmm, set parasitic_parameters, set_voltage)**

```
icc2_shell> source -e ./icc2_common_env/mcmm.tcl
```

- **Read SDC (func, scan_shift, scan_capture)**

```
icc2_shell> current_mode scan_shift
{scan_shift}
icc2_shell> read_sdc $data(shift_sdc)
```

```
icc2_shell> current_mode scan_capture
{scan_capture}
icc2_shell> read_sdc $data(capture_sdc)
```

```
icc2_shell> current_mode func
{func}
icc2_shell> read_sdc $data(func_sdc)
```


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Routing : Use ICC2 (IC Compiler II tool)

4. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/03_set_antenna_rule.tcl

- **Set Antenna Rules for prevent Antenna violation**

```
icc2_shell> source -echo ./icc2_common_env/saed32nm_ant_1p9m.tcl
#Antenna Rule for 32nm TECH
#       Author:         ArtakY
#       @(#) Antenna rules
#       @(#) Revision 1.0.0.0
#       @(#) Date     28-Feb-11
# technology: saed32/28nm_1p9m
#
# Revision history:
# rev            date     who  what
# ------------  --------- ---- ------------------------------
# Rev. 1.0.0.0  28-Feb-11 AY   Initial version
#
# ############################################################
remove_antenna_rules
define_antenna_rule -mode 4 -diode_mode 2 -metal_ratio 1000 -cut_ratio 20
define_antenna_layer_rule -mode 4 -layer "M1" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M2" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M3" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M4" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M5" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M6" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M7" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M8" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M9" -ratio 1000 -diode_ratio {0.06 0 8000 50000}
define_antenna_layer_rule -mode 4 -layer "VIA1" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA2" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA3" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA4" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA5" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA6" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA7" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA8" -ratio 20 -diode_ratio {0.06 0 200 1000}
1
```


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Routing : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/04_set_optimization_cell.tcl

- **Exclude all lib_cells before set optimization cell**

```
icc2_shell> set_lib_cell_purpose -exclude hold [get_lib_cells]
Warning: The 'set_lib_cell_purpose' command cannot be used on library cell 'EXPLORE_physical_only:LSUPX1_HV
T.frame' which has no logical model. (NDMUI-010)
Warning: The 'set_lib_cell_purpose' command cannot be used on library cell 'EXPLORE_physical_only:LSUPX2_HV
T.frame' which has no logical model. (NDMUI-010)
Warning: The 'set_lib_cell_purpose' command cannot be used on library cell 'EXPLORE_physical_only:LSUPX4_HV
T.frame' which has no logical model. (NDMUI-010)
Warning: The 'set_lib_cell_purpose' command cannot be used on library cell 'EXPLORE_physical_only:LSUPX8_HV
T.frame' which has no logical model. (NDMUI-010)
Warning: The 'set_lib_cell_purpose' command cannot be used on library cell 'EXPLORE_physical_only:LSDNSSX8_
HVT.frame' which has no logical model. (NDMUI-010)
Warning: The 'set_lib_cell_purpose' command cannot be used on library cell 'EXPLORE_physical_only:LSDNX1_HV
T.frame' which has no logical model. (NDMUI-010)
```

- **Include & set hold optimization cell list**

```
icc2_shell> set_lib_cell_purpose -include hold [get_lib_cells "*/NBUFFX2_HVT saed32hvt/DELLN1X2_HVT"]
Information: The design specific attribute override for lib_cell 'saed32hvt:NBUFFX2_HVT.timing' is set in t
he current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
Information: The design specific attribute override for lib_cell 'saed32hvt:DELLN1X2_HVT.timing' is set in
the current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
1
```


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Routing : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/05_remove_clock_latency_and_ideal_network.tcl

- **Remove clock latency & ideal network before routing**

```
icc2_shell> set mode "func scan_shift scan_capture"
func scan_shift scan_capture
icc2_shell>
icc2_shell> foreach MODE $mode {
? current_mode $MODE
? echo "current mode is $MODE"
? remove_clock_latency [get_clocks *]
? remove_clock_latency [get_flat_pins -filter "is_clock_pin == true"]
? remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]]]
? set_propagated_clock [all_clocks]
? }
current mode is func
current mode is scan_shift
Warning: Virtual clock 'forcePI_wft1' cannot be made propagated. (UIC-025)
Warning: Virtual clock 'measurePO_wft1' cannot be made propagated. (UIC-025)
current mode is scan_capture
Warning: Virtual clock 'forcePI_wft1' cannot be made propagated. (UIC-025)
Warning: Virtual clock 'measurePO_wft1' cannot be made propagated. (UIC-025)
```

- **Compute_clock_latency after remove clock latency & ideal network**

❖ calculates and updates the I/O latencies of real and virtual clock objects for all active mode and corner combinations

Ex)

```
icc2_shell> compute_clock_latency
Mode:scan_capture     Latency(early)   Latency(late)
Clock        Updated   rise    fall    rise    fall   Corner
-------------------------------------------------------------
pclk_wft1    Yes     0.3095  0.3095  0.3249  0.3249   ff_n40c
pclk_wft1    Yes     0.8063  0.8063  0.8647  0.8647   ss_n40c
pclk_wft1    Yes     0.3808  0.3808  0.3994  0.3994   ff_125c
pclk_wft1    Yes     0.7715  0.7715  0.8270  0.8270   ss_125c
```

This command is used primarily to correct your block's timing constraints after clock tree synthesis

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Routing : Use ICC2 (IC Compiler II tool)

7. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/06_set_max_transition.tcl

- **Set max_transition for all modes & corners**

```
icc2_shell> current_mode func
{func}
icc2_shell> set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
1
icc2_shell>
icc2_shell> current_mode scan_capture
{scan_capture}
icc2_shell> set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
1
icc2_shell>
icc2_shell> current_mode scan_shift
{scan_shift}
icc2_shell> set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
1
```

- **set_max_transition**
  - Specifies a maximum transition on ports, library cell pins, leaf cell pins, clocks or designs for the current scenario
  - If max_transition is set on a port, the port must have a transition time less than the specified transition_value
  - If specified on a design, the default maximum transition for that design is set
  - Library cell pins can also have a max_transition value specified in the library

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Routing : Use ICC2 (IC Compiler II tool)

8. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/07_set_app_option.tcl

- **Set application options before route_auto**

```
icc2_shell> set_app_options -list {opt.tie_cell.add_to_highest_hierarchy {false}}
opt.tie_cell.add_to_highest_hierarchy false
icc2_shell> set_attribute -objects [get_blocks -all -lib_cells saed32hvt:TIEH_HVT.frame] -name dont_touch -
value false
Information: The design specific attribute override for block 'saed32hvt:TIEH_HVT.frame' is set in the curr
ent block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
{saed32hvt:TIEH_HVT.frame}
icc2_shell> set_attribute -objects [get_blocks -all -lib_cells saed32hvt:TIEL_HVT.frame] -name dont_touch -
value false
Information: The design specific attribute override for block 'saed32hvt:TIEL_HVT.frame' is set in the curr
ent block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
{saed32hvt:TIEL_HVT.frame}
icc2_shell> set_app_options -list {route.global.timing_driven {true}}
route.global.timing_driven true
icc2_shell> set_app_options -list {route.global.crosstalk_driven {true}}
route.global.crosstalk_driven true
icc2_shell> set_app_options -list {route.track.timing_driven {true}}
route.track.timing_driven true
icc2_shell> set_app_options -list {route.track.crosstalk_driven {true}}
route.track.crosstalk_driven true
icc2_shell> set_app_options -list {route.detail.timing_driven {true}}
route.detail.timing_driven true
icc2_shell> set_app_options -list {route.detail.force_max_number_iterations {false}}
route.detail.force_max_number_iterations false
icc2_shell> set_app_options -list {route.detail.generate_extra_off_grid_pin_tracks {true}}
route.detail.generate_extra_off_grid_pin_tracks true
icc2_shell> set_app_options -list {route.detail.generate_off_grid_feed_through_tracks {high}}
route.detail.generate_off_grid_feed_through_tracks high
```


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Routing : Use ICC2 (IC Compiler II tool)

9. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/08_create_custom_via_for_routing.tcl

```
icc2_shell> create_via_def -is_default -lower_layer M1 \
?         -upper_layer M2 -cut_size {0.020 0.020} \
?          -lower_enclosure {0.015 0.015} -cut_layer VIA1 \
?         -upper_enclosure {0.015 0.015} VIA12_thin
{VIA12_thin}
```

10. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/09_route_auto.tcl

- **Run route_auto (global routing, track assignment, detail routing)**

```
icc2_shell> sec_StartTimer route
1
icc2_shell> route_auto
icc2_shell> sec_ReportResourceUtil route
icc2_shell>
icc2_shell> set_app_options -list {place.legalize.stream_place {true}}
place.legalize.stream_place true
icc2_shell> optimize_routability
Total DRC Err: 165
Total Cells w/ DRC Err: 87
Total PinDensity DRC Err: 165
Total Cells w/ PinD DRC Err: 87
Off-grid                                          : 165
There are 87 cells with pin density DRC err.
icc2_shell> sec_StartTimer route2
1
icc2_shell> route_auto
icc2_shell> sec_ReportResourceUtil route2
icc2_shell> set_app_options -list {place.legalize.stream_place {false}}
place.legalize.stream_place false
```

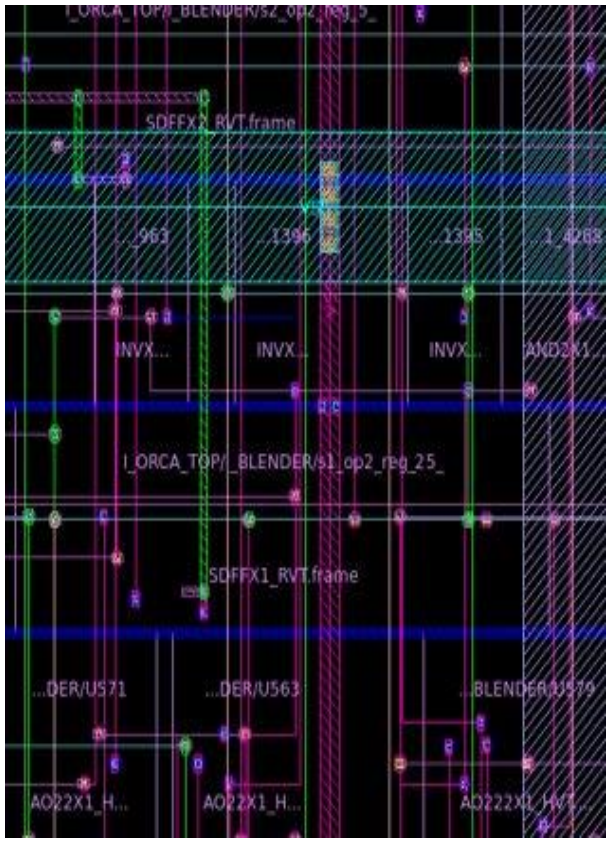
- **Check that the min, max routing Layers, and Antenna Rules on the log are properly applied**

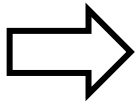
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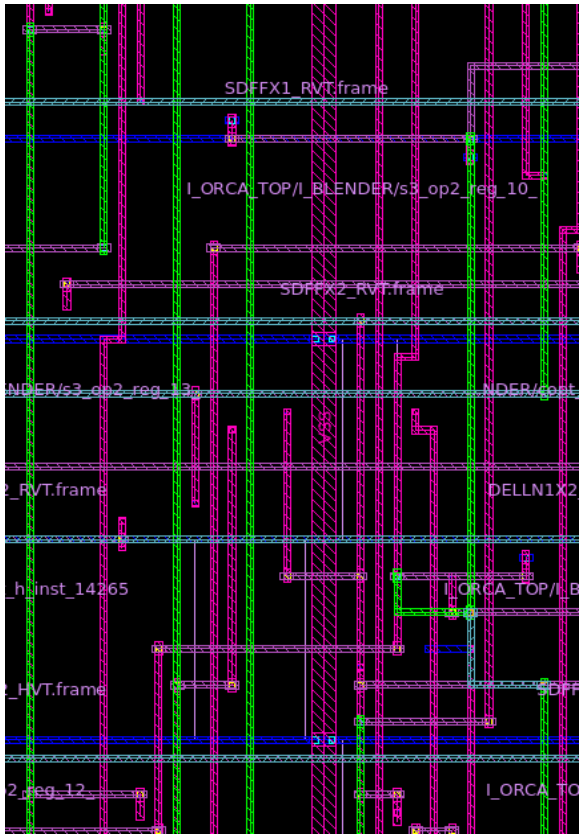
Routing : Use ICC2 (IC Compiler II tool)

10. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/09_route_auto.tcl

- **Check Detailed Routing after route_auto**







PSYN (Global Routing)

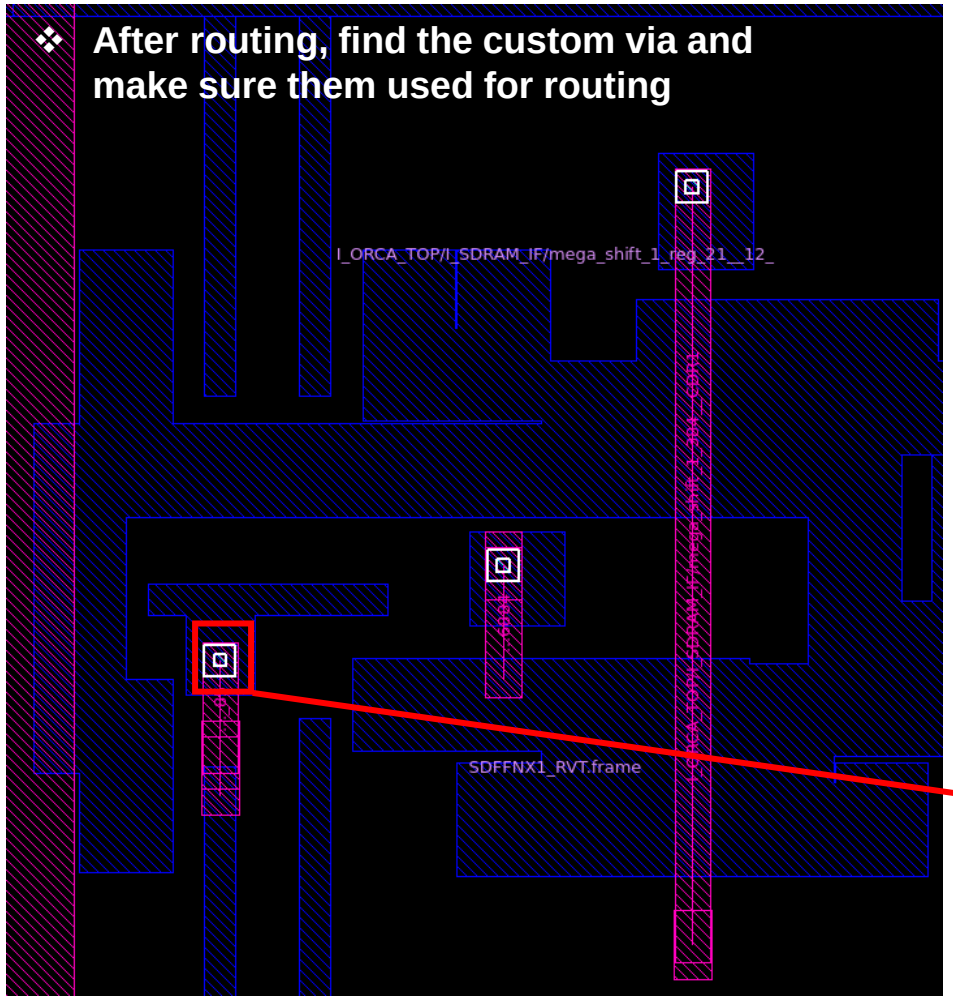
Route (Detailed Routing)

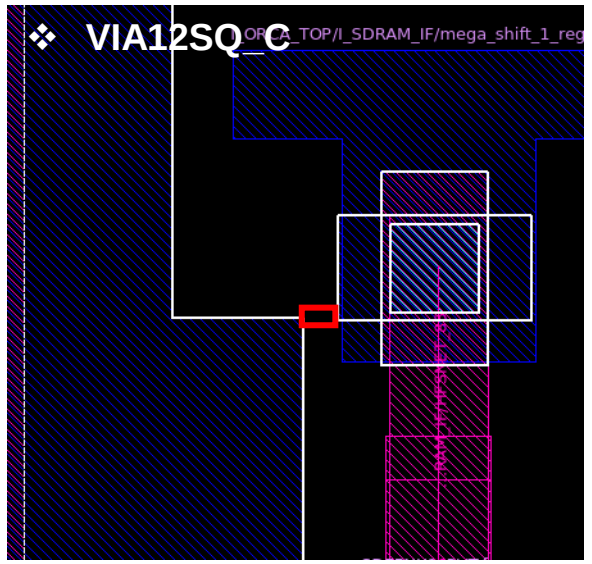
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Routing : Use ICC2 (IC Compiler II tool) check VIA12_thin

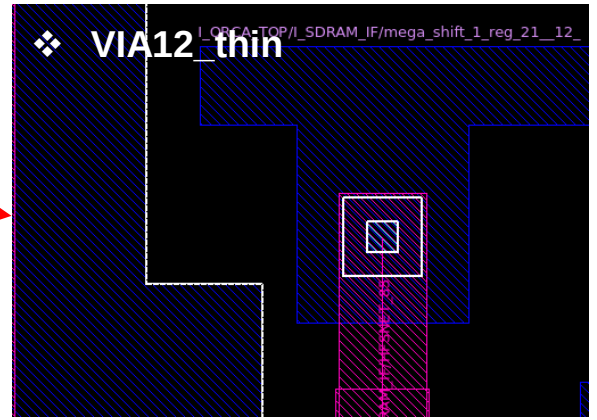
- **Check that custom via (VIA12_thin) is used in the routing step**

```
set custom_via [get_vias -filter "via_def.name  == VIA12_thin"]
change_selection $custom_via
```





Basically used Via can cause Metal spacing Error between the Metal of the Via(M1) and the internal pattern (M1 routing blockage)



Custom Via(VIA12_thin) is used to prevent M1 Metal spacing Error

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Routing : Use ICC2 (IC Compiler II tool)

11. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/10_search_and_repair.tcl

- **Search DRC violation and repair them (Check that the min, max routing Layers, and Antenna Rules on the log are properly applied)**

```
icc2_shell> route_detail -incremental true -initial_drc_from_input true
```

- **Compute clock_latency & update_timing after Routing**

```
icc2_shell> compute_clock_latency
```

```
icc2_shell> update_timing
```

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/11_check_timing_and_DRC.tcl

- **connect_pg_net for CORNER Cell & IO FILLER Cell (VDDIO, VSSIO, VDD, VSS)**

```
icc2_shell>
icc2_shell> #CORNER CELL
icc2_shell> connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_corner_cell_*/VSSIO"]
icc2_shell> connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_corner_cell_*/VSS"]
icc2_shell>
icc2_shell> connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_corner_cell_*/VDDIO"]
icc2_shell> connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_corner_cell_*/VDD"]
icc2_shell> #FILLER CELL
icc2_shell> connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_filler_instance_*/VSSIO"]
icc2_shell> connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_filler_instance_*/VSS"]
icc2_shell>
icc2_shell> connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_filler_instance_*/VDDIO"]
icc2_shell> connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_filler_instance_*/VDD"]
```

- **Filler cells and Corner cells are not included in the Netlist, and are not defined Pg connection in UPF file, so you must define them**
- **If the power connection information is not defined, DRC violation may occur**

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Routing : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/11_check_timing_and_DRC.tcl

- **Check route after Routing**

```
icc2_shell> check_route
```

```
Verify Summary:

Total number of nets = 18030, of which 0 are not extracted
Total number of open nets = 77, of which 0 are frozen
Total number of excluded ports = 0 ports of 0 unplaced cells connected to 0 nets
                                 0 ports without pins of 0 cells connected to 0 nets
                                 0 ports of 0 cover cells connected to 0 non-pg nets
Total number of DRCs = 155
Total number of antenna violations = 0
Total number of tie to rail violations = not checked
Total number of tie to rail directly violations = not checked
```

- **Check LVS after Routing (short, open, floating)**

```
icc2_shell> check_lvs
```

```
[Check Net] All nets are submitted.
[Check Net] 100%        Elapsed =    0:00:08, CPU =    0:00:07

============================================================
    Maximum number of violations is set to 20
    Abort checking when more than 20 violations are found
    All violations might not be found.
============================================================
Total number of input nets is 18134.
Total number of short violations is 0.
Total number of open nets is 20.
Open nets are I_ORCA_TOP/I_RISC_CORE/Oprnd_A[12] I_ORCA_TOP/I_RISC_CORE/Oprnd_A[10] I_ORCA_TOP/I_RISC_CORE/Oprnd_
A[9] I_ORCA_TOP/I_RISC_CORE/Oprnd_A[7] I_ORCA_TOP/I_RISC_CORE/Oprnd_A[0] I_ORCA_TOP/I_RISC_CORE/Oprnd_B[4] I_ORCA
_TOP/I_RISC_CORE/I_ALU/n80_CDR1 I_ORCA_TOP/ZCTSNET_23 I_ORCA_TOP/I_RISC_CORE/I_CONTROL/N57 I_ORCA_TOP/I_SD_W_MUX/
n2 I_ORCA_TOP/I_SD_W_MUX/HFSNET_3 I_ORCA_TOP/I_SD_W_MUX/HFSNET_6 VDDIO VSS VSSIO I_ORCA_TOP/I_RISC_CORE/gre_BUF_2
538_5 I_ORCA_TOP/HFSNET_121 I_ORCA_TOP/HFSNET_203 I_ORCA_TOP/I_RISC_CORE/HFSNET_76 I_ORCA_TOP/I_SD_W_MUX/gre_net_
7691 ...
Total number of floating route violations is 0.

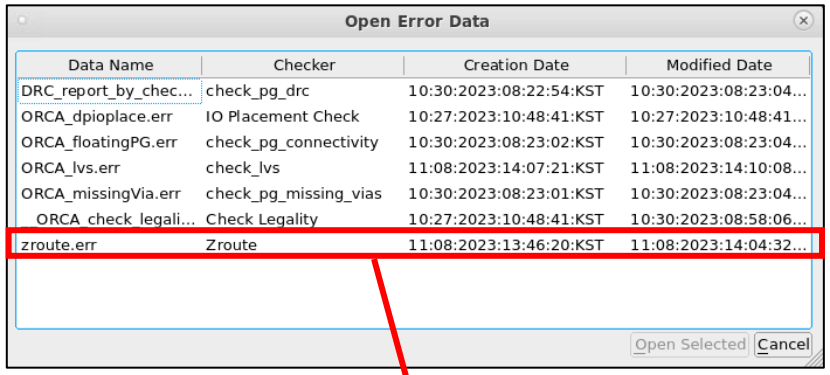
Elapsed =    0:00:08, CPU =    0:00:07
1
```


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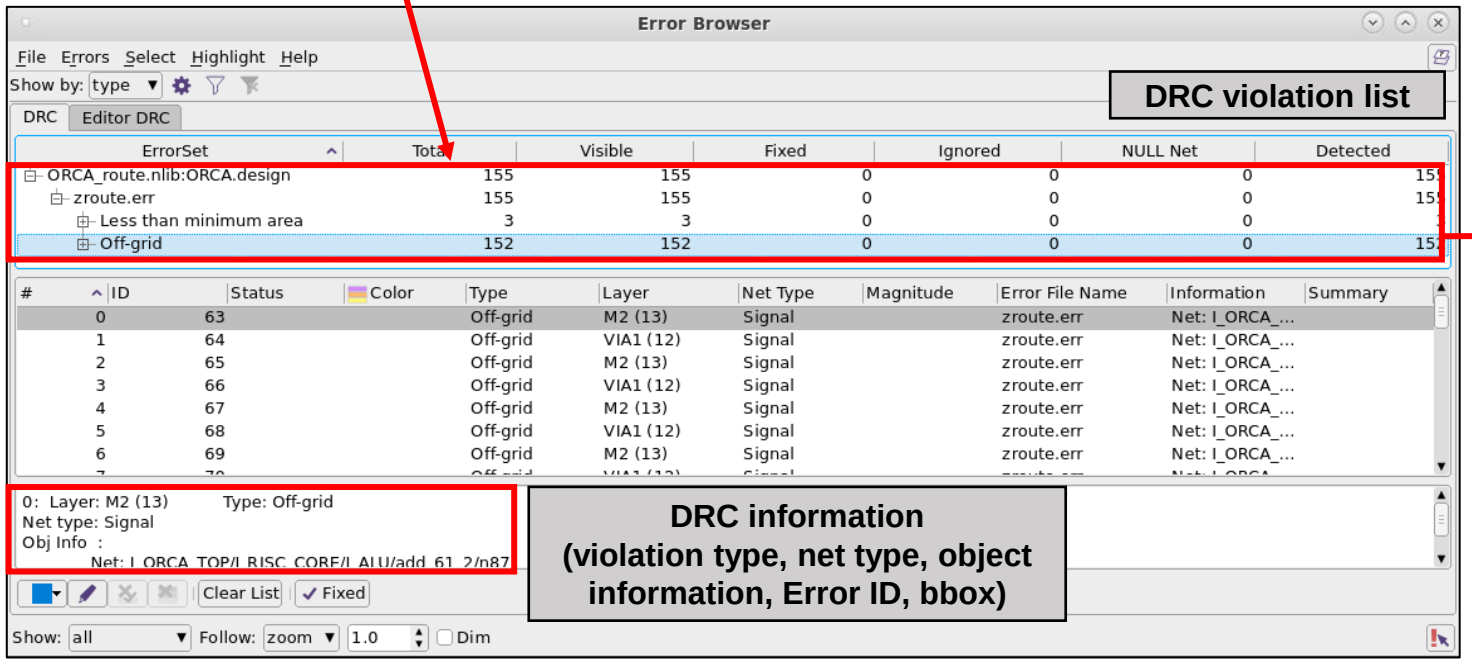
Routing : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/11_check_timing_and_DRC.tcl

- **Analyze check_routes**



- **Analayze Zroute Error**
  1. Open Error Data and choose zroute.err file
  2. Open Error Browser and choose DRC violation list
  3. Check NULL Net & Wave violation list before analyze
  4. Select the DRC violation list, the violations are displayed in the Layout view, and are moved to the selected violations location
  5. Check Net_Type & Net name
  6. Analyze DRC information and fix violation





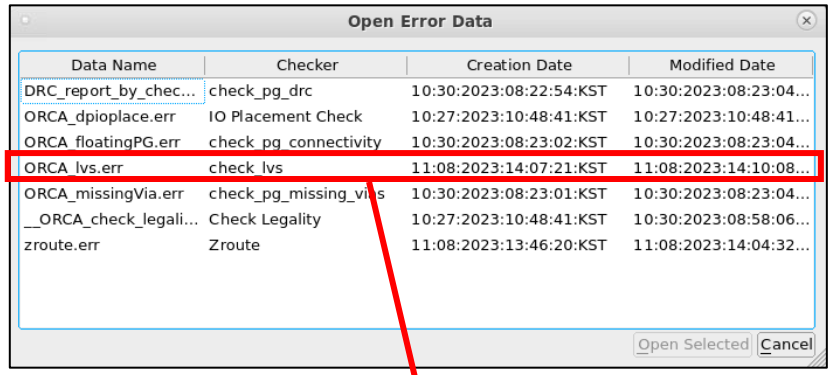
- **Ignore list in ORCA**
  Off-grid Error (M2, VIA1)

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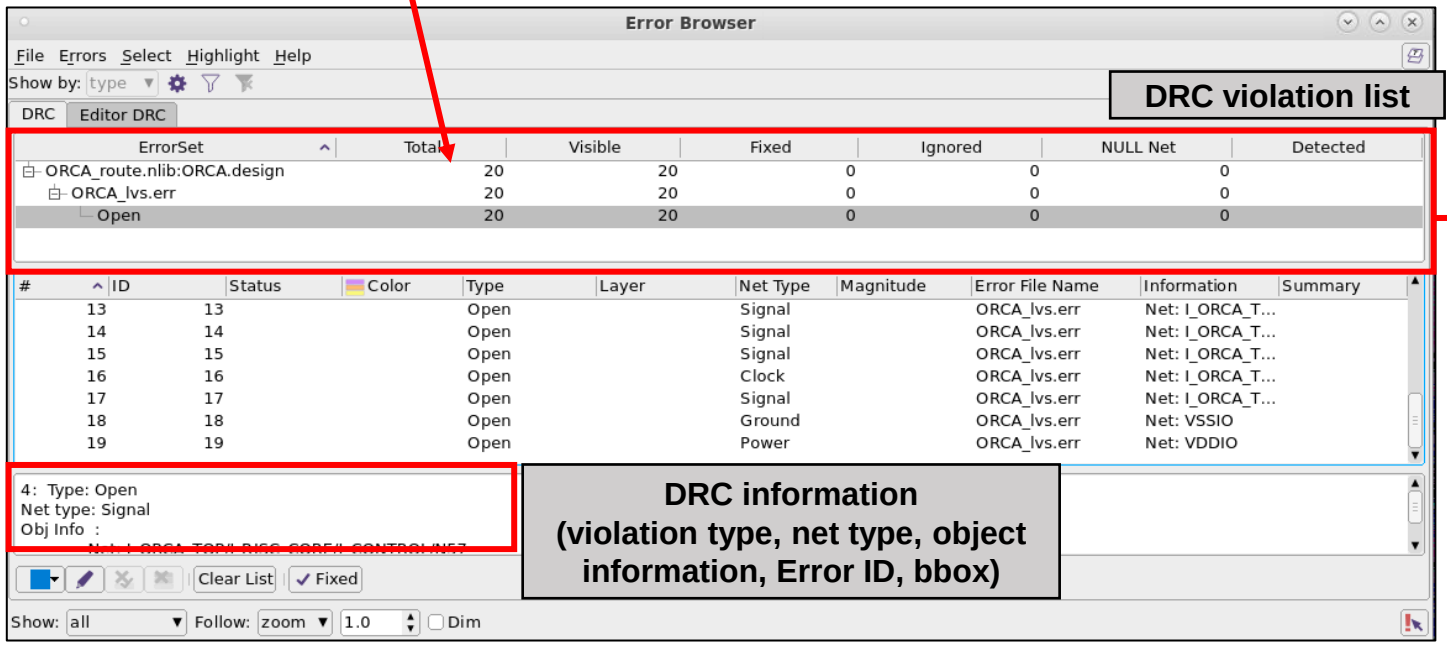
Routing : Use ICC2 (IC Compiler II tool)

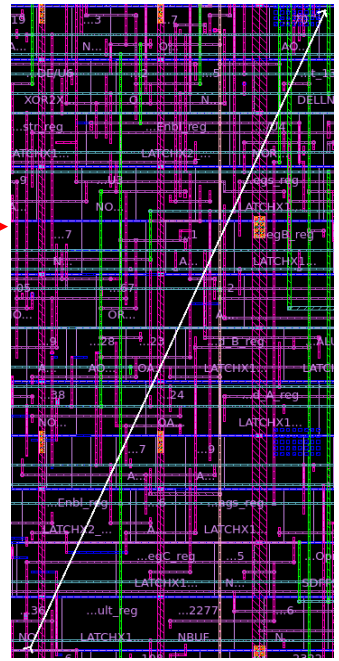
12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/11_check_timing_and_DRC.tcl

- **Analyze check_lvs (In the route_auto step, short,open,floating Errors occur because routing is not optimized yet)**



- **Analayze check_lvs Error**
  1. Open Error Data and choose *_lvs.err file
  2. Open Error Browser and choose DRC violation list
  3. Check NULL Net & Wave violation list before analyze
  4. Select the DRC violation list, the violations are displayed in the Layout view, and are moved to the selected violations location
  5. Analyze DRC information and fix violation (fix all signal,clock net open,short,floating)





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Routing : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/11_check_timing_and_DRC.tcl

- **Check Timing & QoR reports**

```
icc2_shell> report_timing -significant_digits 5
```

DAT

```
Point                                                    Incr         Path
------------------------------------------------------------------------------
clock SYS_2x_CLK (rise edge)                          0.00000      0.00000
clock network delay (propagated)                      1.10945      1.10945

I_ORCA_TOP/I_RISC_CORE/I_DATA_PATH/Oprnd_B_reg_0_/CLK (SDFFX2_LVT)
                                                      0.00000      1.10945 r
I_ORCA_TOP/I_RISC_CORE/I_DATA_PATH/Oprnd_B_reg_0_/Q (SDFFX2_LVT)
                                                      0.17796      1.28740 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/HFSINV_333_2313/Y (INVX0_RVT)
                                                      0.06598      1.35338 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/U149/Y (NAND2X0_RVT)
                                                      0.08810      1.44148 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/U186/Y (OAI21X1_RVT)
                                                      0.16058      1.60206 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/ctmTdsLR_2_3393/Y (INVX0_LVT)
                                                      0.01692      1.61898 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/ctmTdsLR_3_3394/Y (OA221X1_LVT)
                                                      0.08776      1.70673 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/ctmTdsLR_1_3462/Y (INVX0_HVT)
                                                      0.03998      1.74671 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/ctmTdsLR_2_3463/Y (AND2X1_RVT)
                                                      0.05779      1.80450 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/ctmTdsLR_1_3470/Y (AO221X2_LVT)
                                                      0.11908      1.92358 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/ctmTdsLR_2_3792/Y (NAND2X0_HVT)
                                                      0.07664      2.00022 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/ctmTdsLR_3_3793/Y (OAI221X1_RVT)
                                                      0.13811      2.13833 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/U162/Y (XNOR2X1_HVT)
                                                      0.14723      2.28556 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/U234/Y (AO221X1_RVT)
                                                      0.13887      2.42443 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/ctmTdsLR_3_4265/Y (NOR4X1_LVT)
                                                      0.09109      2.51553 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/U86/Y (NAND2X0_LVT)
                                                      0.04536      2.56088 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/ctmTdsLR_2_4603/Y (NAND3X0_LVT)
                                                      0.04105      2.60193 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/Neg_Flag_reg/D (SDFFARX1_RVT)
                                                      0.00002      2.60195 f
data arrival time                                                  2.60195
```

```
Information: Timer using 'CRPR'. (TIM-050)

  Startpoint: I_ORCA_TOP/I_RISC_CORE/I_DATA_PATH/Oprnd_B_reg_0_ (rising edge-trig
gered flip-flop clocked by SYS_2x_CLK)
  Endpoint: I_ORCA_TOP/I_RISC_CORE/I_ALU/Neg_Flag_reg (rising edge-triggered flip
-flop clocked by SYS_2x_CLK)
  Mode: func
  Corner: ss_125c
  Scenario: func.ss_125c
  Path Group: SYS_2x_CLK
  Path Type: max
```


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Routing : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/11_check_timing_and_DRC.tcl

- **Check Timing & QoR reports**

DRT

```
clock SYS_2x_CLK (rise edge)                          2.00000      2.00000
clock network delay (propagated)                      0.86487      2.86487
clock reconvergence pessimism                         0.03641      2.90128
I_ORCA_TOP/I_RISC_CORE/I_ALU/Neg_Flag_reg/CLK (SDFFARX1_RVT)
                                                      0.00000      2.90128 r
clock uncertainty                                    -0.40000      2.50128
library setup time                                   -0.12143      2.37985
data required time                                                 2.37985
---------------------------------------------------------------------------
data required time                                                 2.37985
data arrival time                                                 -2.60195
---------------------------------------------------------------------------
slack (VIOLATED)                                                  -0.22210
```

Slack

```
icc2_shell> report_qor -summary -significant_digits 5
```

```
Timing
------------------------------------------------------------------------------
Context                                WNS            TNS            NVE
------------------------------------------------------------------------------
func.ff_125c       (Setup)          1.20000        0.00000              0
func.ff_n40c       (Setup)          1.39216        0.00000              0
func.ss_125c       (Setup)         -0.22210       -3.21606             19
func.ss_n40c       (Setup)         -0.07315       -0.15556              5
scan_capture.ff_125c
                   (Setup)          9.54089        0.00000              0
scan_capture.ff_n40c
                   (Setup)          9.61873        0.00000              0
scan_capture.ss_125c
                   (Setup)          8.92842        0.00000              0
scan_capture.ss_n40c
                   (Setup)          8.75638        0.00000              0
scan_shift.ff_125c (Setup)          9.87494        0.00000              0
scan_shift.ff_n40c (Setup)          9.88996        0.00000              0
scan_shift.ss_125c (Setup)          9.65909        0.00000              0
scan_shift.ss_n40c (Setup)          9.56667        0.00000              0
Design             (Setup)         -0.22210       -3.32955             22
func.ff_125c       (Hold)          -0.08525       -1.46989             26
func.ff_n40c       (Hold)          -0.06071       -1.02035             24
func.ss_125c       (Hold)          -0.57495       -8.15170             37
func.ss_n40c       (Hold)          -0.04931       -0.31542             11
scan_capture.ff_125c
                   (Hold)           0.10380        0.00000              0
scan_capture.ff_n40c
                   (Hold)           0.09851        0.00000              0
scan_capture.ss_125c
                   (Hold)           0.31075        0.00000              0
scan_capture.ss_n40c
                   (Hold)           0.34200        0.00000              0
scan_shift.ff_125c (Hold)           0.10335        0.00000              0
scan_shift.ff_n40c (Hold)           0.09388        0.00000              0
scan_shift.ss_125c (Hold)           0.29001        0.00000              0
scan_shift.ss_n40c (Hold)           0.30863        0.00000              0
Design             (Hold)          -0.57495       -8.20553             41
------------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------------
Cell Area (netlist):                          768361.36489
Cell Area (netlist and physical only):       1192361.36500
Nets with DRC Violations:       107
1
```


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Routing : Use ICC2 (IC Compiler II tool)

13. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/12_report_route_and_exit.tcl

- **Update timing before report**

```
icc2_shell> update_timing -full
```

- **Report route & exit**

```
icc2_shell> sec_StartTimer report
1
icc2_shell> source ./user_script/route_rpt.tcl
icc2_shell> sec_ReportResourceUtil report
```

```
icc2_shell> save_lib -all
Saving all libraries...
1
icc2_shell> exit
```

- **Analyze Reports about Route**

```
  < user@linux route >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/06_route/reports/route
  < user@linux route >> ls
route.congestion.rpt                  timing_report_max_route_scan_capture.ff_125c  timing_report_min_route_func.ss_125c
route.fanout.rpt                      timing_report_max_route_scan_capture.ff_n40c  timing_report_min_route_func.ss_n40c
route.max_tran.rpt                    timing_report_max_route_scan_capture.ss_125c  timing_report_min_route_scan_capture.ff_125c
route.qor.snap.rpt                    timing_report_max_route_scan_capture.ss_n40c  timing_report_min_route_scan_capture.ff_n40c
route.runtime.rpt                     timing_report_max_route_scan_shift.ff_125c    timing_report_min_route_scan_capture.ss_125c
route.util.rpt                        timing_report_max_route_scan_shift.ff_n40c    timing_report_min_route_scan_capture.ss_n40c
timing_report_max_route_func.ff_125c  timing_report_max_route_scan_shift.ss_125c    timing_report_min_route_scan_shift.ff_125c
timing_report_max_route_func.ff_n40c  timing_report_max_route_scan_shift.ss_n40c    timing_report_min_route_scan_shift.ff_n40c
timing_report_max_route_func.ss_125c  timing_report_min_route_func.ff_125c          timing_report_min_route_scan_shift.ss_125c
timing_report_max_route_func.ss_n40c  timing_report_min_route_func.ff_n40c          timing_report_min_route_scan_shift.ss_n40c
```


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Routing : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/reports/route/* reports**
  - **Route.fanout.rpt (check high fanout)**

```
****************************************
Report : net_fanout
         -high_fanout
Design : ORCA
Version: S-2021.06-SP5
Date   : Wed Nov  8 15:43:01 2023
****************************************
1
```

  - **Route.max_tran.rpt**

```
****************************************
Report : constraint
        -all_violators
        -max_transition
Design : ORCA
Version: S-2021.06-SP5
Date   : Wed Nov  8 15:43:01 2023
****************************************



   Mode: func Corner: ff_n40c
   Scenario: func.ff_n40c
  --------------------------------------------------------------------------------
   Number of max_transition violation(s): 0

   Mode: scan_capture Corner: ff_n40c
   Scenario: scan_capture.ff_n40c
  --------------------------------------------------------------------------------
   Number of max_transition violation(s): 0

   Mode: scan_shift Corner: ff_n40c
   Scenario: scan_shift.ff_n40c
  --------------------------------------------------------------------------------
   Number of max_transition violation(s): 0
```

```
Mode: func Corner: ss_125c
Scenario: func.ss_125c
max_transition
                          Required        Actual
Net                      Transition     Transition        Slack  Violation
-------------------------------------------------------------------------------
sd_CKn                         0.50           0.90        -0.40  (VIOLATED)
  PIN : sdram_CKn_iopad/PADIO
                               0.50           0.90        -0.40  (VIOLATED)
  PORT: sd_CKn                 0.50           0.90        -0.40  (VIOLATED)

sd_CK                          0.50           0.90        -0.40  (VIOLATED)
  PIN : sdram_CK_iopad/PADIO
                               0.50           0.90        -0.40  (VIOLATED)
  PORT: sd_CK                  0.50           0.90        -0.40  (VIOLATED)

pclk                           0.50           0.59        -0.09  (VIOLATED)
  PIN : pclk_iopad/PADIO       0.50           0.59        -0.09  (VIOLATED)
```

- ❖ **Each PAD has a different max transition rule**
- ❖ **Violation occurs in the virtual port because the max transition rules of (sdram_CK_iopad, sdram_CKn_iopad) are different from other PADs**
- ❖ **Therefore, Max Transition violations occuring in virtual PORT of signal PAD can be ignored**

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Routing : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/reports/route/* reports**
  - **Ex) timing_report_min_route_func.ss_125c (DAT)**

```
  Startpoint: I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg (rising edge-triggered flip-flop clocked by SDRAM_CLK)
  Endpoint: I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/syncff_reg_5_ (removal check against rising-edge clock clocked by sdr_clk)
  Mode: func
  Corner: ss_125c
  Scenario: func.ss_125c
  Path Group: sdr_clk
  Path Type: min
```

```
Attributes
    b - black-box (unknown)
    s - size_only
    d - dont_touch
    u - dont_use
    g - generic
    h - hierarchical
    i - ideal
    n - noncombinational
    E - extracted timing model
    Q - Quick timing model
```

- **Check Startpoint & Endpoint**
- **Check Scenario (include Mode & Corner)**
- **Check Path Group**
- **Check Path Type (max – setup min - hold)**
- **Check Attribues and refer to the analysis**

Setup violation > DRT – RAT = Slack

Hold violation > DAT – DRT = Slack

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Routing : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/reports/route/* reports**
  - **Ex) timing_report_min_route_func.ss_125c (DAT)**

Point	Fanout	Trans	Derate	Delta	Incr	Path	Location	Attributes
clock SDRAM_CLK (rise edge)					0.000	0.000		
source latency					0.000	0.000		
I_CLOCK_GEN/I_PLL_SD/CLK_1X (PLL)		0.026	1.000		0.000	0.000 r	(901.30,664.39)	s, n
I_CLOCK_GEN/net_sdram_clk (net)	1							d
I_CLOCK_GEN/cts_trgdly_8692/A (INVX2_RVT)		0.026	1.000		0.000	0.000 r	(895.52,666.59)	
I_CLOCK_GEN/cts_trgdly_8692/Y (INVX2_RVT)		0.045	1.000		0.039	0.039 f	(895.39,666.49)	
I_CLOCK_GEN/cts4 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8696/A (INVX2_RVT)		0.045	1.000		0.000	0.039 f	(895.75,753.54)	
I_CLOCK_GEN/cts_trgdly_8696/Y (INVX2_RVT)		0.045	1.000		0.046	0.085 r	(895.88,753.44)	
I_CLOCK_GEN/cts9 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8698/A (INVX8_RVT)		0.045	1.000		0.000	0.085 r	(895.29,798.48)	
I_CLOCK_GEN/cts_trgdly_8698/Y (INVX8_RVT)		0.027	1.000		0.021	0.106 f	(895.42,798.59)	
I_CLOCK_GEN/cts11 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8697/A (INVX8_RVT)		0.027	1.000		0.000	0.106 f	(893.93,755.01)	
I_CLOCK_GEN/cts_trgdly_8697/Y (INVX8_RVT)		0.020	1.000		0.021	0.128 r	(894.05,755.12)	
I_CLOCK_GEN/cts10 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8695/A (INVX2_RVT)		0.020	1.000		0.000	0.128 r	(894.53,666.59)	
I_CLOCK_GEN/cts_trgdly_8695/Y (INVX2_RVT)		0.016	1.000		0.016	0.145 f	(894.66,666.49)	
I_CLOCK_GEN/cts8 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8691/A (INVX2_RVT)		0.016	1.000		0.000	0.145 f	(896.43,663.25)	
I_CLOCK_GEN/cts_trgdly_8691/Y (INVX2_RVT)		0.012	1.000		0.014	0.159 r	(896.30,663.15)	
I_CLOCK_GEN/cts2 (net)	1							d
I_CLOCK_GEN/macro_guide_buf_5/A (NBUFFX2_RVT)		0.012	1.000		0.000	0.159 r	(896.52,664.73)	
I_CLOCK_GEN/macro_guide_buf_5/Y (NBUFFX2_RVT)		0.028	1.000		0.045	0.204 r	(896.00,664.83)	
I_CLOCK_GEN/guide_buf_5 (net)	1							d
I_CLOCK_GEN/U9/A3 (AO22X1_RVT)		0.028	1.000		0.000	0.204 r	(894.85,694.24)	s
I_CLOCK_GEN/U9/Y (AO22X1_RVT)		0.053	1.000		0.084	0.288 r	(894.41,694.94)	s
I_CLOCK_GEN/o_sdram_clk (net)	1							d
I_ORCA_TOP/cto_inv_drc_7140/A (INVX8_RVT)		0.053	1.000		0.000	0.289 r	(897.73,694.82)	
I_ORCA_TOP/cto_inv_drc_7140/Y (INVX8_RVT)		0.037	1.000		0.031	0.319 f	(897.85,694.92)	
I_ORCA_TOP/cts4 (net)	2							d
I_ORCA_TOP/ZCTSINV_8164_7128/A (INVX16_RVT)		0.037	1.000		0.000	0.319 f	(894.38,693.35)	
I_ORCA_TOP/ZCTSINV_8164_7128/Y (INVX16_RVT)		0.036	1.000		0.032	0.351 r	(894.51,693.24)	
I_ORCA_TOP/ZCTSNET_37 (net)	4							d
I_ORCA_TOP/ZCTSINV_8095_7100/A (INVX8_RVT)		0.037	1.000		0.004	0.355 r	(800.90,674.76)	
I_ORCA_TOP/ZCTSINV_8095_7100/Y (INVX8_RVT)		0.045	1.000		0.038	0.393 f	(801.03,674.86)	
I_ORCA_TOP/ZCTSNET_32 (net)	5							d
I_ORCA_TOP/cto_inv_drc_7197/A (INVX2_RVT)		0.044	1.000		0.002	0.395 f	(812.30,607.88)	
I_ORCA_TOP/cto_inv_drc_7197/Y (INVX2_RVT)		0.026	1.000		0.025	0.420 r	(812.43,607.98)	

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Routing : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/reports/route/* reports**
  - **Ex) timing_report_min_route_func.ss_125c (DAT)**

I_ORCA_TOP/cts12 (net)	1					d
I_ORCA_TOP/cto_inv_drc_7201/A (INVX2_RVT)		0.026	1.000	0.000	0.420 r	(811.54,611.22)
I_ORCA_TOP/cto_inv_drc_7201/Y (INVX2_RVT)		0.033	1.000	0.030	0.451 f	(811.67,611.32)
I_ORCA_TOP/cts15 (net)	2					d
I_ORCA_TOP/cto_inv_drc_7200/A (INVX4_RVT)		0.033	1.000	0.000	0.451 f	(814.89,604.53)
I_ORCA_TOP/cto_inv_drc_7200/Y (INVX4_RVT)		0.040	1.000	0.040	0.490 r	(815.01,604.64)
I_ORCA_TOP/cts14 (net)	16					d
I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg/CLK (SDFFARX1_HVT)		0.040	1.000	0.000	0.491 r	(820.97,608.64) s, n
I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg/CLK (SDFFARX1_HVT)		0.040	1.000	0.000	0.491 r	(820.97,608.64) s, n
I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg/Q (SDFFARX1_HVT)		0.044	1.000	0.259	0.750 r	(817.79,607.99) s, n
I_ORCA_TOP/I_RESET_BLOCK/dftopt23 (net)	2					
I_ORCA_TOP/I_RESET_BLOCK/U14/A1 (AND2X1_LVT)		0.044	1.000	0.000	0.750 r	(816.63,606.23)
I_ORCA_TOP/I_RESET_BLOCK/U14/Y (AND2X1_LVT)		0.022	1.000	0.038	0.787 r	(817.14,606.30)
I_ORCA_TOP/I_RESET_BLOCK/N4 (net)	1					
I_ORCA_TOP/I_RESET_BLOCK/C52/A1 (OR2X1_HVT)		0.022	1.000	0.000	0.787 r	(813.27,607.94)
I_ORCA_TOP/I_RESET_BLOCK/C52/Y (OR2X1_HVT)		0.037	1.000	0.078	0.866 r	(814.03,608.01)
I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n (net)	1					
I_ORCA_TOP/HFSBUF_2333_2249/A (NBUFFX8_LVT)		0.037	1.000	0.000	0.866 r	(810.10,606.40)
I_ORCA_TOP/HFSBUF_2333_2249/Y (NBUFFX8_LVT)		0.046	1.000	0.057	0.922 r	(809.06,606.30)
I_ORCA_TOP/HFSNET_194 (net)	7					
I_ORCA_TOP/HFSINV_732_2246/A (INVX4_HVT)		0.048	1.000	0.010	0.932 r	(559.83,546.21)
I_ORCA_TOP/HFSINV_732_2246/Y (INVX4_HVT)		0.035	1.000	0.038	0.970 f	(559.96,546.11)
I_ORCA_TOP/HFSNET_193 (net)	3					
I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/HFSINV_252_2245/A (INVX4_HVT)		0.035	1.000	0.000	0.970 f	(550.48,546.21)
I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/HFSINV_252_2245/Y (INVX4_HVT)		0.031	1.000	0.035	1.005 r	(550.35,546.11)
I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/HFSNET_10 (net)	8					
I_ORCA_TOP/I_SDRAM_READ_FIFO/SD_RFIFO_CTL/U1/syncff_reg_5_/RSTB (SDFFARX1_HVT)		0.031	1.000	0.000	1.005 r	(542.55,533.42) s, n
data arrival time					1.005	

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Routing : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/reports/route/* reports**
  - **Ex) timing_report_min_route_func.ss_125c (DRT & Slack)**

```
clock sdr_clk (rise edge)                                                
```

LAB WEEK 2 : Route Optimization (route_opt)

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Route_opt: Use ICC2 (IC Compiler II tool)

It is also good to Read Design by source 02_floorplan.tcl, but when you proceed for the first time, it is recommended that you open the script and apply it one phrase at a time

1. **Invoke tool icc2_shell**
   - **Recommend the icc2/2021.06-SP5 version**

>> icc2_shell –gui | tee -i ./08_route_opt.log ;# (| tee -I ./08_route_opt.log) is the command to output log

```
< user@linux 06_route >> icc2_shell -gui | tee -i ./08_route_opt.log



                            IC Compiler II (TM)

                Version S-2021.06-SP5 for linux64 - Jan 14, 2022 -SLE

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Loading user preference file /u/kjinsoo/.synopsys_icc2_gui/preferences.tcl
icc2_shell>
```

2. **Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/13_design_setup_for_route_opt.tcl**
   - **set step & source design_setup.tcl**

```
icc2_shell> set step route_opt
route_opt
icc2_shell> set before_step route
route
icc2_shell> source -e ./icc2_common_env/design_setup.tcl
```


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Route_opt: Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/13_design_setup_for_route_opt.tcl

- **Open_lib & copy lib**

```
icc2_shell> open_lib ./nlib/ORCA_${before_step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/06_route/nli
b/ORCA_route.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32sram.ndm' (FILE-007)
{ORCA_route.nlib}
copy_lib -from ./nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
Information: Copying from library "ORCA_route.nlib" to "./nlib/ORCA_route_opt.nlib".
{ORCA_route_opt.nlib}
icc2_shell> close_lib -all
Closing all libraries...
1
```


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Route_opt: Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/13_design_setup_for_route_opt.tcl

- **Close reference lib & open & link current block**

```
icc2_shell> open_lib ./nlib/ORCA_${step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/06_route/nli
b/ORCA_route_opt.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/00_create_nd
m/ndm/saed32sram.ndm' (FILE-007)
{ORCA_route_opt.nlib}
icc2_shell> open_block ./nlib/ORCA_${step}.nlib:ORCA.design
Information: User units loaded from library 'saed32hvt' (LNK-040)
Opening block 'ORCA_route_opt.nlib:ORCA.design' in edit mode
{ORCA_route_opt.nlib:ORCA.design}
```

```
icc2_shell> link_block
Using libraries: ORCA_route_opt.nlib EXPLORE_physical_only saed32hvt saed32io_fc saed32lvt saed32pll saed32
rvt saed32sram
Visiting block ORCA_route_opt.nlib:ORCA.design


icc2_shell> link_block -rebind
Warning: Block 'ORCA_route_opt.nlib:ORCA.design' is already linked. (LNK-067)
0
```


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Route_opt: Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/13_design_setup_for_route_opt.tcl

- **Connect_pg_net for CORNER Cell & IO FILLER Cell (VDDIO, VSSIO, VDD, VSS)**

```
icc2_shell>
icc2_shell> #CORNER CELL
icc2_shell> connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_corner_cell_*/VSSIO"]
icc2_shell> connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_corner_cell_*/VSS"]
icc2_shell>
icc2_shell> connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_corner_cell_*/VDDIO"]
icc2_shell> connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_corner_cell_*/VDD"]
icc2_shell> #FILLER CELL
icc2_shell> connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_filler_instance_*/VSSIO"]
icc2_shell> connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_filler_instance_*/VSS"]
icc2_shell>
icc2_shell> connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_filler_instance_*/VDDIO"]
icc2_shell> connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_filler_instance_*/VDD"]
```

- **Filler cells and Corner cells are not included in the Netlist, and are not defined Pg connection in UPF file, so you must define them**
- **If the power connection information is not defined, DRC violation may occur**

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Route_opt : Use ICC2 (IC Compiler II tool)

3. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/14_mcmm.tcl (same as placement step)

- **Remove all (scenarios, modes, corners) and recreate mcmm**

```
icc2_shell> remove_scenarios -all
12
icc2_shell> remove_modes -all
3
icc2_shell> remove_corners -all
4
```

- **Read TLU+ files**

```
icc2_shell> read_parasitic_tech -name {minTLU} -tlup $data(tech_tlupbc)
1
icc2_shell> read_parasitic_tech -name {maxTLU} -tlup $data(tech_tlupwc)
1
```

- **Source mcmm.tcl (create mcmm, set parasitic_parameters, set_voltage)**

```
icc2_shell> source -e ./icc2_common_env/mcmm.tcl
```

- **Read SDC (func, scan_shift, scan_capture)**

```
icc2_shell> current_mode scan_shift
{scan_shift}
icc2_shell> read_sdc $data(shift_sdc)
```

```
icc2_shell> current_mode scan_capture
{scan_capture}
icc2_shell> read_sdc $data(capture_sdc)
```

```
icc2_shell> current_mode func
{func}
icc2_shell> read_sdc $data(func_sdc)
```


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Route_opt : Use ICC2 (IC Compiler II tool)

4. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/15_set_antenna_rule.tcl

- **Set Antenna Rules for prevent Antenna violation**

```
icc2_shell> source -echo ./icc2_common_env/saed32nm_ant_1p9m.tcl
#Antenna Rule for 32nm TECH
#       Author:         ArtakY
#       @(#) Antenna rules
#       @(#) Revision 1.0.0.0
#       @(#) Date     28-Feb-11
# technology: saed32/28nm_1p9m
#
# Revision history:
# rev            date     who  what
# ------------  --------- ---- ------------------------------
# Rev. 1.0.0.0  28-Feb-11 AY   Initial version
#
# ############################################################
remove_antenna_rules
define_antenna_rule -mode 4 -diode_mode 2 -metal_ratio 1000 -cut_ratio 20
define_antenna_layer_rule -mode 4 -layer "M1" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M2" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M3" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M4" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M5" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M6" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M7" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M8" -ratio 1000 -diode_ratio {0.06 0 400 40000}
define_antenna_layer_rule -mode 4 -layer "M9" -ratio 1000 -diode_ratio {0.06 0 8000 50000}
define_antenna_layer_rule -mode 4 -layer "VIA1" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA2" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA3" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA4" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA5" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA6" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA7" -ratio 20 -diode_ratio {0.06 0 200 1000}
define_antenna_layer_rule -mode 4 -layer "VIA8" -ratio 20 -diode_ratio {0.06 0 200 1000}
1
```


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Route_opt : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/16_set_optimization_cell.tcl

- **Exclude all lib_cells before set optimization cell**

```
icc2_shell> set_lib_cell_purpose -exclude hold [get_lib_cells]
Warning: The 'set_lib_cell_purpose' command cannot be used on library cell 'EXPLORE_physical_only:LSUPX1_HV
T.frame' which has no logical model. (NDMUI-010)
Warning: The 'set_lib_cell_purpose' command cannot be used on library cell 'EXPLORE_physical_only:LSUPX2_HV
T.frame' which has no logical model. (NDMUI-010)
Warning: The 'set_lib_cell_purpose' command cannot be used on library cell 'EXPLORE_physical_only:LSUPX4_HV
T.frame' which has no logical model. (NDMUI-010)
Warning: The 'set_lib_cell_purpose' command cannot be used on library cell 'EXPLORE_physical_only:LSUPX8_HV
T.frame' which has no logical model. (NDMUI-010)
Warning: The 'set_lib_cell_purpose' command cannot be used on library cell 'EXPLORE_physical_only:LSDNSSX8_
HVT.frame' which has no logical model. (NDMUI-010)
Warning: The 'set_lib_cell_purpose' command cannot be used on library cell 'EXPLORE_physical_only:LSDNX1_HV
T.frame' which has no logical model. (NDMUI-010)
```

- **Include & set hold optimization cell list**

```
icc2_shell> set_lib_cell_purpose -include hold [get_lib_cells "*/NBUFFX2_HVT saed32hvt/DELLN1X2_HVT"]
Information: The design specific attribute override for lib_cell 'saed32hvt:NBUFFX2_HVT.timing' is set in t
he current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
Information: The design specific attribute override for lib_cell 'saed32hvt:DELLN1X2_HVT.timing' is set in
the current block 'ORCA', because the actual library setting may not be overwritten. (ATTR-12)
1
```


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Route_opt : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/17_remove_clock_latency_and_ideal_network.tcl

- **Remove clock latency & ideal network before routing**

```
icc2_shell> set mode "func scan_shift scan_capture"
func scan_shift scan_capture
icc2_shell>
icc2_shell> foreach MODE $mode {
? current_mode $MODE
? echo "current mode is $MODE"
? remove_clock_latency [get_clocks *]
? remove_clock_latency [get_flat_pins -filter "is_clock_pin == true"]
? remove_ideal_network [remove_from_collection [all_fanout -flat -clock_tree] [get_ports [all_output]]]
? set_propagated_clock [all_clocks]
? }
current mode is func
current mode is scan_shift
Warning: Virtual clock 'forcePI_wft1' cannot be made propagated. (UIC-025)
Warning: Virtual clock 'measurePO_wft1' cannot be made propagated. (UIC-025)
current mode is scan_capture
Warning: Virtual clock 'forcePI_wft1' cannot be made propagated. (UIC-025)
Warning: Virtual clock 'measurePO_wft1' cannot be made propagated. (UIC-025)
```

- **Compute_clock_latency after remove clock latency & ideal network**

❖ calculates and updates the I/O latencies of real and virtual clock objects for all active mode and corner combinations

Ex)

```
icc2_shell> compute_clock_latency
Mode:scan_capture     Latency(early)   Latency(late)
Clock        Updated   rise    fall    rise    fall   Corner
-------------------------------------------------------------
pclk_wft1    Yes     0.3095  0.3095  0.3249  0.3249   ff_n40c
pclk_wft1    Yes     0.8063  0.8063  0.8647  0.8647   ss_n40c
pclk_wft1    Yes     0.3808  0.3808  0.3994  0.3994   ff_125c
pclk_wft1    Yes     0.7715  0.7715  0.8270  0.8270   ss_125c
```

This command is used primarily to correct your block's timing constraints after clock tree synthesis

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Route_opt : Use ICC2 (IC Compiler II tool)

7. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/18_set_max_transition.tcl

- **Set max_transition for all modes & corners**

```
icc2_shell> current_mode func
{func}
icc2_shell> set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
1
icc2_shell>
icc2_shell> current_mode scan_capture
{scan_capture}
icc2_shell> set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
1
icc2_shell>
icc2_shell> current_mode scan_shift
{scan_shift}
icc2_shell> set_max_transition 0.5 -clock_path [get_clocks] -corners [all_corners]
1
```

- **set_max_transition**
  - Specifies a maximum transition on prots, library cell pins, leaf cell pins, clocks or designs for the current scenario
  - If max_transition is set on a port, the port must have a transition time less than the specified transition_value
  - If specified on a design, the default maximum transition for that design is set
  - Library cell pins can also have a max_transition value specified in the library

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Route_opt : Use ICC2 (IC Compiler II tool)

8. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/19_set_app_option.tcl

- **Set appplication options before route_auto**

```
icc2_shell> set_app_options -list {route.global.timing_driven {true}}
route.global.timing_driven true
icc2_shell> set_app_options -list {route.global.crosstalk_driven {true}}
route.global.crosstalk_driven true
icc2_shell> set_app_options -list {route.track.timing_driven {true}}
route.track.timing_driven true
icc2_shell> set_app_options -list {route.track.crosstalk_driven {true}}
route.track.crosstalk_driven true
icc2_shell> set_app_options -list {route.detail.timing_driven {true}}
route.detail.timing_driven true
icc2_shell> set_app_options -list {route.detail.force_max_number_iterations {false}}
route.detail.force_max_number_iterations false
icc2_shell> set_app_options -list {route.detail.generate_extra_off_grid_pin_tracks {true}}
route.detail.generate_extra_off_grid_pin_tracks true
icc2_shell> set_app_options -list {route.detail.generate_off_grid_feed_through_tracks {high}}
route.detail.generate_off_grid_feed_through_tracks high
```

- **Optimize route before routing optimization**

```
icc2_shell> optimize_routes
Cell Min-Routing-Layer = M1
Cell Max-Routing-Layer = M6
Found antenna rule mode 4, diode mode 2:
```


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Route_opt : Use ICC2 (IC Compiler II tool)

9. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/20_route_opt.tcl

```
icc2_shell> sec_StartTimer route_opt
1
icc2_shell> route_opt

icc2_shell> sec_StartTimer route_opt
1
icc2_shell> optimize_routes
Cell Min-Routing-Layer = M1
Cell Max-Routing-Layer = M6
Found antenna rule mode 4, diode mode 2:
```

- **Check that the min, max routing Layers, and Antenna Rules on the log are properly applied**

10. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/21_search_and_repair.tcl

- **Search DRC violation and repair them (Check that the min, max routing Layers, and Antenna Rules on the log are properly applied)**

```
icc2_shell> route_detail -incremental true -initial_drc_from_input true
```

- **Compute clock_latency & update_timing after Routing optimization**

```
icc2_shell> compute_clock_latency
```

```
icc2_shell> update_timing
```


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Routing : Use ICC2 (IC Compiler II tool)

11. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/22_check_timing_and_DRC.tcl

- **Connect_pg_net for CORNER Cell & IO FILLER Cell (VDDIO, VSSIO, VDD, VSS)**

```
icc2_shell>
icc2_shell> #CORNER CELL
icc2_shell> connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_corner_cell_*/VSSIO"]
icc2_shell> connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_corner_cell_*/VSS"]
icc2_shell>
icc2_shell> connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_corner_cell_*/VDDIO"]
icc2_shell> connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_corner_cell_*/VDD"]
icc2_shell> #FILLER CELL
icc2_shell> connect_pg_net -net VSSIO [get_pins -filter "full_name =~ __added_filler_instance_*/VSSIO"]
icc2_shell> connect_pg_net -net VSS [get_pins -filter "full_name =~ __added_filler_instance_*/VSS"]
icc2_shell>
icc2_shell> connect_pg_net -net VDDIO [get_pins -filter "full_name =~ __added_filler_instance_*/VDDIO"]
icc2_shell> connect_pg_net -net VDD [get_pins -filter "full_name =~ __added_filler_instance_*/VDD"]
```

- **Filler cells and Corner cells are not included in the Netlist, and are not defined Pg connection in UPF file, so you must define them**
- **If the power connection information is not defined, DRC violation may occur**

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Route_opt : Use ICC2 (IC Compiler II tool)

11. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/22_check_timing_and_DRC.tcl

- **Check Route after Routing optimization**

```
icc2_shell> check_route
```

```
Verify Summary:

Total number of nets = 13956, of which 0 are not extracted
Total number of open nets = 0, of which 0 are frozen
Total number of excluded ports = 0 ports of 0 unplaced cells connected to 0 nets
                                 0 ports without pins of 0 cells connected to 0 nets
                                 0 ports of 0 cover cells connected to 0 non-pg nets
Total number of DRCs = 164
Total number of antenna violations = 0
Total number of tie to rail violations = not checked
Total number of tie to rail directly violations = not checked
```

- **Check LVS after Routing optimization(short, open, floating)**

```
icc2_shell> check_lvs
```

```
============================================================
    Maximum number of violations is set to 20
    Abort checking when more than 20 violations are found
    All violations might not be found.
============================================================
Total number of input nets is 14060.
Total number of short violations is 0.
Total number of open nets is 7.
Open nets are AVDD AVSS DVDD VDD VDDIO VSS VSSIO
Total number of floating route violations is 0.

Elapsed =    0:00:06, CPU =    0:00:05
1
```


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Route_opt : Use ICC2 (IC Compiler II tool)

11. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/22_check_timing_and_DRC.tcl

- **Check_pg_drc after Routing optimization**

```
icc2_shell> check_pg_drc
```

```
Command check_pg_drc started  at Thu Nov  9 11:28:40 2023
Command check_pg_drc finished at Thu Nov  9 11:28:57 2023
CPU usage for check_pg_drc: 16.52 seconds ( 0.00 hours)
Elapsed time for check_pg_drc: 17.29 seconds ( 0.00 hours)
Total number of errors found: 3484
   567 illegal overlaps on M1
   729 illegal overlaps on M6
   729 illegal overlaps on M7
   729 illegal overlaps on M8
   729 illegal overlaps on M9
   1 insufficient spacing on M1
```

- **Report_timing after Routing optimization**

```
icc2_shell> report_timing -significant_digits 5
```

```
Information: Timer using 'CRPR'. (TIM-050)

  Startpoint: I_ORCA_TOP/I_RISC_CORE/I_DATA_PATH/Oprnd_B_reg_3_ (rising edge-triggered flip-flop clocked by SYS_2x_C
LK)
  Endpoint: I_ORCA_TOP/I_RISC_CORE/I_ALU/Neg_Flag_reg (rising edge-triggered flip-flop clocked by SYS_2x_CLK)
  Mode: func
  Corner: ss_125c
  Scenario: func.ss_125c
  Path Group: SYS_2x_CLK
  Path Type: max
```


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Route_opt : Use ICC2 (IC Compiler II tool)

11. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/22_check_timing_and_DRC.tcl

- **Report_timing after Routing optimization**

DAT

```
Point                                                  Incr         Path
-----------------------------------------------------------------------------
clock SYS_2x_CLK (rise edge)                         0.00000      0.00000
clock network delay (propagated)                     1.10937      1.10937

I_ORCA_TOP/I_RISC_CORE/I_DATA_PATH/Oprnd_B_reg_3_/CLK (SDFFX1_LVT)
                                                     0.00000      1.10937 r
I_ORCA_TOP/I_RISC_CORE/I_DATA_PATH/Oprnd_B_reg_3_/Q (SDFFX1_LVT)
                                                     0.15312      1.26249 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/ZBUF_69_inst_8808/Y (NBUFFX2_LVT)
                                                     0.04585      1.30835 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/r307/U108/Y (AND2X1_RVT)
                                                     0.05508      1.36343 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/r307/ctmTdsLR_1_3368/Y (AND3X1_LVT)
                                                     0.06308      1.42651 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/r307/U117/Y (AND2X1_RVT)
                                                     0.06796      1.49446 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/r307/U49/Y (XNOR2X1_LVT)
                                                     0.09720      1.59166 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/U175/Y (NOR2X0_LVT)
                                                     0.06695      1.65861 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/U206/Y (NOR2X0_RVT)
                                                     0.08553      1.74414 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/ctmTdsLR_2_3463/Y (AND2X1_RVT)
                                                     0.05428      1.79842 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/ctmTdsLR_1_3470/Y (AO221X2_LVT)
                                                     0.12051      1.91893 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/ctmTdsLR_2_3792/Y (NAND2X0_HVT)
                                                     0.07605      1.99497 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/ctmTdsLR_3_3793/Y (OAI221X1_RVT)
                                                     0.13767      2.13264 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/add_61_2/U162/Y (XNOR2X1_HVT)
                                                     0.14748      2.28012 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/U234/Y (AO221X1_RVT)
                                                     0.13836      2.41848 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/ctmTdsLR_3_4265/Y (NOR4X1_LVT)
                                                     0.09068      2.50916 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/U86/Y (NAND2X0_LVT)
                                                     0.04513      2.55428 r
I_ORCA_TOP/I_RISC_CORE/I_ALU/ctmTdsLR_2_4603/Y (NAND3X0_LVT)
                                                     0.04101      2.59529 f
I_ORCA_TOP/I_RISC_CORE/I_ALU/Neg_Flag_reg/D (SDFFARX1_RVT)
                                                     0.00002      2.59531 f
data arrival time                                                 2.59531
```

DRT & Slack

```
clock SYS_2x_CLK (rise edge)                          2.00000      2.00000
clock network delay (propagated)                      0.86487      2.86487
clock reconvergence pessimism                         0.03632      2.90118
I_ORCA_TOP/I_RISC_CORE/I_ALU/Neg_Flag_reg/CLK (SDFFARX1_RVT)
                                                      0.00000      2.90118 r
clock uncertainty                                    -0.40000      2.50118
library setup time                                   -0.12141      2.37978
data required time                                                 2.37978
-----------------------------------------------------------------------------
data required time                                                 2.37978
data arrival time                                                 -2.59531
-----------------------------------------------------------------------------
slack (VIOLATED)                                                  -0.21553
```


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Route_opt : Use ICC2 (IC Compiler II tool)

11. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/22_check_timing_and_DRC.tcl

- **Report_qor_summary after Routing optimization**

```
Timing
------------------------------------------------------------------------------
Context                                 WNS             TNS             NVE
------------------------------------------------------------------------------
func.ff_125c       (Setup)          1.19931         0.00000               0
func.ff_n40c       (Setup)          1.39167         0.00000               0
func.ss_125c       (Setup)         -0.21553        -0.41611               2
func.ss_n40c       (Setup)         -0.04171        -0.06973               2
scan_capture.ff_125c
                   (Setup)          9.53269         0.00000               0
scan_capture.ff_n40c
                   (Setup)          9.60462         0.00000               0
scan_capture.ss_125c
                   (Setup)          8.88385         0.00000               0
scan_capture.ss_n40c
                   (Setup)          8.69506         0.00000               0
scan_shift.ff_125c (Setup)          9.87520         0.00000               0
scan_shift.ff_n40c (Setup)          9.89013         0.00000               0
scan_shift.ss_125c (Setup)          9.65963         0.00000               0
scan_shift.ss_n40c (Setup)          9.56707         0.00000               0
Design             (Setup)         -0.21553        -0.41611               2

func.ff_125c       (Hold)          -0.08555        -0.72599              93
func.ff_n40c       (Hold)          -0.06104        -0.40734              13
func.ss_125c       (Hold)          -0.18343        -2.68476              23
func.ss_n40c       (Hold)          -0.05031        -0.10061               2
scan_capture.ff_125c
                   (Hold)           0.02525         0.00000               0
scan_capture.ff_n40c
                   (Hold)           0.03148         0.00000               0
scan_capture.ss_125c
                   (Hold)           0.12296         0.00000               0
scan_capture.ss_n40c
                   (Hold)           0.11593         0.00000               0
scan_shift.ff_125c (Hold)           0.00760         0.00000               0
scan_shift.ff_n40c (Hold)           0.01111         0.00000               0
scan_shift.ss_125c (Hold)           0.07076         0.00000               0
scan_shift.ss_n40c (Hold)           0.06302         0.00000               0
Design             (Hold)          -0.18343        -2.87456             105
------------------------------------------------------------------------------

Miscellaneous
------------------------------------------------------------------------------
Cell Area (netlist):                      753083.49833
Cell Area (netlist and physical only):    1177083.49800

Nets with DRC Violations:       81
1
```


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Route_opt : Use ICC2 (IC Compiler II tool)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/23_report_route_opt_and_exit.tcl

- **Update timing before report**

```
icc2_shell> update_timing -full
```

- **Report route & exit**

```
icc2_shell> sec_StartTimer report
1
icc2_shell> source ./user_script/route_opt_rpt.tcl
icc2_shell> sec_ReportResourceUtil report
```

```
icc2_shell> save_lib -all
Saving all libraries...
1
icc2_shell> exit
```

- **Analyze Reports about route_opt**

```
  < user@linux route_opt >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/06_route/reports/route_opt
  < user@linux route_opt >> ls
route_opt.congestion.rpt                  timing_report_max_route_opt_scan_capture.ff_125c  timing_report_min_route_opt_func.ss_125c
route_opt.fanout.rpt                      timing_report_max_route_opt_scan_capture.ff_n40c  timing_report_min_route_opt_func.ss_n40c
route_opt.max_tran.rpt                    timing_report_max_route_opt_scan_capture.ss_125c  timing_report_min_route_opt_scan_capture.ff_125c
route_opt.qor.snap.rpt                    timing_report_max_route_opt_scan_capture.ss_n40c  timing_report_min_route_opt_scan_capture.ff_n40c
route_opt.runtime.rpt                     timing_report_max_route_opt_scan_shift.ff_125c    timing_report_min_route_opt_scan_capture.ss_125c
route_opt.util.rpt                        timing_report_max_route_opt_scan_shift.ff_n40c    timing_report_min_route_opt_scan_capture.ss_n40c
timing_report_max_route_opt_func.ff_125c  timing_report_max_route_opt_scan_shift.ss_125c    timing_report_min_route_opt_scan_shift.ff_125c
timing_report_max_route_opt_func.ff_n40c  timing_report_max_route_opt_scan_shift.ss_n40c    timing_report_min_route_opt_scan_shift.ff_n40c
timing_report_max_route_opt_func.ss_125c  timing_report_min_route_opt_func.ff_125c          timing_report_min_route_opt_scan_shift.ss_125c
timing_report_max_route_opt_func.ss_n40c  timing_report_min_route_opt_func.ff_n40c          timing_report_min_route_opt_scan_shift.ss_n40c
```


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Route_opt : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/reports/route_opt/* reports**
  - **route_opt.fanout.rpt (check high fanout)**

```
****************************************
Report : net_fanout
         -high_fanout
Design : ORCA
Version: S-2021.06-SP5
Date   : Wed Nov  8 15:43:01 2023
****************************************
1
```

  - **route_opt.max_tran.rpt**

```
****************************************
Report : constraint
        -all_violators
        -max_transition
Design : ORCA
Version: S-2021.06-SP5
Date   : Wed Nov  8 15:43:01 2023
****************************************



  Mode: func Corner: ff_n40c
  Scenario: func.ff_n40c
 --------------------------------------------------------------------------------
  Number of max_transition violation(s): 0

  Mode: scan_capture Corner: ff_n40c
  Scenario: scan_capture.ff_n40c
 --------------------------------------------------------------------------------
  Number of max_transition violation(s): 0

  Mode: scan_shift Corner: ff_n40c
  Scenario: scan_shift.ff_n40c
 --------------------------------------------------------------------------------
  Number of max_transition violation(s): 0
```

```
Mode: func Corner: ss_125c
Scenario: func.ss_125c
max_transition
                          Required        Actual
Net                      Transition      Transition        Slack  Violation
------------------------------------------------------------------------------
sd_CKn                        0.50            0.90        -0.40  (VIOLATED)
  PIN : sdram_CKn_iopad/PADIO
                              0.50            0.90        -0.40  (VIOLATED)
  PORT: sd_CKn                0.50            0.90        -0.40  (VIOLATED)

sd_CK                         0.50            0.90        -0.40  (VIOLATED)
  PIN : sdram_CK_iopad/PADIO
                              0.50            0.90        -0.40  (VIOLATED)
  PORT: sd_CK                 0.50            0.90        -0.40  (VIOLATED)

pclk                          0.50            0.59        -0.09  (VIOLATED)
  PIN : pclk_iopad/PADIO      0.50            0.59        -0.09  (VIOLATED)
```

- ❖ **Each PAD has a different max transition rule**
- ❖ **Violation occurs in the virtual port because the max transition rules of (sdram_CK_iopad, sdram_CKn_iopad) are different from other PADs**
- ❖ **Therefore, Max Transition violations occuring in virtual PORT of signal PAD can be ignored**

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Route_opt : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/reports/route_opt/* reports**
  - **Ex) timing_report_min_route_opt_func.ss_125c (DAT)**

```
  Startpoint: I_CLOCK_GEN/I_PLL_SD (internal path startpoint clocked by SDRAM_CLK)
  Endpoint: sd_DQ[12] (output port clocked by SD_DDR_CLK)
  Mode: func
  Corner: ss_125c
  Scenario: func.ss_125c
  Path Group: OUTS
  Path Type: min
```

```
Attributes
    b - black-box (unknown)
    s - size_only
    d - dont_touch
    u - dont_use
    g - generic
    h - hierarchical
    i - ideal
    n - noncombinational
    E - extracted timing model
    Q - Quick timing model
```

- **Check Startpoint & Endpoint**
- **Check Scenario (include Mode & Corner)**
- **Check Path Group**
- **Check Path Type (max – setup min - hold)**
- **Check Attribues and refer to the analysis**

Setup violation > DRT – RAT = Slack

Hold violation > DAT – DRT = Slack

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Route_opt : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/reports/route_opt/* reports**
  - **Ex) timing_report_min_route_func.ss_125c (DAT)**

Point	Fanout	Trans	Derate	Delta	Incr	Path	Location	Attributes
clock SDRAM_CLK (fall edge)					2.000	2.000		
source latency					0.000	2.000		
I_CLOCK_GEN/I_PLL_SD/CLK_1X (PLL)		0.026	1.000		0.000	2.000 f	(901.30,664.39)	s, n
I_CLOCK_GEN/net_sdram_clk (net)	1							d
I_CLOCK_GEN/I_PLL_SD/CLK_1X (PLL)		0.026	1.000		0.000	2.000 f	(901.30,664.39)	s, n
I_CLOCK_GEN/net_sdram_clk (net)	1							d
I_CLOCK_GEN/cts_trgdly_8724/A (INVX2_RVT)		0.026	1.000		0.000	2.000 f	(895.60,663.25)	
I_CLOCK_GEN/cts_trgdly_8724/Y (INVX2_RVT)		0.043	1.000		0.040	2.040 r	(895.73,663.15)	
I_CLOCK_GEN/cts7 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8728/A (INVX2_RVT)		0.043	1.000		0.000	2.040 r	(896.21,744.98)	
I_CLOCK_GEN/cts_trgdly_8728/Y (INVX2_RVT)		0.043	1.000		0.039	2.079 f	(896.33,745.08)	
I_CLOCK_GEN/cts9 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8730/A (INVX8_RVT)		0.043	1.000		0.000	2.079 f	(855.01,743.51)	
I_CLOCK_GEN/cts_trgdly_8730/Y (INVX8_RVT)		0.025	1.000		0.024	2.103 r	(855.14,743.40)	
I_CLOCK_GEN/cts11 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8729/A (INVX8_RVT)		0.025	1.000		0.000	2.104 r	(895.75,736.82)	
I_CLOCK_GEN/cts_trgdly_8729/Y (INVX8_RVT)		0.020	1.000		0.018	2.122 f	(895.88,736.71)	
I_CLOCK_GEN/cts10 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8727/A (INVX2_RVT)		0.020	1.000		0.000	2.122 f	(895.29,664.72)	
I_CLOCK_GEN/cts_trgdly_8727/Y (INVX2_RVT)		0.016	1.000		0.018	2.141 r	(895.42,664.83)	
I_CLOCK_GEN/cts8 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8723/A (INVX2_RVT)		0.016	1.000		0.000	2.141 r	(896.21,669.94)	
I_CLOCK_GEN/cts_trgdly_8723/Y (INVX2_RVT)		0.013	1.000		0.014	2.155 f	(896.33,669.84)	
I_CLOCK_GEN/cts6 (net)	1							d
I_CLOCK_GEN/macro_guide_buf_5/A (NBUFFX2_RVT)		0.013	1.000		0.000	2.155 f	(897.34,664.73)	
I_CLOCK_GEN/macro_guide_buf_5/Y (NBUFFX2_RVT)		0.028	1.000		0.046	2.201 f	(897.85,664.83)	
I_CLOCK_GEN/guide_buf_5 (net)	1							d
I_CLOCK_GEN/U9/A3 (AO22X1_RVT)		0.028	1.000		0.000	2.201 f	(899.26,697.28)	s
I_CLOCK_GEN/U9/Y (AO22X1_RVT)		0.056	1.000		0.078	2.279 f	(898.82,696.57)	s
I_CLOCK_GEN/o_sdram_clk (net)	2							d
I_ORCA_TOP/ZCTSINV_17099_7271/A (INVX8_RVT)		0.056	1.000		0.000	2.279 f	(899.40,698.16)	
I_ORCA_TOP/ZCTSINV_17099_7271/Y (INVX8_RVT)		0.033	1.000		0.030	2.309 r	(899.53,698.27)	
I_ORCA_TOP/ZCTSNET_42 (net)	1							d
I_ORCA_TOP/ZCTSINV_15998_7270/A (INVX16_RVT)		0.033	1.000		0.000	2.310 r	(896.66,673.28)	
I_ORCA_TOP/ZCTSINV_15998_7270/Y (INVX16_RVT)		0.033	1.000		0.028	2.338 f	(896.79,673.18)	
I_ORCA_TOP/ZCTSNET_41 (net)	6							d

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Route_opt : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/reports/route_opt/* reports**
  - **Ex) timing_report_min_route_func.ss_125c (DAT)**

I_ORCA_TOP/ZCTSINV_4984_7269/A (INVX16_RVT)		0.033	1.000	0.003	2.341 f	(799.23,617.91)	
I_ORCA_TOP/ZCTSINV_4984_7269/Y (INVX16_RVT)		0.030	1.000	0.028	2.369 r	(799.36,618.01)	
I_ORCA_TOP/ZCTSNET_40 (net)	3						d
I_ORCA_TOP/ZCTSINV_1914_7267/A (INVX2_RVT)		0.031	1.000	0.003	2.372 r	(720.19,577.78)	
I_ORCA_TOP/ZCTSINV_1914_7267/Y (INVX2_RVT)		0.055	1.000	0.047	2.420 f	(720.32,577.88)	
I_ORCA_TOP/ZCTSNET_39 (net)	2						d
I_ORCA_TOP/ZCTSINV_1239_f_7266/A (INVX8_RVT)		0.055	1.000	0.000	2.420 f	(705.60,562.93)	
I_ORCA_TOP/ZCTSINV_1239_f_7266/Y (INVX8_RVT)		0.051	1.000	0.048	2.468 r	(705.73,562.83)	
I_ORCA_TOP/ZCTSNET_38 (net)	6						d
I_ORCA_TOP/I_SDRAM_IF/ZCTSINV_367_7256/A (INVX4_RVT)		0.051	1.000	0.005	2.473 r	(541.59,442.55)	
I_ORCA_TOP/I_SDRAM_IF/ZCTSINV_367_7256/Y (INVX4_RVT)		0.049	1.000	0.044	2.517 f	(541.72,442.44)	
I_ORCA_TOP/I_SDRAM_IF/ZCTSNET_75 (net)	16						d
I_ORCA_TOP/I_SDRAM_IF/genblk1_12__sd_mux_dq_out/S0 (MUX21X1_RVT)		0.049	1.000	0.001	2.517 f	(530.61,427.45)	s
I_ORCA_TOP/I_SDRAM_IF/genblk1_12__sd_mux_dq_out/Y (MUX21X1_RVT)		0.043	1.000	0.096	2.613 r	(529.68,427.41)	s
I_ORCA_TOP/I_SDRAM_IF/sd_DQ_out[12] (net)	1						
ZBUF_4_inst_9157/A (NBUFFX8_RVT)		0.043	1.000	0.000	2.613 r	(535.13,427.30)	
ZBUF_4_inst_9157/Y (NBUFFX8_RVT)		0.064	1.000	0.074	2.687 r	(534.09,427.40)	
ZBUF_4_51 (net)	1						
SDRAM_DQ_Bus_12__sdram_DQ_iopad/DIN (B16I1025_NS)		0.068	1.000	0.013	2.701 r	(530.29,199.58)	s, d
SDRAM_DQ_Bus_12__sdram_DQ_iopad/PADIO (B16I1025_NS)		0.000	1.000	1.392	4.093 r	(540.00,2.10)	s, d
sd_DQ[12] (net)	2						
sd_DQ[12] (inout)		0.000	1.000	0.000	4.093 r	(540.00,2.10)	
data arrival time					4.093		

  - **Ex) timing_report_min_route_func.ss_125c (DRT & Slack)**

clock SD_DDR_CLK (fall edge)				2.000	2.000		
source latency				0.000	2.000		
I_CLOCK_GEN/I_PLL_SD/CLK_1X (PLL)		0.029	1.000	0.000	2.000 f	(901.30,664.39)	s, n
I_CLOCK_GEN/net_sdram_clk (net)	1						d
I_CLOCK_GEN/cts_trgdly_8724/A (INVX2_RVT)		0.029	1.000	0.000	2.000 f	(895.60,663.25)	
I_CLOCK_GEN/cts_trgdly_8724/Y (INVX2_RVT)		0.046	1.000	0.043	2.043 r	(895.73,663.15)	
I_CLOCK_GEN/cts7 (net)	1						d
I_CLOCK_GEN/cts_trgdly_8728/A (INVX2_RVT)		0.046	1.000	0.000	2.043 r	(896.21,744.98)	
I_CLOCK_GEN/cts_trgdly_8728/Y (INVX2_RVT)		0.050	1.000	0.045	2.089 f	(896.33,745.08)	
I_CLOCK_GEN/cts9 (net)	1						d
I_CLOCK_GEN/cts_trgdly_8730/A (INVX8_RVT)		0.050	1.000	0.000	2.089 f	(855.01,743.51)	
I_CLOCK_GEN/cts_trgdly_8730/Y (INVX8_RVT)		0.030	1.000	0.028	2.117 r	(855.14,743.40)	
I_CLOCK_GEN/cts11 (net)	1						d
I_CLOCK_GEN/cts_trgdly_8729/A (INVX8_RVT)		0.030	1.000	0.000	2.117 r	(895.75,736.82)	
I_CLOCK_GEN/cts_trgdly_8729/Y (INVX8_RVT)		0.022	1.000	0.020	2.137 f	(895.88,736.71)	
I_CLOCK_GEN/cts10 (net)	1						d
I_CLOCK_GEN/cts_trgdly_8727/A (INVX2_RVT)		0.022	1.000	0.000	2.138 f	(895.29,664.72)	
I_CLOCK_GEN/cts_trgdly_8727/Y (INVX2_RVT)		0.019	1.000	0.021	2.159 r	(895.42,664.83)	
I_CLOCK_GEN/cts8 (net)	1						d
I_CLOCK_GEN/cts_trgdly_8723/A (INVX2_RVT)		0.019	1.000	0.000	2.159 r	(896.21,669.94)	
I_CLOCK_GEN/cts_trgdly_8723/Y (INVX2_RVT)		0.015	1.000	0.015	2.174 f	(896.33,669.84)	

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Route_opt : Use ICC2 (IC Compiler II tool)

- **Analyze $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/reports/route_opt/* reports**
  - **Ex) timing_report_min_route_func.ss_125c (DRT & Slack)**

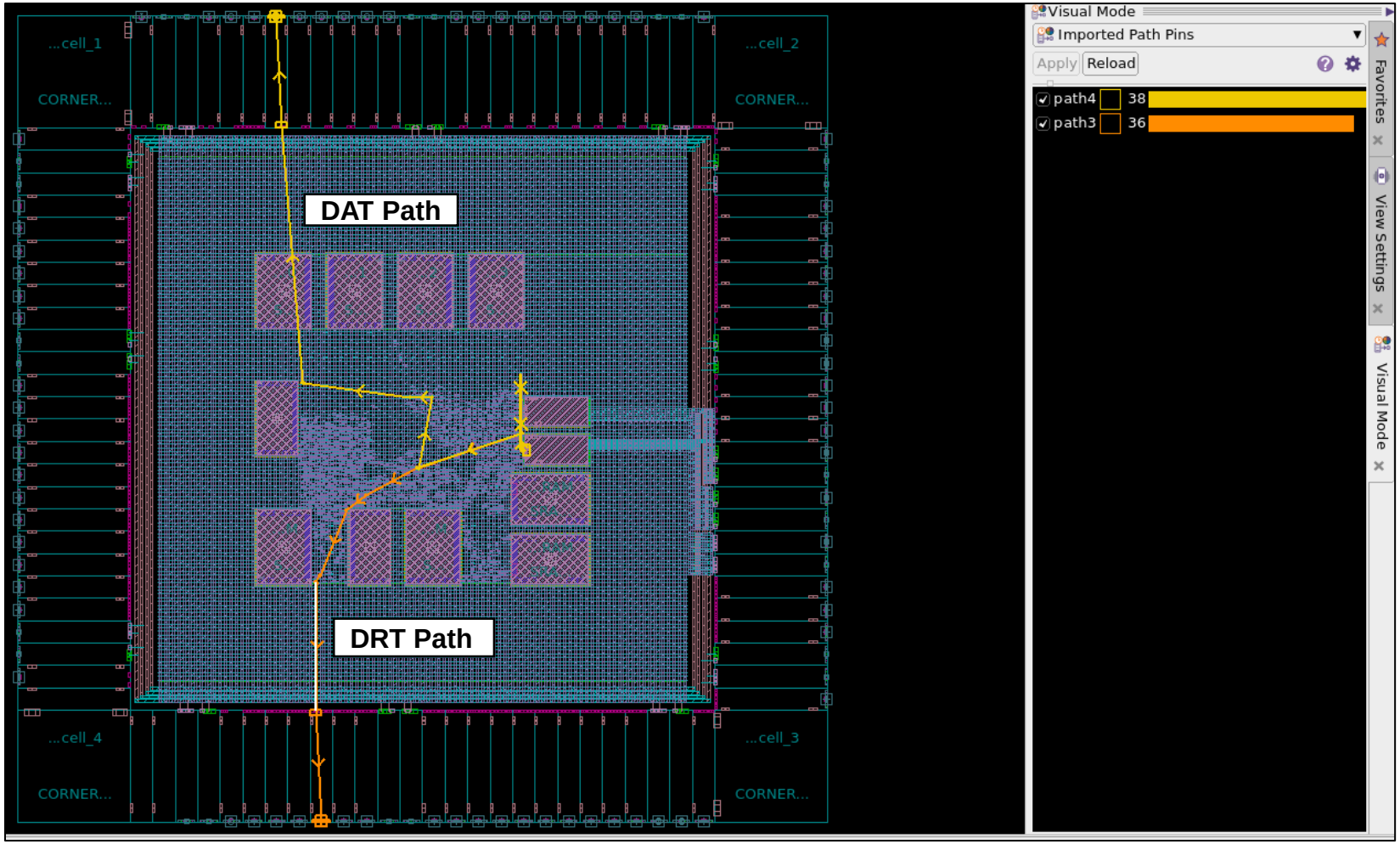
```
I_CLOCK_GEN/cts6 (net)                                    1                                                                                  d
I_CLOCK_GEN/macro_guide_buf_5/A (NBUFFX2_RVT)                      0.015      1.000            0.000      2.174 f    (897.34,664.73)
I_CLOCK_GEN/macro_guide_buf_5/Y (NBUFFX2_RVT)                      0.029      1.000            0.047      2.221 f    (897.85,664.83)
I_CLOCK_GEN/guide_buf_5 (net)                             1                                                                                  d
I_CLOCK_GEN/U9/A3 (AO22X1_RVT)                                     0.029      1.000            0.000      2.221 f    (899.26,697.28)   s
I_CLOCK_GEN/U9/Y (AO22X1_RVT)                                      0.076      1.000            0.092      2.313 f    (898.82,696.57)   s
I_CLOCK_GEN/o_sdram_clk (net)                             2                                                                                  d
I_ORCA_TOP/ZCTSINV_17099_7271/A (INVX8_RVT)                        0.076      1.000            0.000      2.314 f    (899.40,698.16)
I_ORCA_TOP/ZCTSINV_17099_7271/Y (INVX8_RVT)                        0.043      1.000            0.038      2.352 r    (899.53,698.27)
I_ORCA_TOP/ZCTSNET_42 (net)                               1                                                                                  d
I_ORCA_TOP/ZCTSINV_15998_7270/A (INVX16_RVT)                       0.043      1.000            0.000      2.352 r    (896.66,673.28)
I_ORCA_TOP/ZCTSINV_15998_7270/Y (INVX16_RVT)                       0.044      1.000            0.035      2.387 f    (896.79,673.18)
I_ORCA_TOP/ZCTSNET_41 (net)                               6                                                                                  d
I_ORCA_TOP/ZCTSINV_4984_7269/A (INVX16_RVT)                        0.044      1.000            0.004      2.392 f    (799.23,617.91)
I_ORCA_TOP/ZCTSINV_4984_7269/Y (INVX16_RVT)                        0.037      1.000            0.033      2.425 r    (799.36,618.01)
I_ORCA_TOP/ZCTSNET_40 (net)                               3                                                                                  d
I_ORCA_TOP/inv_drc_cln8538/A (INVX16_RVT)                          0.038      1.000            0.004      2.429 r    (724.29,577.78)
I_ORCA_TOP/inv_drc_cln8538/Y (INVX16_RVT)                          0.046      1.000            0.031      2.460 f    (724.42,577.88)
I_ORCA_TOP/cts3 (net)                                    14                                                                                  d
I_ORCA_TOP/I_SDRAM_IF/sd_mux_CK/S0 (MUX21X1_RVT)                   0.053      1.000            0.016      2.476 f    (508.72,781.91)   s
I_ORCA_TOP/I_SDRAM_IF/sd_mux_CK/Y (MUX21X1_RVT)                    0.075      1.000            0.140      2.616 f    (509.65,781.88)   s
I_ORCA_TOP/I_SDRAM_IF/sd_CK (net)                         1                                                                                  d
ZCTSINV_160_7279/A (INVX8_RVT)                                     0.075      1.000            0.000      2.616 f    (511.11,780.29)
ZCTSINV_160_7279/Y (INVX8_RVT)                                     0.041      1.000            0.036      2.652 r    (510.98,780.19)
ZCTSNET_5 (net)                                           1                                                                                  d
ZCTSINV_83_7278/A (INVX16_RVT)                                     0.041      1.000            0.000      2.652 r    (508.15,780.29)
ZCTSINV_83_7278/Y (INVX16_RVT)                                     0.051      1.000            0.033      2.685 f    (508.28,780.19)
ZCTSNET_4 (net)                                           1                                                                                  d
sdram_CK_iopad/DIN (D8I1025_NS)                                    0.059      1.000            0.017      2.703 f    (469.71,1240.42)  d
sdram_CK_iopad/PADIO (D8I1025_NS)                                  0.891      1.000            1.391      4.094 f    (460.00,1437.90)  d
sd_CK (net)                                               2                                                                                  d
sd_CK (out)                                                        0.891      1.000            0.000      4.094 f    (460.00,1437.90)
sd_CK (out) (gclock 'SD_DDR_CLK' source)                           0.891      1.000            0.000      4.094 f    (460.00,1437.90)

clock reconvergence pessimism                                                                 -0.000      4.094
output external delay                                                                          0.100      4.194
data required time                                                                                       4.194
-------------------------------------------------------------------------------------------------------------------------------------------
data required time                                                                                       4.194
data arrival time                                                                                       -4.093
-------------------------------------------------------------------------------------------------------------------------------------------
slack (VIOLATED)                                                                                        -0.102
```


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Route_opt : Use ICC2 (IC Compiler II tool) Analyze Timing Path

- **Analyze VIOLATED Timing Path use Imported Path Pins (Visual Mode)** ***Analyze the timing path from pages 131 to 134***
  - **Start_point : I_CLOCK_GEN/I_PLL_SD**
  - **End_point : sd_DQ[12]**

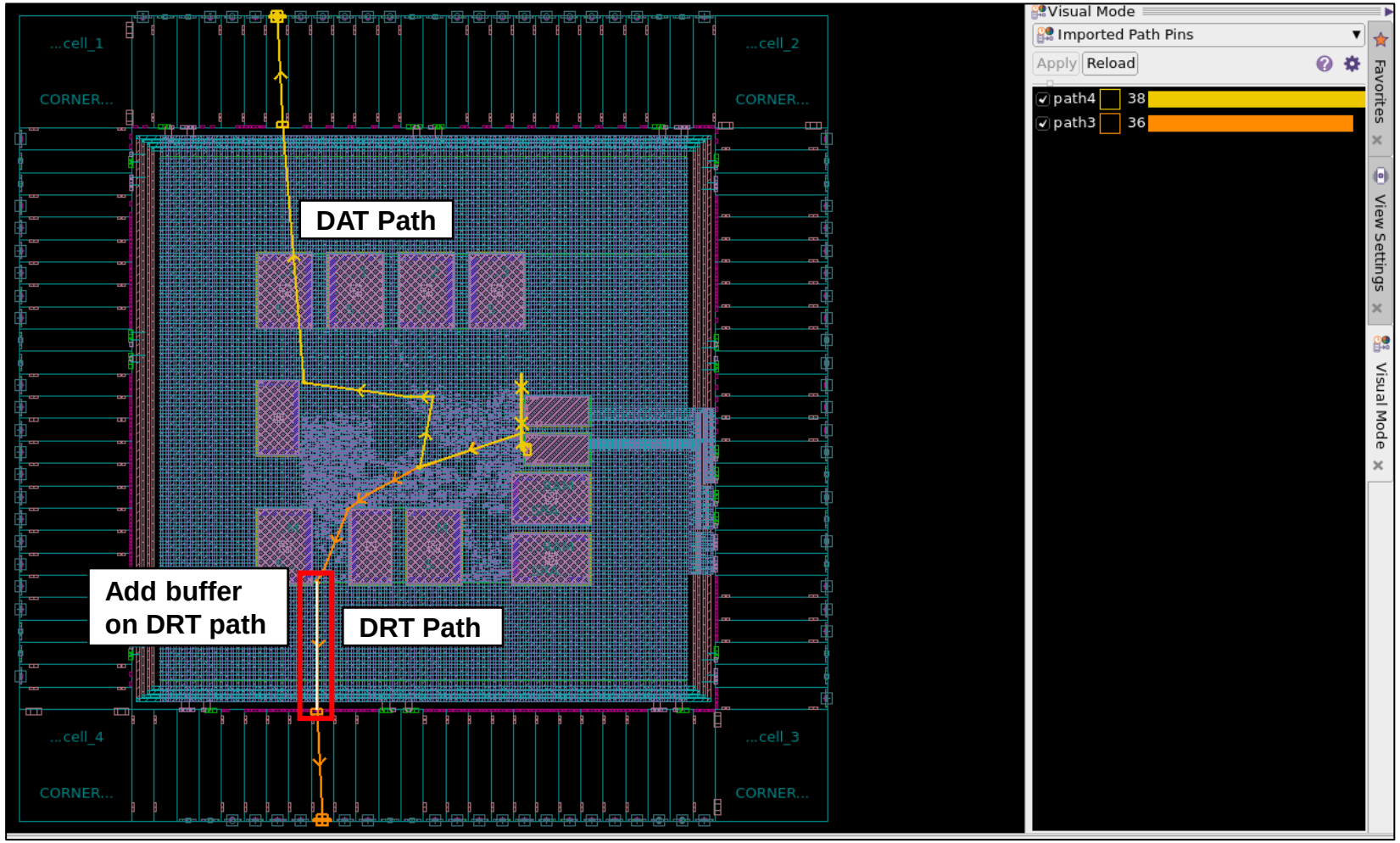


DAT = 4.078
DRT = 4.261
Slack = -0.183

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Route_opt : Use ICC2 (IC Compiler II tool) Analyze Timing Path

- **Fix VIOLATED Timing Path use Imported Path Pins (Visual Mode)** ***Analyze the timing path from pages 131 to 134***
  - **Start_point : I_CLOCK_GEN/I_PLL_SD**
  - **End_point : sd_DQ[12]**



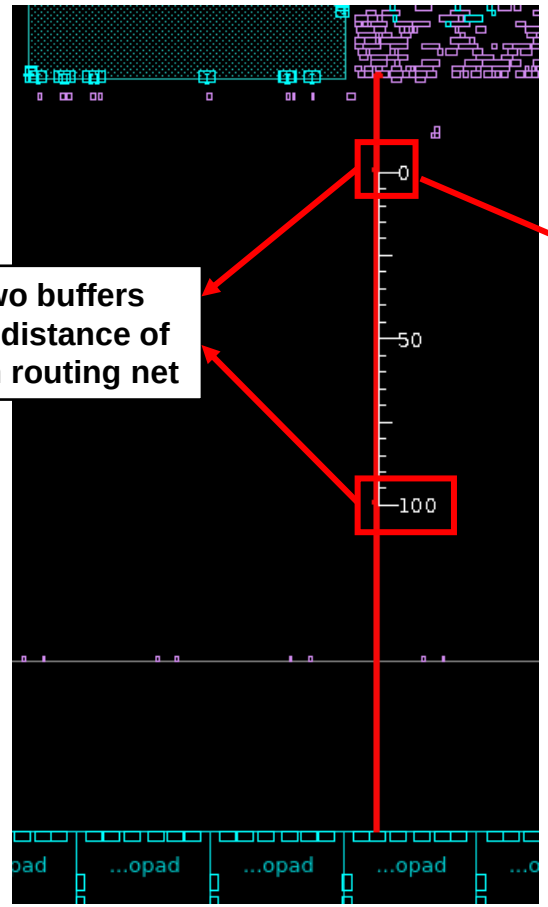
DAT = 4.078
DRT = 4.261
Slack = -0.183

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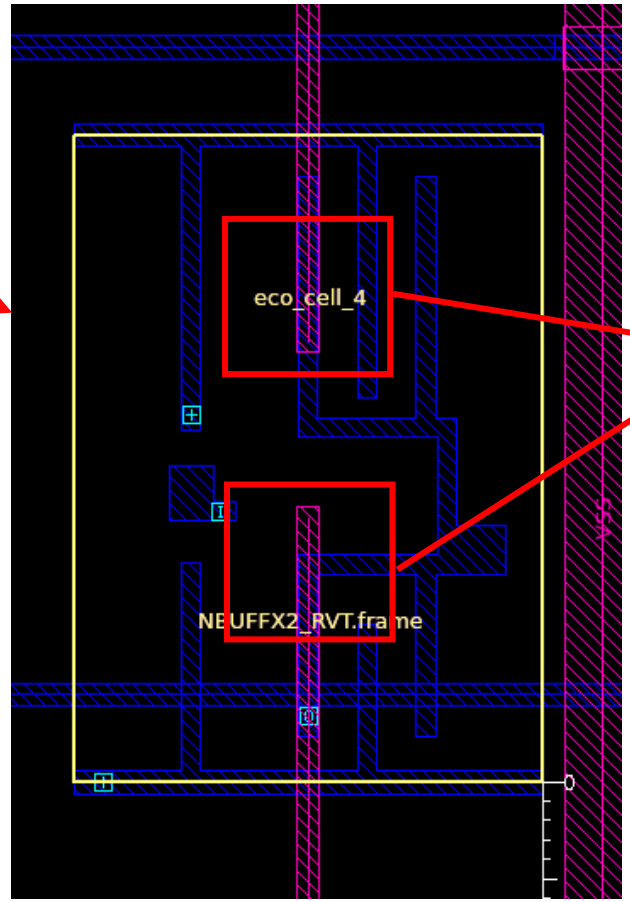
Route_opt : Use ICC2 (IC Compiler II tool) Analyze Timing Path

- **Fix VIOLATED Timing Path use Imported Path Pins (Visual Mode)** ***Analyze the timing path from pages 131 to 134***
  - **Add Buffer on DRT path**

```
add_buffer_on_route "ZBUF_4_51" \
        -first_distance 30 -repeater_distance 100 \
        -lib_cell saed32rvt/NBUFFX2_RVT
```



Add two buffers with a distance of 100 on routing net



Buffers input, output pins are not connected, so routing is required for these cells

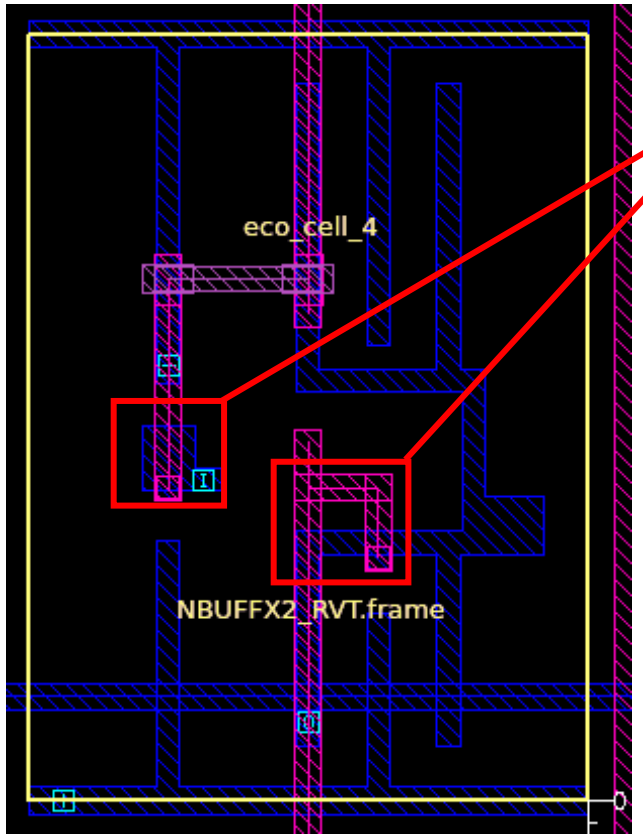
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Route_opt : Use ICC2 (IC Compiler II tool) Analyze Timing Path

- **Fix VIOLATED Timing Path use Imported Path Pins (Visual Mode)** ***Analyze the timing path from pages 131 to 134***
  - **Routing for added buffers, after search & repair after "route_eco"**

```
route_eco -cells {*eco_cell*}

route_detail -incremental true -initial_drc_from_input true
compute_clock_latency ;# This command is used primarily to correct your block's timing constraints after CTS
update_timing
```



- **Only cells added using the route_eco command were routed**
- **After routing, proceed with DRC violation fix through search & repair**

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Route_opt : Use ICC2 (IC Compiler II tool) Analyze Timing Path

- **Fix VIOLATED Timing Path use Imported Path Pins (Visual Mode)** ***Analyze the timing path from pages 131 to 134***
  - **Write Timing report and analyze it**

```
report_timing -scenarios func.ss_125c \
              -significant_digits 3 -delay_type min \
              -input_pins -physical -attributes -nosplit \
              -derate -crosstalk_delta -path_type full_clock \
              -from I_CLOCK_GEN/I_PLL_SD -to sd_DQ[12] \
              > hold_fix2.rpt
```

File name

  - **Check "hold_fix.rpt" timing report**

```
Information: Timer using 'CRPR'. (TIM-050)

  Startpoint: I_CLOCK_GEN/I_PLL_SD (internal path startpoint clocked by SDRAM_CLK)
  Endpoint: sd_DQ[12] (output port clocked by SD_DDR_CLK)
  Mode: func
  Corner: ss_125c
  Scenario: func.ss_125c
  Path Group: OUTS
  Path Type: min


Attributes
    b - black-box (unknown)
    s - size_only
    d - dont_touch
    u - dont_use
    g - generic
    h - hierarchical
    i - ideal
    n - noncombinational
    E - extracted timing model
    Q - Quick timing model
```


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Route_opt : Use ICC2 (IC Compiler II tool) Analyze Timing Path

- **Fix VIOLATED Timing Path use Imported Path Pins (Visual Mode)** ***Analyze the timing path from pages 131 to 134***
  - **Check "hold_fix.rpt" timing report**

DAT

Point	Fanout	Trans	Derate	Delta	Incr	Path	Location	Attributes
clock SDRAM_CLK (fall edge)					2.000	2.000		
source latency					0.000	2.000		
I_CLOCK_GEN/I_PLL_SD/CLK_1X (PLL)		0.026	1.000		0.000	2.000 f	(901.30,664.39)	s, n
I_CLOCK_GEN/net_sdram_clk (net)	1							d
I_CLOCK_GEN/I_PLL_SD/CLK_1X (PLL)		0.026	1.000		0.000	2.000 f	(901.30,664.39)	s, n
I_CLOCK_GEN/net_sdram_clk (net)	1							d
I_CLOCK_GEN/cts_trgdly_8724/A (INVX2_RVT)		0.026	1.000		0.000	2.000 f	(895.60,663.25)	
I_CLOCK_GEN/cts_trgdly_8724/Y (INVX2_RVT)		0.043	1.000		0.040	2.040 r	(895.73,663.15)	
I_CLOCK_GEN/cts7 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8728/A (INVX2_RVT)		0.043	1.000		0.000	2.040 r	(896.21,744.98)	
I_CLOCK_GEN/cts_trgdly_8728/Y (INVX2_RVT)		0.043	1.000		0.039	2.079 f	(896.33,745.08)	
I_CLOCK_GEN/cts9 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8730/A (INVX8_RVT)		0.043	1.000		0.000	2.079 f	(855.01,743.51)	
I_CLOCK_GEN/cts_trgdly_8730/Y (INVX8_RVT)		0.025	1.000		0.024	2.103 r	(855.14,743.40)	
I_CLOCK_GEN/cts11 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8729/A (INVX8_RVT)		0.025	1.000		0.000	2.104 r	(895.75,736.82)	
I_CLOCK_GEN/cts_trgdly_8729/Y (INVX8_RVT)		0.020	1.000		0.018	2.122 f	(895.88,736.71)	
I_CLOCK_GEN/cts10 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8727/A (INVX2_RVT)		0.020	1.000		0.000	2.122 f	(895.29,664.72)	
I_CLOCK_GEN/cts_trgdly_8727/Y (INVX2_RVT)		0.016	1.000		0.018	2.141 r	(895.42,664.83)	
I_CLOCK_GEN/cts8 (net)	1							d
I_CLOCK_GEN/cts_trgdly_8723/A (INVX2_RVT)		0.016	1.000		0.000	2.141 r	(896.21,669.94)	
I_CLOCK_GEN/cts_trgdly_8723/Y (INVX2_RVT)		0.013	1.000		0.014	2.155 f	(896.33,669.84)	
I_CLOCK_GEN/cts6 (net)	1							d
I_CLOCK_GEN/macro_guide_buf_5/A (NBUFFX2_RVT)		0.013	1.000		0.000	2.155 f	(897.34,664.73)	
I_CLOCK_GEN/macro_guide_buf_5/Y (NBUFFX2_RVT)		0.028	1.000		0.046	2.201 f	(897.85,664.83)	
I_CLOCK_GEN/guide_buf_5 (net)	1							d
I_CLOCK_GEN/U9/A3 (AO22X1_RVT)		0.028	1.000		0.000	2.201 f	(899.26,697.28)	s
I_CLOCK_GEN/U9/Y (AO22X1_RVT)		0.056	1.000		0.078	2.279 f	(898.82,696.57)	s
I_CLOCK_GEN/o_sdram_clk (net)	2							d
I_ORCA_TOP/ZCTSINV_17099_7271/A (INVX8_RVT)		0.056	1.000		0.000	2.279 f	(899.40,698.16)	
I_ORCA_TOP/ZCTSINV_17099_7271/Y (INVX8_RVT)		0.033	1.000		0.030	2.309 r	(899.53,698.27)	
I_ORCA_TOP/ZCTSNET_42 (net)	1							d
I_ORCA_TOP/ZCTSINV_15998_7270/A (INVX16_RVT)		0.033	1.000		0.000	2.310 r	(896.66,673.28)	
I_ORCA_TOP/ZCTSINV_15998_7270/Y (INVX16_RVT)		0.033	1.000		0.028	2.338 f	(896.79,673.18)	
I_ORCA_TOP/ZCTSNET_41 (net)	6							d
I_ORCA_TOP/ZCTSINV_4984_7269/A (INVX16_RVT)		0.033	1.000		0.003	2.341 f	(799.23,617.91)	
I_ORCA_TOP/ZCTSINV_4984_7269/Y (INVX16_RVT)		0.030	1.000		0.028	2.369 r	(799.36,618.01)	
I_ORCA_TOP/ZCTSNET_40 (net)	3							d

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Route_opt : Use ICC2 (IC Compiler II tool) Analyze Timing Path

- **Fix VIOLATED Timing Path use Imported Path Pins (Visual Mode)** ***Analyze the timing path from pages 131 to 134***
  - **Check "hold_fix.rpt" timing report**

DAT

I_ORCA_TOP/ZCTSINV_1914_7267/A (INVX2_RVT)		0.031	1.000	0.003	2.372 r	(720.19,577.78)	
I_ORCA_TOP/ZCTSINV_1914_7267/Y (INVX2_RVT)		0.055	1.000	0.047	2.420 f	(720.32,577.88)	
I_ORCA_TOP/ZCTSNET_39 (net)	2						d
I_ORCA_TOP/ZCTSINV_1239_f_7266/A (INVX8_RVT)		0.055	1.000	0.000	2.420 f	(705.60,562.93)	
I_ORCA_TOP/ZCTSINV_1239_f_7266/Y (INVX8_RVT)		0.051	1.000	0.048	2.468 r	(705.73,562.83)	
I_ORCA_TOP/ZCTSNET_38 (net)	6						d
I_ORCA_TOP/I_SDRAM_IF/ZCTSINV_367_7256/A (INVX4_RVT)		0.051	1.000	0.005	2.473 r	(541.59,442.55)	
I_ORCA_TOP/I_SDRAM_IF/ZCTSINV_367_7256/Y (INVX4_RVT)		0.049	1.000	0.044	2.517 f	(541.72,442.44)	
I_ORCA_TOP/I_SDRAM_IF/ZCTSNET_75 (net)	16						d
I_ORCA_TOP/I_SDRAM_IF/genblk1_12__sd_mux_dq_out/S0 (MUX21X1_RVT)		0.049	1.000	0.001	2.517 f	(530.61,427.45)	s
I_ORCA_TOP/I_SDRAM_IF/genblk1_12__sd_mux_dq_out/Y (MUX21X1_RVT)		0.043	1.000	0.096	2.613 r	(529.68,427.41)	s
I_ORCA_TOP/I_SDRAM_IF/sd_DQ_out[12] (net)	1						
ZBUF_4_inst_9157/A (NBUFFX8_RVT)		0.043	1.000	0.000	2.613 r	(535.13,427.30)	
ZBUF_4_inst_9157/Y (NBUFFX8_RVT)		0.022	1.000	0.049	2.662 r	(534.09,427.40)	
ZBUF_4_51 (net)	1						
eco_cell_1/A (NBUFFX2_RVT)		0.022	1.000	0.000	2.662 r	(531.17,400.61)	
eco_cell_1/Y (NBUFFX2_RVT)		0.046	1.000	0.062	2.725 r	(531.69,400.70)	
eco_net_1_ZBUF_4_51 (net)	1						
eco_cell_0/A (NBUFFX2_RVT)		0.046	1.000	0.001	2.725 r	(531.17,300.61)	
eco_cell_0/Y (NBUFFX2_RVT)		0.150	1.000	0.139	2.864 r	(531.69,300.70)	
eco_net_0_ZBUF_4_51 (net)	1						
SDRAM_DQ_Bus_12__sdram_DQ_iopad/DIN (B16I1025_NS)		0.150	1.000	0.005	2.869 r	(530.29,199.58)	s, d
SDRAM_DQ_Bus_12__sdram_DQ_iopad/PADIO (B16I1025_NS)		0.000	1.000	1.434	4.303 r	(540.00,2.10)	s, d
sd_DQ[12] (net)	2						
sd_DQ[12] (inout)		0.000	1.000	0.000	4.303 r	(540.00,2.10)	
data arrival time					4.303		

Added Buffer

DRT

Write Timing report and analyze it

clock SD_DDR_CLK (fall edge)				2.000	2.000		
source latency				0.000	2.000		
I_CLOCK_GEN/I_PLL_SD/CLK_1X (PLL)		0.029	1.000	0.000	2.000 f	(901.30,664.39)	s, n
I_CLOCK_GEN/net_sdram_clk (net)	1						d
I_CLOCK_GEN/cts_trgdly_8724/A (INVX2_RVT)		0.029	1.000	0.000	2.000 f	(895.60,663.25)	
I_CLOCK_GEN/cts_trgdly_8724/Y (INVX2_RVT)		0.046	1.000	0.043	2.043 r	(895.73,663.15)	
I_CLOCK_GEN/cts7 (net)	1						d
I_CLOCK_GEN/cts_trgdly_8728/A (INVX2_RVT)		0.046	1.000	0.000	2.043 r	(896.21,744.98)	
I_CLOCK_GEN/cts_trgdly_8728/Y (INVX2_RVT)		0.050	1.000	0.045	2.089 f	(896.33,745.08)	
I_CLOCK_GEN/cts9 (net)	1						d
I_CLOCK_GEN/cts_trgdly_8730/A (INVX8_RVT)		0.050	1.000	0.000	2.089 f	(855.01,743.51)	
I_CLOCK_GEN/cts_trgdly_8730/Y (INVX8_RVT)		0.030	1.000	0.028	2.117 r	(855.14,743.40)	
I_CLOCK_GEN/cts11 (net)	1						d
I_CLOCK_GEN/cts_trgdly_8729/A (INVX8_RVT)		0.030	1.000	0.000	2.117 r	(895.75,736.82)	
I_CLOCK_GEN/cts_trgdly_8729/Y (INVX8_RVT)		0.022	1.000	0.020	2.137 f	(895.88,736.71)	
I_CLOCK_GEN/cts10 (net)	1						d
I_CLOCK_GEN/cts_trgdly_8727/A (INVX2_RVT)		0.022	1.000	0.000	2.138 f	(895.29,664.72)	
I_CLOCK_GEN/cts_trgdly_8727/Y (INVX2_RVT)		0.019	1.000	0.021	2.159 r	(895.42,664.83)	
I_CLOCK_GEN/cts8 (net)	1						d
I_CLOCK_GEN/cts_trgdly_8723/A (INVX2_RVT)		0.019	1.000	0.000	2.159 r	(896.21,669.94)	
I_CLOCK_GEN/cts_trgdly_8723/Y (INVX2_RVT)		0.015	1.000	0.015	2.174 f	(896.33,669.84)	
I_CLOCK_GEN/cts6 (net)	1						d

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Route_opt : Use ICC2 (IC Compiler II tool) Analyze Timing Path

- **Fix VIOLATED Timing Path use Imported Path Pins (Visual Mode)** ***Analyze the timing path from pages 131 to 134***
  - **Check "hold_fix.rpt" timing report**

DRT & Slack

```
I_CLOCK_GEN/macro_guide_buf_5/A (NBUFFX2_RVT)                                0.015      1.000              0.000      2.174 f    (897.34,664.73)
I_CLOCK_GEN/macro_guide_buf_5/Y (NBUFFX2_RVT)                                0.029      1.000              0.047      2.221 f    (897.85,664.83)
I_CLOCK_GEN/guide_buf_5 (net)                                  1                                                                                 d
I_CLOCK_GEN/U9/A3 (AO22X1_RVT)                                               0.029      1.000              0.000      2.221 f    (899.26,697.28)   s
I_CLOCK_GEN/U9/Y (AO22X1_RVT)                                                0.076      1.000              0.092      2.313 f    (898.82,696.57)   s
I_CLOCK_GEN/o_sdram_clk (net)                                  2                                                                                 d
I_ORCA_TOP/ZCTSINV_17099_7271/A (INVX8_RVT)                                  0.076      1.000              0.000      2.314 f    (899.40,698.16)
I_ORCA_TOP/ZCTSINV_17099_7271/Y (INVX8_RVT)                                  0.043      1.000              0.038      2.352 r    (899.53,698.27)
I_ORCA_TOP/ZCTSNET_42 (net)                                    1                                                                                 d
I_ORCA_TOP/ZCTSINV_15998_7270/A (INVX16_RVT)                                 0.043      1.000              0.000      2.352 r    (896.66,673.28)
I_ORCA_TOP/ZCTSINV_15998_7270/Y (INVX16_RVT)                                 0.044      1.000              0.035      2.387 f    (896.79,673.18)
I_ORCA_TOP/ZCTSNET_41 (net)                                    6                                                                                 d
I_ORCA_TOP/ZCTSINV_4984_7269/A (INVX16_RVT)                                  0.044      1.000              0.004      2.392 f    (799.23,617.91)
I_ORCA_TOP/ZCTSINV_4984_7269/Y (INVX16_RVT)                                  0.037      1.000              0.033      2.425 r    (799.36,618.01)
I_ORCA_TOP/ZCTSNET_40 (net)                                    3                                                                                 d
I_ORCA_TOP/inv_drc_cln8538/A (INVX16_RVT)                                    0.038      1.000              0.004      2.429 r    (724.29,577.78)
I_ORCA_TOP/inv_drc_cln8538/Y (INVX16_RVT)                                    0.046      1.000              0.031      2.460 f    (724.42,577.88)
I_ORCA_TOP/cts3 (net)                                         14                                                                                 d
I_ORCA_TOP/I_SDRAM_IF/sd_mux_CK/S0 (MUX21X1_RVT)                             0.053      1.000              0.016      2.476 f    (508.72,781.91)   s
I_ORCA_TOP/I_SDRAM_IF/sd_mux_CK/Y (MUX21X1_RVT)                              0.075      1.000              0.140      2.616 f    (509.65,781.88)   s
I_ORCA_TOP/I_SDRAM_IF/sd_CK (net)                              1                                                                                 d
ZCTSINV_160_7279/A (INVX8_RVT)                                               0.075      1.000              0.000      2.616 f    (511.11,780.29)
ZCTSINV_160_7279/Y (INVX8_RVT)                                               0.041      1.000              0.036      2.652 r    (510.98,780.19)
ZCTSNET_5 (net)                                                1                                                                                 d
ZCTSINV_83_7278/A (INVX16_RVT)                                               0.041      1.000              0.000      2.652 r    (508.15,780.29)
ZCTSINV_83_7278/Y (INVX16_RVT)                                               0.051      1.000              0.033      2.685 f    (508.28,780.19)
ZCTSNET_4 (net)                                                1                                                                                 d
sdram_CK_iopad/DIN (D8I1025_NS)                                              0.059      1.000              0.017      2.703 f    (469.71,1240.42)  d
sdram_CK_iopad/PADIO (D8I1025_NS)                                            0.891      1.000              1.391      4.094 f    (460.00,1437.90)  d
sd_CK (net)                                                    2                                                                                 d
sd_CK (out)                                                                  0.891      1.000              0.000      4.094 f    (460.00,1437.90)
sd_CK (out) (gclock 'SD_DDR_CLK' source)                                     0.891      1.000              0.000      4.094 f    (460.00,1437.90)

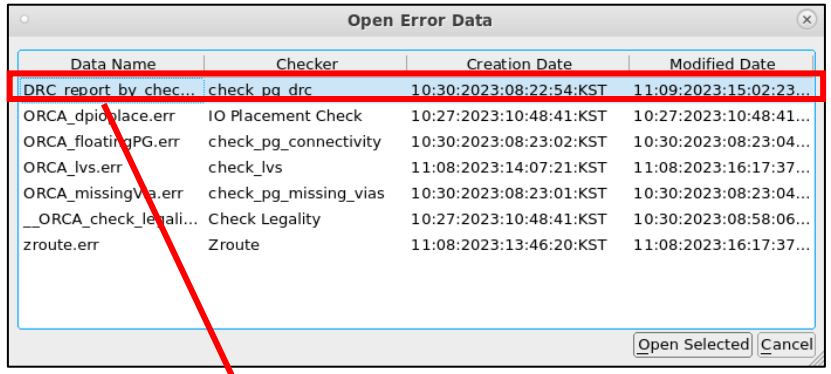
clock reconvergence pessimism                                                                              -0.000      4.094
output external delay                                                                                       0.100      4.194
data required time                                                                                                     4.194
-------------------------------------------------------------------------------------------------------------------------------------------------
data required time                                                                                                     4.194
data arrival time                                                                                                     -4.303
-------------------------------------------------------------------------------------------------------------------------------------------------
slack (MET)                                                                                                            0.109
```

DAT > DRT = MET

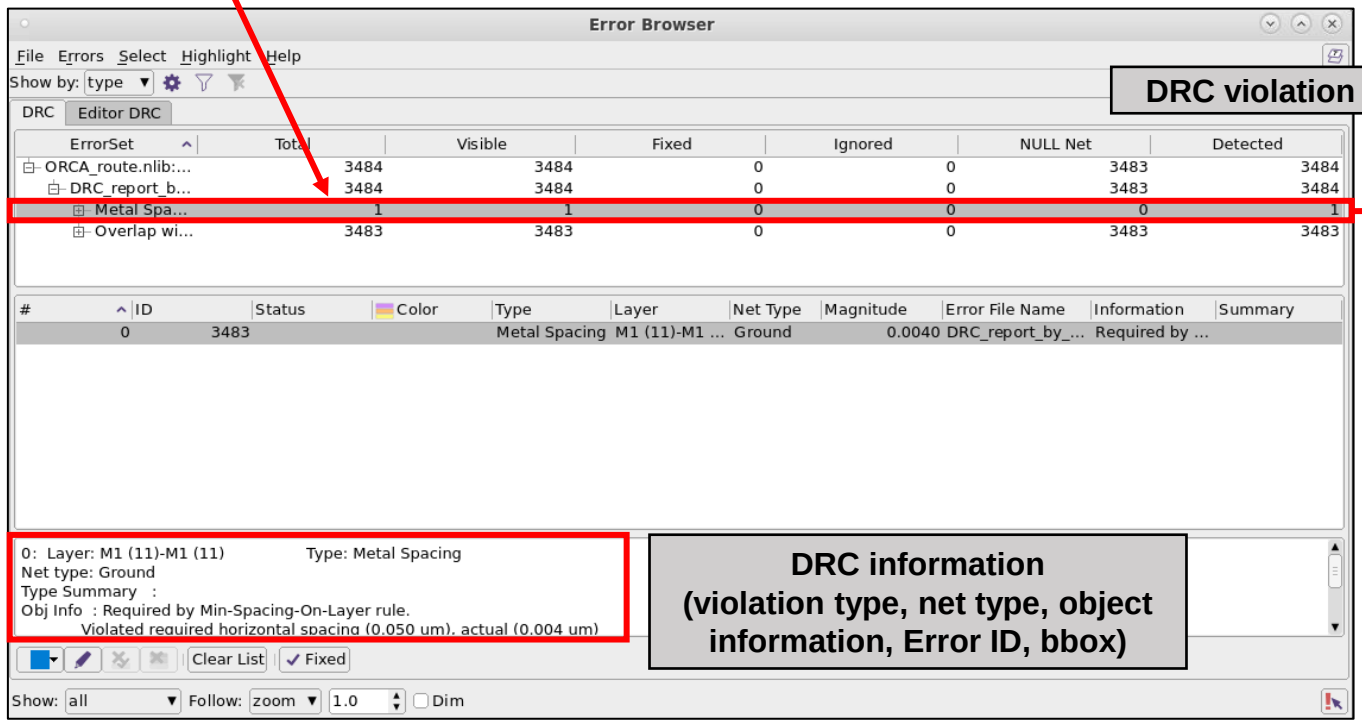
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Route_opt : Use ICC2 (IC Compiler II tool) check_pg_drc

- **Analyze check_pg_drc After "route_opt"**



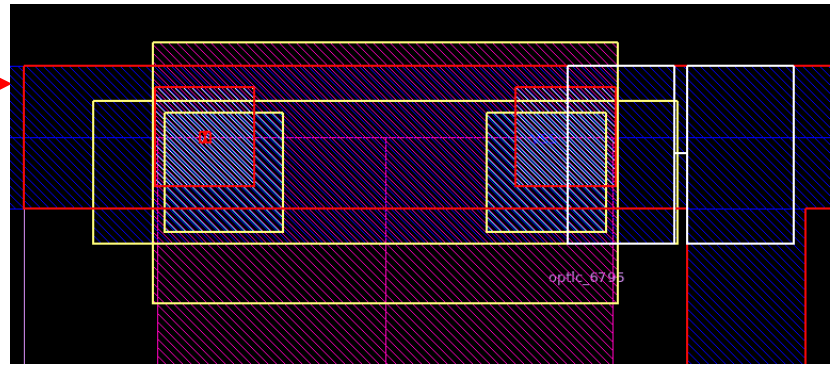
- **Analayze Zroute Error**
  1. Open Error Data and choose zroute.err file
  2. Open Error Browser and choose DRC violation list
  3. Check NULL Net & Wave violation list before analyze
  4. Select the DRC violation list, the violations are displayed in the Layout view, and are moved to the selected violations location
  5. Check Net_Type & Net name
  6. Analyze DRC information and fix violation



DRC violation list

DRC information (violation type, net type, object information, Error ID, bbox)

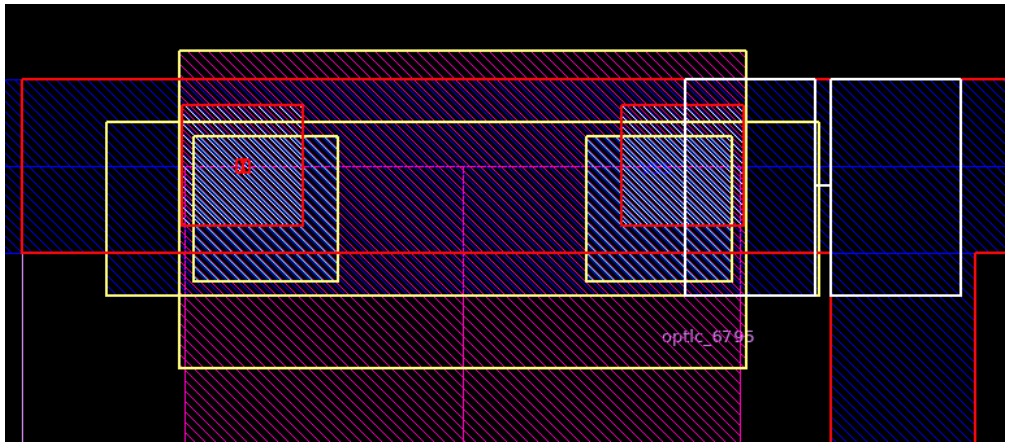
- **Wave list in ORCA Overlap with Blockage Error**



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Route_opt : Use ICC2 (IC Compiler II tool) check_pg_drc

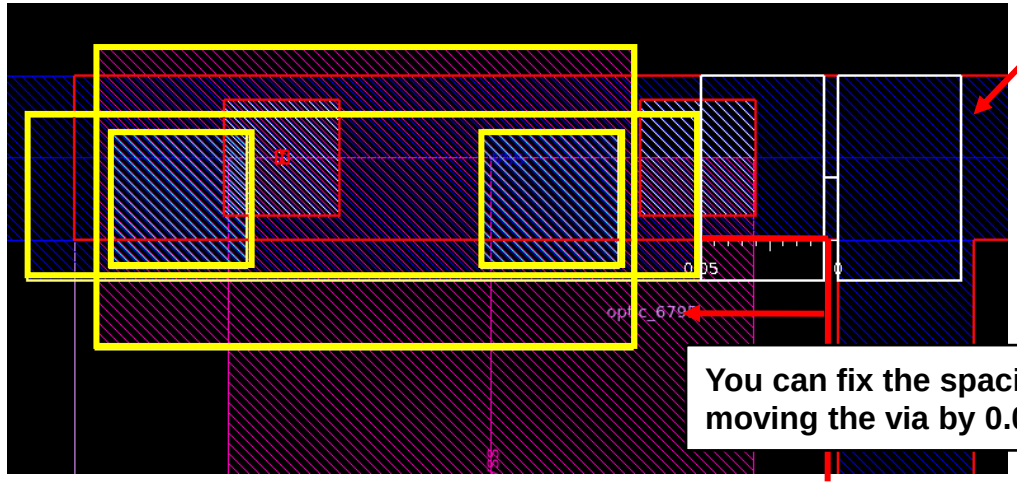
- **Fix "check_pg_drc" Error (Overlap Blockage Error)**



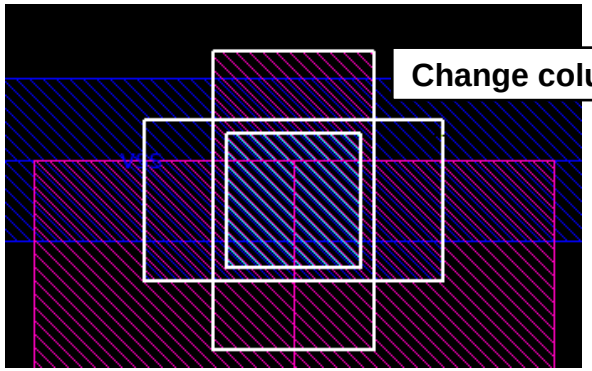
```
0:  Layer: M1 (11)-M1 (11)              Type: Metal Spacing
Net type: Ground
Type Summary   :
Obj Info  : Required by Min-Spacing-On-Layer rule.
            Violated required horizontal spacing (0.050 um), actual (0.004 um)
             Net1: VSS
             Net2: VSS
Error ID: 3483              Status: Error
Bbox      : (410.529 1188.951) (410.625 1189.026)
```

- **This is a spacing Error between Pin(VSS, M1) and VIA(VSS,M1)**
- **The distance between pin and VIA shall be greater than 0.004(um) in the Horizontal direction**

***Solution 1**



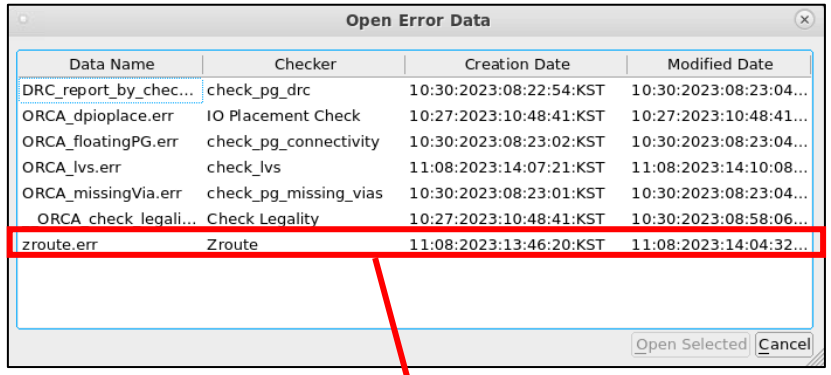
***Solution 2**



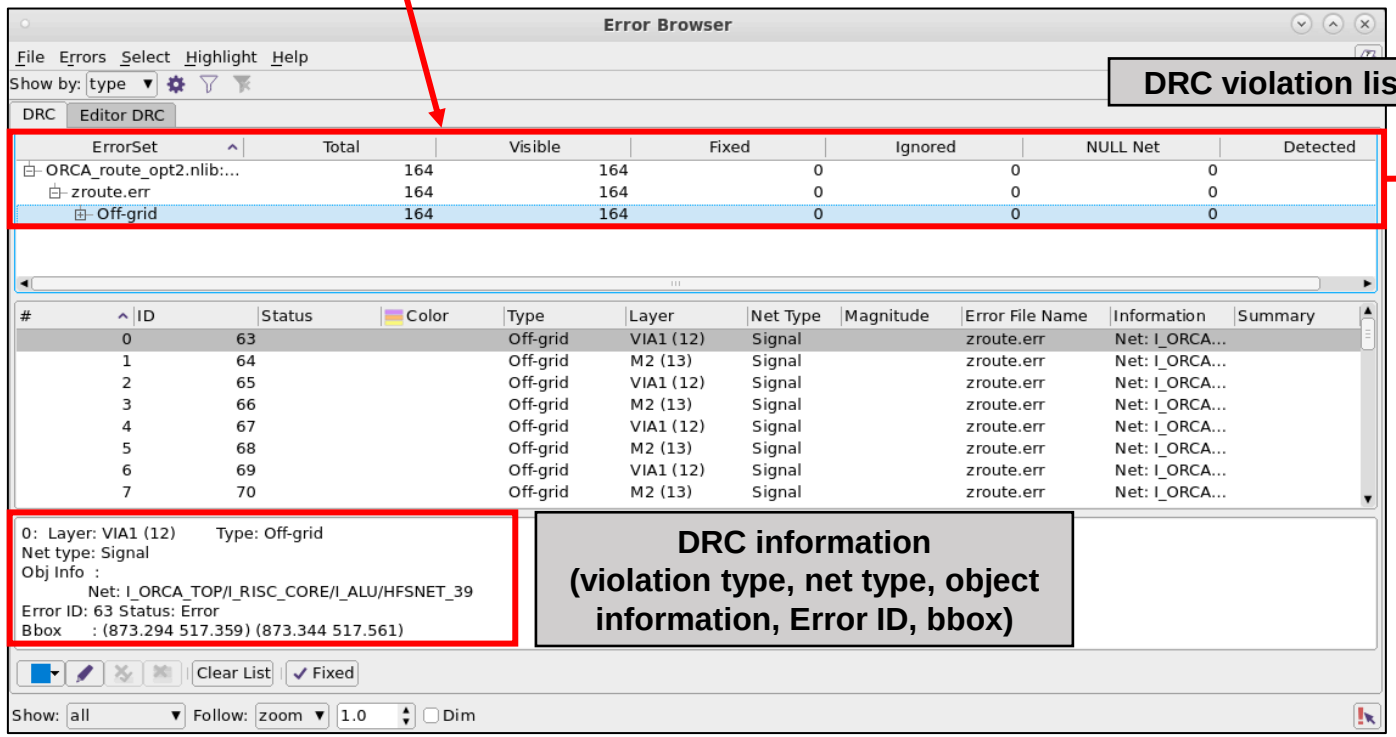
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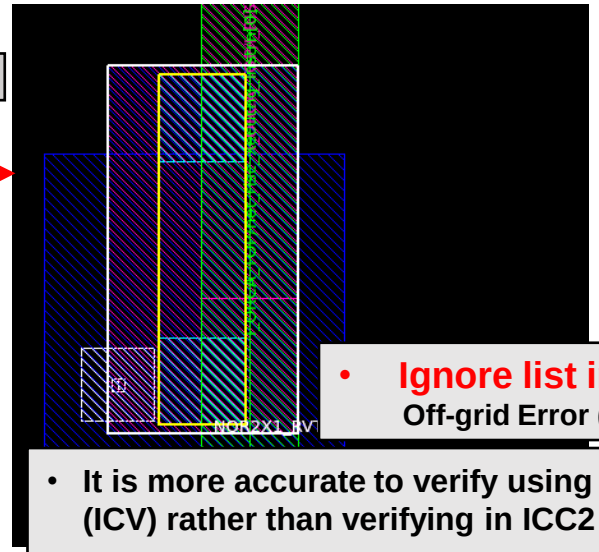
Route_opt : Use ICC2 (IC Compiler II tool) check_routes

- **Analyze check_routes After "route_opt"**



- **Analayze Zroute Error**
  1. Open Error Data and choose zroute.err file
  2. Open Error Browser and choose DRC violation list
  3. Check NULL Net & Wave violation list before analyze
  4. Select the DRC violation list, the violations are displayed in the Layout view, and are moved to the selected violations location
  5. Check Net_Type & Net name
  6. Analyze DRC information and fix violation





- **Ignore list in ORCA**
  Off-grid Error (M2, VIA1)

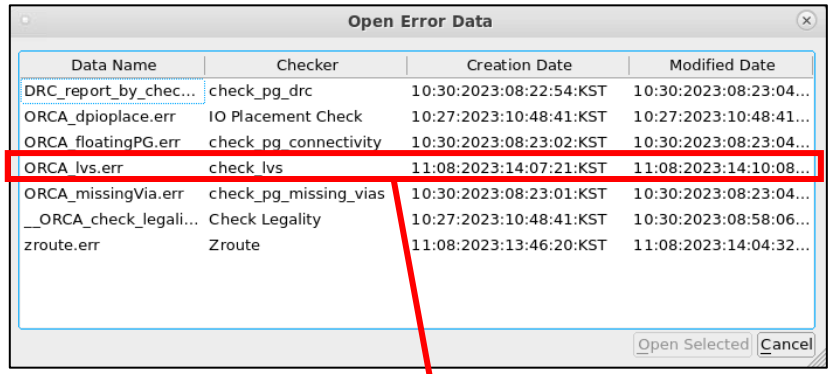
- **It is more accurate to verify using PV Tool (ICV) rather than verifying in ICC2**
- **Although it may be Error in ICC2, it may not be Error in PV Tool (ICV)**

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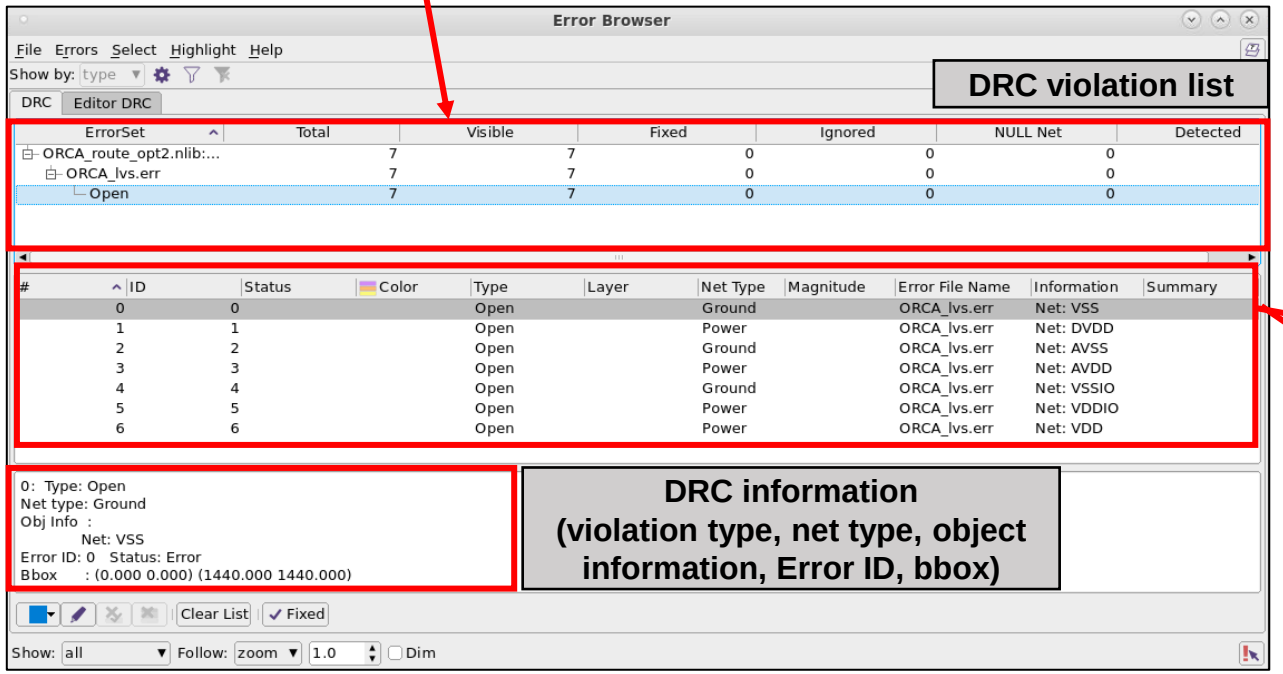
Route : Use ICC2 (IC Compiler II tool) Check_lvs (open, short)

12. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/06_route/script/11_check_timing_and_DRC.tcl

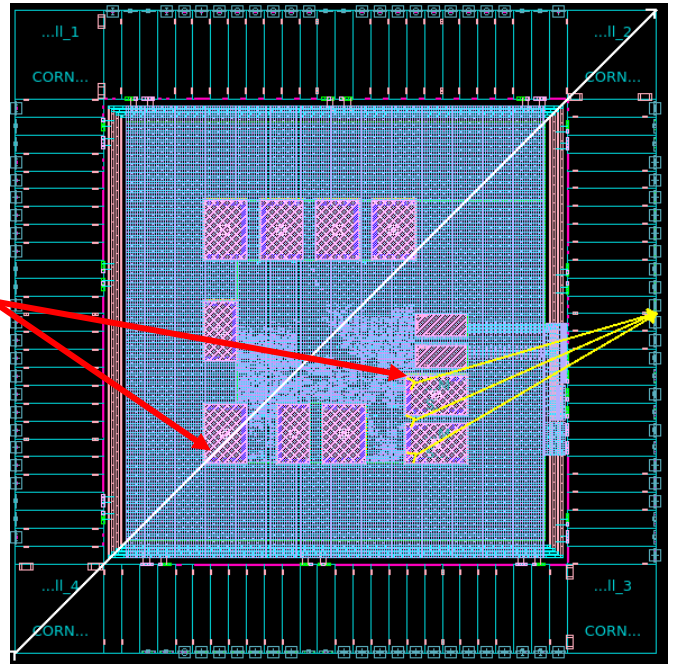
- **Analyze check_lvs (In the route_auto step, short,open,floating Errors occur because routing is not optimized yet)**



- **Analayze check_lvs Error**
  1. Open Error Data and choose *_lvs.err file
  2. Open Error Browser and choose DRC violation list
  3. Check NULL Net & Wave violation list before analyze
  4. Select the DRC violation list, the violations are displayed in the Layout view, and are moved to the selected violations location
  5. Analyze DRC information and fix violation (fix all signal, clock net open, short, floating)



Check if the clock, signal net is open or short



- **You can ignore This Error because it's not a real power, ground open**
- **Clock, signal net open, short are must be fixed**

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WEEK 13 : Practice for Routing

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Routing: Use ICC2 (IC Compiler II tool) HomeWork

- **In the "Exercise for routing", Run the routing under several case, and check the results"**

  - **Using script & psyn.nlib**

    routing script - ${WORK_DIR}/2_BACKEND/1_pnr/06_route/script/*

    psyn.nlib - ${WORK_DIR}/2_BACKEND/1_pnr/05_cts/nlib/ORCA_psyn.nlib

    psyn.nlib - ${WORK_DIR}/2_BACKEND/1_pnr/05_cts/nlib/ORCA_psyn_homework.nlib

  - **Guidelines**

  - Exercise 1 – Compare Two Timing Path : route & route_opt
  - Exercise 2 – Compare Two DRC violation : route & route_opt
  - Exercise 3 – run route & route_opt with ORCA_psyn_homework.nlib
  - Exercise 4– Compare the results between ORCA_route_opt_homework.nlib and ORCA_route_opt.nlib

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Route & Route_opt : Use ICC2 (IC Compiler II tool) HomeWork

- **Exercise 1 – Compare Two Timing Path in route & route_opt**
  - **Analyze violated paths in route first**

    Timing reports path : ${WORK_DIR}/2_BACKEND/1_pnr/06_route/reports/route/*

  - **Timing path(VIOLATED) to compare**

```
Start_point - I_ORCA_TOP/I_RESET_BLOCK/sdram_rst_n_buf_reg
End_point - I_ORCA_TOP/I_SDRAM_IF/sd_DQ_en_reg_5_
```

  - **Use report_timing command to compare two paths (use –from –to option) after refine opt**

```
report_timing -scenarios func.ss_125c \
              -significant_digits 3 -delay_type min \
              -input_pins -physical -attributes -nosplit \
              -derate -crosstalk_delta -path_type full_clock \
              -from $StartPoint -to $EndPoint \
             > homework_timing.rpt
```

  - **Compare Timing Path use "imported path pins" in ICC2 (use Homework_timing_report)**
  - **Analyze how the violated path changed after route_opt**
  - **IS this Path Optimized in route_opt?**

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Route & Route_opt : Use ICC2 (IC Compiler II tool) HomeWork

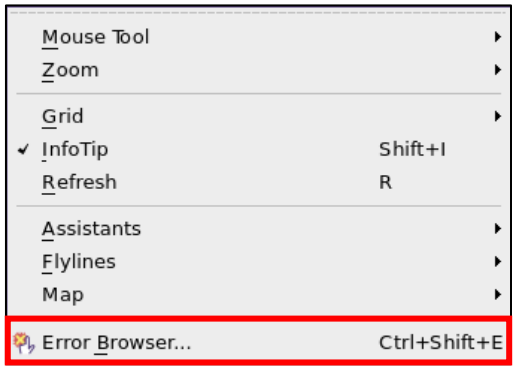
- **Exercise 2 – Compare Two DRC violation in route & route_opt**
  - **DRC violation to compare**

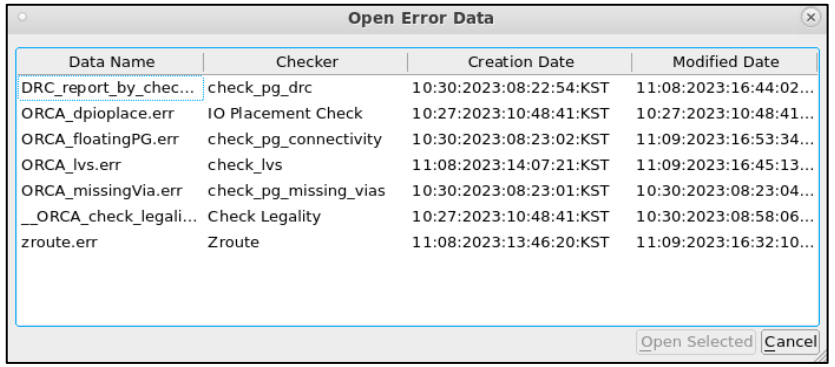
❖Check_routes

❖Check_lvs

❖Check_pg_drc

  - **Use Error browser to compare route & route_opt**





  - **Has DRC violation occurred during routing been resolved through routing optimization?**

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Route & Route_opt : Use ICC2 (IC Compiler II tool) HomeWork

- **Exercise 3 – run CTS & PSYN with ORCA_refine_opt_homework.nlib**

  - **Run CTS & PSYN with ORCA_refine_opt_homework.nlib**

    Psyn.nlib path : ${WORK_DIR}/2_BACKEND/1_pnr/05_cts/nlib/ORCA_psyn_homework.nlib

    Route & Route_opt nlib path : ${WORK_DIR}/2_BACKEND/1_pnr/06_route/nlib/*

    Save nlib name : ORCA_route_homework.nlib & ORCA_route_opt_homework.nlib

- **Exercise 4 – Compare the results between ORCA_route_opt_homework.nlib and ORCA_route_opt. nlib**

  - **Compare reports after Routing & Route_opt**

    Routing reports path : ${WORK_DIR}/2_BACKEND/1_pnr/06_route/reports/route or route_opt/*

    ❖ *Please Backup the report files, report during LAB before proceeding with Homeworks separately*

  - **Which floorplan has better results?**

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Thank You

